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Company Relationship Modeling and Graph Neural Networks for Financial Market Forecasting

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Abstract

This thesis presents a series of studies for financial market forecasting, leveraging various graph neural networks based on a novel company relationship modeling scheme.

Departing from the conventional view of treating companies as standalone entities, this thesis models them as interconnected nodes within a Semantic Company Relationship Graph (SCRG). To achieve this, statistics on the co-occurrence of company names are compiled from a comprehensive financial news corpus, reflecting patterns of frequent business interactions collectively. These statistics are then used to create vector embeddings for each company, thus positioning all companies within the same semantic relationship space. The cosine similarities between these vectors are employed to define the numerical interrelationships among companies, thereby constructing the SCRG. This innovative relationship modeling scheme is grounded in the principles of statistical semantics and the distributional hypothesis in linguistics, which posit that patterns of word co-occurrence in a large corpus can effectively delineate semantic interconnections.

Building on the SCRG’s relationship foundation, this thesis explores the adaptation of spatial-temporal graph neural networks for predicting stock movements. A key innovation is the introduction of the Non-Independent and Identically Distributed Spatial-Temporal Graph Neural Network (NIST-GNN). This model is uniquely designed to integrate features from neighboring companies and domestic historical time-series data. It effectively addresses the temporal non-IID characteristics of stock data, enabling a more nuanced analysis of each stock’s temporal dynamics. Empirical results demonstrate that this methodology significantly outperforms existing benchmarks in profitability with better risk management. The experimental findings reveal insights into the dynamics of information spread within the US market, uncovering a typical one-day delay in the diffusion of public information among interrelated companies, thus challenging traditional views on market efficiency.

Secondly, this thesis investigates the inference of absent news sentiment during periods with no media coverage, extending the use of the SCRG. News sentiment is a crucial proxy for investor sentiment and is widely used in asset pricing. However, consistent media coverage is not guaranteed for all companies, many of which experience “media silent” periods, especially as media attention shifts towards more sensational business news. An analysis of 14 years of news data reveals that even well-known companies lack daily news coverage on almost half of the trading days. Traditional missing value imputation (MVI) methods are abundant but generally insufficient for the finance

context, characterized by complex spatial and temporal interconnections among companies. To address these challenges, this thesis proposes a Non-IID Spatial-Temporal Chebyshev Network (NIST-Cheb) to leverage these relationships for inferring non-existent news sentiment. A masked semi-supervised training approach is introduced to enhance the utility of the available sentiment data. The efficacy of this method is systematically validated through error-based metrics and empirical trading results. Experimental findings indicate that asset pricing models incorporating NIST-Cheb’s estimated sentiment scores significantly outperform traditional baselines. Theoretical contributions also discuss the spillover effects of news sentiment, emphasizing the importance and feasibility of using spatial and temporal sentiment information to infer absent news sentiment.

The concluding part of this thesis focuses on the prediction of intraday market index movements, utilizing the SCRG as a foundational relationship prior for industry hierarchical analysis. It is known that previous studies on market index predictions have leaned heavily on machine learning strategies that predominantly targeted the temporal dynamics of market indices, often overlooking the valuable insights from the market microstructures of the underlying industrial clusters. The emergence of hierarchical graph pooling techniques marks a new direction in this field. This thesis pioneers the use of these techniques by framing market index prediction as a graph classification task and introduces a FinPool graph pooling operator, designed for the hierarchical feature representation of industrial clusters in the financial market. To optimally apply FinPool operators for index prediction, two innovative prediction frameworks, Stacked FinPool and Multi-tier Attention FinPool, are proposed, based on the insights of the Global Industry Classification Standard (GICS). Empirical trading evaluations indicate a notable improvement in profits and risk-adjusted returns, significantly outperforming conventional benchmarks. These findings not only challenge the Efficient Market Hypothesis but also demonstrate the untapped predictive power inherent in the microstructural details of market constituents.

Lay Summary

This thesis explores how advanced machine learning techniques, specifically graph neural networks (GNNs), can improve the forecasting of financial markets. It introduces a new way to understand how companies are related to one another in the market, using a concept called the Semantic Company Relationship Graph (SCRG). This model represents companies not as isolated entities, but as interconnected nodes within a network, capturing how frequently and significantly they interact based on shared business events, as reported in financial news. These connections are then used to assess the strength of relationships between companies. The aim is to reflect how the performance of one company can affect others, which traditional prediction models often overlook.

Building on this new relationship modeling framework, the thesis introduces an innovative model called the Non-Independent and Identically Distributed Spatial-Temporal Graph Neural Network (NIST-GNN). This model takes into account not only the individual performance of companies but also how they influence each other over time, addressing an important aspect of financial data: the fact that stock market data does not behave in a simple, predictable manner. The results show that this model can more accurately predict stock movements, outperforming existing models and improving risk management.

A key challenge in the financial markets is the lack of consistent news coverage for many companies, which can lead to missing or incomplete sentiment data. To tackle this, the thesis also proposes the NIST-Cheb model, which uses related companies' news sentiment to predict sentiment for companies with no coverage, thus filling in the gaps. The model is validated through empirical tests and trading simulations, showing that it significantly enhances prediction accuracy compared to traditional methods.

Finally, the thesis shifts to predicting market index movements, traditionally done by analyzing market quotes alone. It introduces the FinPool model, which treats the prediction of market indices as a graph classification task, using hierarchical information from industry clusters. This model shows how the interactions within industries — such as companies in the same sector — can provide deeper insights into market movements, improving forecasting accuracy. The FinPool model also performs better than conventional models in real-world trading simulations.

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Chapter 1

Backgrounds and Introduction

This thesis explores the two research questions of Company Relationship Modeling and Graph Neural Networks for Financial Market Forecasting. As the beginning of this thesis, this chapter provides the necessary background to equip readers with the foundational knowledge required to understand the core concepts of the thesis. This includes an overview of graphs and graph neural networks (section 1.1), as well as a review of financial time series modeling (section 1.2). We then introduce the first research question — Semantic Company Relationship Modeling — in section 1.3, where we propose the innovative Semantic Company Relationship Graph (SCRG) that serves as a powerful framework for representing complex interconnections between companies within the financial market. Building on this foundation, section 1.4 demonstrates how our proposed Graph Neural Network techniques can be applied to three specific market forecasting applications by leveraging the relational insights derived from the SCRG framework. After that, section 1.5 outlines the specific research contributions of this thesis, providing a roadmap for the subsequent chapters. Finally, section 1.6 presents the notations that will be used throughout the thesis.

1.1 Graphs and Graph Neural Networks

1.1.1 Graphs

Graphs are a powerful and versatile data structure used to represent complex systems through a set of objects (nodes) and their relationships (edges) (Hamilton, 2020). They are widely used across various domains, such as social networks, where nodes represent individuals and edges represent friendships or interactions (Wu et al., 2019a),

and in transportation systems, where cities are nodes and routes are edges (Yu et al., 2018). Similarly, in biological networks, graphs model interactions between proteins or genes, offering insights into complex biological processes (Strokach et al., 2020).

In the context of this thesis, graphs serve as a crucial framework for modeling the intricate relationships between companies within the financial market. They allow for the capture of relational data, emphasizing interactions rather than isolated properties of individual companies. For instance, companies might be connected by shared market sectors, cooperation, supply chain relationships, or investment dependencies, each of which can be represented as edges in a graph (Lozano et al., 2013; Cohen and Frazzini, 2008). By focusing on these interconnections, graphs can offer a more holistic view of the financial market, enabling better analysis of systemic risks, market dynamics, and company behaviors (Cheng et al., 2019; Kim et al., 2019).

At the most basic level, a graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$ consists of a set of nodes $\mathcal{N}$, representing entities such as companies, and a set of edges $\mathcal{E}$, representing the relationships between these entities. In practice, graph data is represented through an adjacency matrix A , where each entry A_{uv} indicates the presence or absence of a relationship between two company nodes u and v . Depending on the application, these edges can be directed or undirected, weighted or unweighted. Beyond edges, graphs can also incorporate additional feature information at the node level: each node $v \in \mathcal{N}$ is associated with a feature vector $x_v \in \mathbb{R}^d$, and the entire set of node features is represented by $X \in \mathbb{R}^{N \times d}$, where N is the number of nodes and d is the dimensionality of the feature vectors.

It should be noted that, in some academic communities, the term "network" is preferred over "graph". For the purposes of this thesis, we will use these terms interchangeably.

1.1.2 Measures for Network Analysis

Since graph is a complex data structure with various nodes and connections, in order to grasp a comprehensive understanding for graphs, especially the larger ones, various graph analysis measures have been proposed for Exploratory Data Analysis (EDA) (Hevey, 2018).

In this thesis, we use graph analysis measures from the angles of degree, centrality, density and community structure.

Degree is the most basic measure of nodes in a graph. Degree of a node v , $d(v)$ represents the number of neighboring nodes it is connected to. Centrality is another

fundamental concept in network analysis, measuring the importance and influence of nodes in a graph (Borgatti and Everett, 2006). Since importance has a wide number of meanings, this leads to many different types of centrality. In this thesis, we adopt the widely used degree centrality and betweenness centrality for EDA.

Degree centrality of a given node v , $C_D(v)$, is a count of the number of edges incident upon a given node. It is calculated as below:

$$C_D(v) = \frac{d(v)}{N-1} \quad (1.1)$$

which is normalized by dividing by the maximum possible degree in a graph, where N is the number of nodes in G and $d(v)$ denotes the node degree (Hagberg et al., 2008).

Betweenness centrality $C_B(v)$ measures the number of times a given node v acts as a bridge along the shortest path between two other nodes. A path is a sequence of edges that joins a sequence of nodes. Nodes with high betweenness centrality often control the flow of information in the network. It is formally defined as the sum of the fraction of all-pairs shortest paths that pass through v :

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (1.2)$$

where σ_{st} is the total number of shortest paths between nodes s and t , and $\sigma_{st}(v)$ is the number of shortest paths that pass through node v (Brandes, 2008).

Density is a measure that quantifies how close a network is to being a complete graph (every node is connected to every other node). It is the ratio of the number of edges present in the network to the maximum possible number of edges. The density value ranges between 0 and 1, where a density of 1 indicates a complete graph. Mathematically, density D is defined as,

$$D = \frac{2E}{N(N-1)}, \quad (1.3)$$

where E is the total number of edges in the graph, and the denominator represents the maximum number of possible edges given N nodes.

A graph can be divided into communities based on the principle that nodes within the same community have a higher density of connections among themselves than with nodes in other communities, reflecting structural or functional groupings within the network. There are currently plenty of algorithms that can divide nodes in a graph into communities, such as the widely used Louvain Community Detection Algorithm (Blondel et al., 2008).

Modularity is a measure used to evaluate the quality of a division of a network into communities (Newman, 2018). It quantifies the strength of the division by comparing the actual density of edges within communities to the expected density of edges if the edges were distributed at random, while preserving the node degrees. The modularity Q of a graph partition is defined as,

$$Q = \frac{1}{2E} \sum_{i,j} \left[A_{ij} - \frac{d(i)d(j)}{2E} \right] \delta(c_i, c_j), \quad (1.4)$$

where $\delta(c_i, c_j)$ is defined as:

$$\delta(c_i, c_j) = \begin{cases} 1 & \text{if nodes } i \text{ and } j \text{ belong to the same community} \\ 0 & \text{otherwise} \end{cases} \quad (1.5)$$

A high modularity value (closer to 1) indicates a strong community structure, where nodes within communities are densely connected, and nodes between communities are sparsely connected; while a low value suggests that the division of the graph into communities is not better than a random configuration, indicating weak or no community structure.

1.1.3 Graph Neural Networks

Graph Neural Networks (GNNs) represent a powerful class of neural networks designed to directly operate on graphs, enabling the capture of dependencies among nodes and edges within graph-structured data. This capability makes GNNs particularly suitable for a wide range of applications, from social network analysis and recommendation systems to bio-informatics and financial predictions (Wu et al., 2020a).

The core idea behind GNNs lies in the neural message passing paradigm, where nodes exchange information with their neighbors through layers of neural networks (Gilmer et al., 2017). This process allows for the aggregation of feature information from local neighborhoods, which is then used to update the node representations iteratively. As a result, GNNs can learn complex patterns in the data that are not accessible to traditional neural network architectures.

Under the neural message passing framework, a GNN operates on an input graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ to iteratively update the representation of each node by aggregating information from its neighbors (Gilmer et al., 2017). This process can be formalized over T iterations (or layers) as follows:

Message Passing. At each iteration l , every node v receives messages from its neighboring nodes $u \in \mathcal{N}(v)$, where $\mathcal{N}(v)$ denotes the set of neighbors of v . The message $m_{uv}^{(l)}$ from node u to node v at layer l is computed based on the features of u and v . The general form of the message function can be:

$$m_{uv}^{(l)} = f\left(h_u^{(l-1)}, h_v^{(l-1)}\right), \quad (1.6)$$

where $h_u^{(l-1)}$ and $h_v^{(l-1)}$ are the feature representations of nodes u and v at layer $l-1$, respectively, and f is a differentiable function, such as a Multi-layer Neural Network (Hamilton, 2020).

Aggregation. The messages received by node v are then aggregated to form a single unified message $m_v^{(l)}$. The aggregation function, AGG, can take various forms, such as sum, mean, or max (Hamilton, 2020):

$$m_v^{(l)} = \text{AGG}\left(\{m_{uv}^{(l)}, \forall u \in \mathcal{N}(v)\}\right). \quad (1.7)$$

Feature Update. Finally, the node v 's new feature representation $h_v^{(l)}$ is updated by combining its previous feature representation $h_v^{(l-1)}$ with the aggregated message $m_v^{(l)}$, using an update function g :

$$h_v^{(l)} = g\left(h_v^{(l-1)}, m_v^{(l)}\right) \quad (1.8)$$

The update function involves a non-linear transformation, such as a neural network layer with an activation function like ReLU.

After T iterations of message passing, the final node representations $h_v^{(T)}$ can be used for various downstream tasks such as node classification, graph classification, or link prediction. The power of GNNs lies in their ability to learn complex functions f , AGG, and g that can effectively capture the structural and feature information of the graph.

For instance, Graph Convolutional Network (GCN) proposed by Kipf and Welling (2017) is one of the most popular and widely used types of GNNs. It can be described using the neural message passing framework, specifically through the functions, f , AGG and g .

In GCN, the message function f is a linear transformation applied to the features of neighboring nodes. For a node u sending a message to node v , the message can be represented as:

$$m_{uv}^{(l)} = W^{(l)} h_u^{(l-1)}, \quad (1.9)$$

where $W^{(l)}$ is the weight matrix for layer l and $h_u^{(l-1)}$ is the feature vector of node u from the previous layer $l - 1$. This linear transformation is the same for all edges and does not explicitly depend on edge features.

GCN adopts a simple yet effective mean aggregation function. For a node v , the aggregated message from its neighborhood $\mathcal{N}(v)$ is:

$$m_v^{(l)} = \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} m_{uv}^{(l)}, \quad (1.10)$$

This mean aggregator takes the average of the incoming messages (after transformation by $W^{(l)}$), ensuring that the update for each node is influenced equally by all its neighbors.

The update function g combines the node's existing features with the aggregated messages to produce a new set of features. In GCN, this is done through the following function:

$$h_v^{(l)} = \text{ReLU} \left(W^{(l)} h_v^{(l-1)} + b^{(l)} + m_v^{(l)} \right), \quad (1.11)$$

where $b^{(l)}$ is a bias term and ReLU provides the non-linearity necessary for learning complex patterns. This formula effectively updates each node's features based on both its own features and the aggregated features from its neighbors.

An important aspect of GCN is its normalization of the adjacency matrix to account for the varying degrees of nodes, which helps in stabilizing the learning process. This is often achieved by using the symmetric normalization, leading to the modified aggregation step:

$$m_v^{(l)} = \sum_{u \in \mathcal{N}(v)} \frac{1}{\sqrt{|\mathcal{N}(v)| |\mathcal{N}(u)|}} W^{(l)} h_u^{(l-1)} \quad (1.12)$$

This normalization ensures that the influence of a node's neighbors is scaled inversely by their degree, preventing high-degree nodes from dominating the feature representation.

1.2 Financial Time Series Modeling

1.2.1 Literature and Methodological Backgrounds

Study	Year	Modeling subject	Market	Prediction horizon	Prediction Method	Features	Prediction target	
							Classification	Regression
french1980stock	1980	Stock	US	Daily	LR	Numerical		✓
banz1981relationship	1981	Stock	US	Monthly	LR	Numerical, technical, fundamental		✓
fama1993common	1993	Stock	US	Monthly	LR	Factors		✓
kim2003financial	2003	Index	Korea	Daily	SVM	Technical	✓	
kim2004stock	2004	Index	Korea	Daily	MLP	Technical	✓	
tetlock2007giving	2007	Stock	US	Daily	VAR	News, technical, fundamental		✓
campbell2008predicting	2008	Index	US	Monthly	LR	Technical		✓
guresen2011using	2011	Index	US	Daily	MLP, Hybrid Models	Numerical		✓
fama2012size	2012	Stock	US, Europe, Japan, Asia Pacific	Monthly	LR	Technical		✓
lin2013svm	2013	Index, Stock	Taiwan	Daily	SVM	Technical	✓	
adebiyi2014comparison	2014	Stock	US	Daily	ARIMA	Numerical		✓
bisoi2014hybrid	2014	Index, Stock	US, India	Daily	MLP	Technical		✓
ballings2015evaluating	2015	Stock	Europe	Yearly	RF, MLP, SVM	Numerical, technical, fundamental	✓	
chen2015lstm	2015	Stock	China	Daily	LSTM	Numerical	✓	
ding2015deep	2015	Index, Stock	US	Daily	DL, CNN	News	✓	
nelson2017stock	2017	Stock	Brazil	Daily	LSTM	Numerical	✓	
bao2017deep	2017	Index	China, India, Hong Kong, Japan, US	Daily	LSTM	Numerical, technical, fundamental		✓
fischer2018deep	2018	Stock	US	Daily	LSTM	Numerical	✓	
li2018stock	2018	Stock	China	Daily	Attention	Numerical		✓
li2019multi	2019	Stock	China	Daily	Attention	Numerical		✓

Table 1.1: Summary of selected financial time series modeling research. Numerical feature represents price related features, including open, high, low, close, volume, etc.

Financial time series modeling is a notoriously difficult task, drawing attention from both the econometrics and machine learning communities. These efforts can be broadly categorized into statistical methods, classical machine learning methods and sequential deep learning methods. To provide a comprehensive review on the research ideas and methodologies employed in this field, we have reviewed and summarized 20 key studies. A summary of these studies is presented in Table 1.1.

Common econometric techniques include linear regression - LR (French, 1980; Banz, 1981; Fama and French, 1993; Campbell and Thompson, 2008; Fama and French, 2012), autoregressive integrated moving average - ARIMA (Adebiyi et al., 2014), and vector autoregression - VAR (Tetlock, 2007). While these models are built upon strong statistical foundations, they rely on linear assumptions, limiting their ability to capture complex, nonlinear relationships often present in financial data (Granger, 1993; Tsay, 2005; Cheng et al., 2022). Moreover, the performance of these models is highly sensitive to the selection and combination of financial variables (factors), a process that is both challenging and often driven by expert judgment (Ahmed et al., 2019; Michaelides, 2024).

Classical machine learning methods treat the financial time series as flat feature vectors without position aware structures. There is no inherent distinction between the order of input financial data. Representative machine learning methods include Support Vector Machine (SVM), Random Forest (RF) and Multi-layer Perceptron (MLP). For instance, the study by Lin et al. (2013) presents an SVM-based approach for stock market trend prediction, focusing on feature selection and prediction model development using historical data from the Taiwan stock market. Their study demonstrates improved hit ratios across various SVM kernels. Ballings et al. (2015) presents a comprehensive benchmark of ensemble methods, including Random Forest, against traditional classifiers such as MLP and SVM, for stock price direction prediction. Utilizing yearly data from 5,767 European companies, the authors demonstrate the superiority of Random Forest in enhancing predictive accuracy, thereby addressing a notable gap in the stock price direction prediction literature where ensemble techniques remain underutilized. Kim and Lee (2004) present an innovative application of artificial neural networks (MLP) for stock market prediction, emphasizing the importance of optimal feature transformation. By integrating genetic algorithms (GAs), the authors address the limitations of conventional MLP models in handling noisy and non-stationary financial data. Their approach enhances the learning efficiency of MLPs, resulting in improved predictive accuracy when applied to daily KOSPI index data.

In recent years, sequential deep learning methods have gained prominence. The widely adopted methods include the family of recurrent neural networks - LSTM (Hochreiter and Schmidhuber, 1997) and GRU (Cho et al., 2014), 1D convolutional neural networks (Kiranyaz et al., 2021), and attention based neural networks (Vaswani et al., 2017). Sequential deep learning treats the stock prices as temporally related sequences rather than a flat feature vector with no temporal hierarchy. Specifically, LSTM uses a gate mechanism, working as dynamic thresholds, to memorize and update the concatenation between feature information from previous time stamps and the current step (Chen et al., 2015; Nelson et al., 2017; Bao et al., 2017; Fischer and Krauss, 2018). Convolutional neural networks process the stock sequence in a way of sliding window. Instead of process the sequence in one shot, convolutional neural networks take snapshots in a fixed length at one step, and move along the sequence step by step (Chen and He, 2018). Attention based neural networks using attention scores, ranging from 0 to 1 and summing up to 1, to represent the dynamic temporal relationships of the entries in the sequence (Li et al., 2018b, 2019a). Thanks to the massive amount of model parameters, researchers also incorporate multimodal features to enhance the prediction performance of the sequential deep learning models. Bao et al. (2017) explores the performance of widely recognized technical and fundamental factors. Wu et al. (2019b) states that news embeddings can boost the prediction power of LSTM based prediction models. These studies prove that sequential deep learning methods can effectively capture the temporal relationships inside the stock price sequence.

In addition to methodological distinctions, Table 1.1 highlights significant variation in experimental settings across different studies. These studies cover different modeling subjects, from stock prices to index quotes, across major global markets. Some studies use daily level data, and some do monthly or yearly. Some utilize daily data, while others focus on weekly or monthly intervals. Similarly, the task formulation differs, with some studies treating time series prediction as a regression problem and others framing it as a binary classification task.

The features employed in these studies also vary significantly. The introduction of innovative features often drives new trends in financial time series research. For example, the Fama-French Three-Factor Model Fama and French (1993) extends the traditional Capital Asset Pricing Model (CAPM) (Sharpe, 1964) by incorporating two additional factors (features) — size (SMB) and value (HML) — to better explain stock returns. Empirical evidence suggests that smaller firms tend to outperform larger

ones, reflecting a risk premium (SMB), while the value factor (HML) captures the performance differential between high book-to-market (value) and low book-to-market (growth) stocks. The Fama-French model has been highly influential in empirical asset pricing and has inspired numerous subsequent studies that build upon or extend it (Carhart, 1997; Fama and French, 1996; Davis et al., 2000; Griffin, 2002; Asness et al., 2013).

All these methodological and experimental diversities reflect the inherent complexity of financial time series modeling.

1.2.2 Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH) first formalized by (Fama, 1970) is a cornerstone of financial time series modeling, asserting that asset prices in financial markets fully reflect all available information at any given point in time. This thesis works on the challenge of EMH by incorporating relational information between companies to assist price forecasting in Chapter 3 and Chapter 5.

Fama categorized the EMH into three distinct forms — weak, semi-strong, and strong — based on the type of information that is assumed to be fully reflected in market prices, each with significant implications for how financial data is modeled and interpreted.

1.2.2.1 Weak Form Efficiency

The weak form of the EMH posits that all past trading information, such as historical prices, volumes, and returns, is already incorporated into current stock prices. Consequently, under weak form efficiency, technical analysis, i.e. the use of past price patterns to predict future movements, would not provide any consistent advantage. In terms of financial time series modeling, weak form efficiency suggests that price changes should follow a random walk, with no serial correlation between past and future prices. Models such as ARIMA or moving averages, which rely on historical data for forecasting, would thus be ineffective in generating excess returns, as any patterns observed in past data are assumed to be unexploitable for future price prediction.

1.2.2.2 Semi-Strong Form Efficiency

The semi-strong form of the EMH extends this concept to include not only historical price data but also all publicly available information, such as financial statements, earn-

ings reports, macroeconomic news, and other forms of publicly disclosed information. In a semi-strong efficient market, prices adjust instantaneously to new public information, rendering fundamental analysis, i.e. the examination of financial statements and economic data to forecast future price movements, ineffective in consistently outperforming the market. For financial time series modeling, this form of efficiency implies that any model incorporating publicly available macroeconomic or company-specific variables would have limited predictive power. Researchers in this domain may focus on volatility models or event studies to examine how new information impacts prices, though they would expect any price reaction to be immediate and comprehensive.

1.2.2.3 Strong Form Efficiency

The strong form of the EMH asserts that all information, both public and private (including insider information), is fully reflected in asset prices. In this scenario, even insiders with access to non-public information would be unable to consistently achieve abnormal returns. This version of market efficiency is the most stringent and implies that no participant, regardless of their information advantage, can outperform the market in the long run. For financial time series models, the strong form of EMH further challenges the efficacy of any predictive modeling based on information asymmetries. It suggests that prices reflect all relevant information at every moment, meaning that predictive modeling in any form, whether using technical, fundamental, or proprietary data, would be futile in generating abnormal returns.

1.2.3 Long short-term memory (LSTM)

Among all the sequential deep learning methods, Long Short-Term Memory (LSTM) networks play a central role in this thesis. Therefore, we specifically introduce the architecture of LSTM and its application in financial research. Detailed discussions on LSTM architecture are provided in Chapters 3 and 4.

LSTM is a type of Recurrent Neural Network (RNN) specifically designed to handle sequence learning tasks by overcoming the vanishing and exploding gradient problems that hinder traditional RNNs (Hochreiter and Schmidhuber, 1997). Introduced by Hochreiter and Schmidhuber in 1997, LSTM networks have become one of the most advanced architectures for modeling long-term dependencies in sequential data, making them particularly suitable for time series predictions, including financial time series (Yan et al., 2020).

LSTM networks differ from standard feed-forward neural networks by incorporating memory cells that store information over extended periods. These cells are controlled by three gates: the forget gate, which determines what information to discard, the input gate, which adds new information, and the output gate, which regulates the information flow from the memory cell. This unique structure allows LSTMs to maintain and update memory states, making them highly effective for capturing temporal dependencies in financial data (Fischer and Krauss, 2018).

In financial applications, LSTM networks have been applied to predict stock prices (Nelson et al., 2017), market indices (Bao et al., 2017), and other financial variables (Liu, 2019) by treating price sequences as time series data. Unlike traditional econometric models, which often rely on linear assumptions, LSTMs can capture complex, non-linear relationships between financial variables (Fischer and Krauss, 2018). Studies have shown that LSTMs outperform memory-free models like random forests and logistic regression in predicting stock price movements, by identifying patterns such as momentum, mean-reversion, and volatility clustering, which are critical for making informed trading decisions (Fischer and Krauss, 2018).

The ability of LSTMs to process sequences of financial data, coupled with their flexibility in modeling complex temporal patterns, makes them a powerful tool for financial market forecasting.

1.3 Semantic Company Relationship Modeling

It is known in the literature that the relationships between companies in stock markets are inherently complex, characterized by their dynamic and interconnected nature (Xiang et al., 2022; Song et al., 2023). These relationships, which include supply chains, investments, competitive dynamics and more, evolve frequently over time and can be influenced by market trends, sector-specific events, and individual company actions. Capturing and predicting market behavior amidst such complexity requires advanced methodologies capable of modeling these intricate and ever-changing connections effectively.

Graph Neural Networks have emerged as a transformative tool in financial predictions (Chen et al., 2018; Matsunaga et al., 2019), addressing critical limitations in traditional time-series analytical methods. Conventional approaches often treat stocks as isolated entities, relying solely on time-series data to predict future movements (Sawhney et al., 2021). However, such methods fail to account for the intricate interdepen-

dencies between companies, which can profoundly influence stock price behavior (Xiang et al., 2022). By leveraging GNNs, models can integrate historical price data with relational information, capturing not only individual stock trends but also the dynamic impact of one company’s performance on others within a broader network.

Before applying GNNs for various financial market forecasting tasks, one critical step is to conceptualize the financial market as a graph of interconnected entities, where companies are nodes linked by edges representing company relationships. For domains like traffic forecasting (Yu et al., 2018; Guo et al., 2019), biomedical engineering (Rhee et al., 2018; Zitnik et al., 2018) and recommendation systems (Fan et al., 2019), the underlying relationship graph between entities is standard and readily-given. For instance, traffic forecasting features well-defined and concrete traffic graphs (Ye et al., 2021; Ren et al., 2021; Chen and Tsourakakis, 2022), such as sensor networks (Ye et al., 2021) or road maps (Ren et al., 2021). These graphs provide a clear framework for representing spatial relationships between traffic nodes, such as traffic sensors, road intersections, landmarks, and other critical points of interest.

However, this is not the case for company relationships in financial domain, which is implicit and complex, lacking a unified way of modeling (Xiang et al., 2022).

1.3.1 Graphs in Financial Domain

In the current literature, company relationship graphs can be categorized into three types: correlation graphs, which rely on stock price correlations (Li et al., 2020a); WikiData graphs, which use some hand-coded rules to parse binary connections from WikiData (Feng et al., 2019); and vector distance graphs, which measure the distances between company embeddings.

1.3.1.1 Correlation Graph

Some previous literature uses historical correlations between individual stock price sequences to build company graphs since it is the most straightforward approach (Li et al., 2020a; Keskin et al., 2021; Yin et al., 2021; Xiang et al., 2022). For instance, Keskin et al. (2021) constructs a stock correlation graph to predict percentage changes in individual stocks, but their method limits edge weights to discrete values of +1 or -1, decided by the sign of the correlation value. In contrast, Li et al. (2020a) opts for continuous edge weights, underlining the belief that positively correlated stocks exhibit similar behavior. Yin et al. (2021) builds a stock correlation graph by calculating

correlation coefficients between historical price sequences and applying a threshold to retain significant edges. This graph is then used to integrate correlation-related features into a Graph Convolutional Network (GCN), which propagates market signals and enhances predictive accuracy in a GRU-based sequence forecasting model. Similarly, Xiang et al. (2022) builds dynamic company graphs based on correlation matrices derived from historical market signals, using rolling windows to calculate correlation coefficients and thresholds to filter significant relationships. Although not explicitly constructing company graphs, Zhang et al. (2018) leverages a stock correlation matrix to model stock relationships, integrating it into a tensor-based prediction framework to improve predictive accuracy and address data sparsity issues.

However, correlation graphs based on historical prices typically suffer from an indirect and delayed capture of company interrelationships. These models lack direct access to semantic information like financial events and news stories that more accurately reflect the dynamics of company relationships and potentially influence the public perception of the dynamics (Rönnqvist and Sarlin, 2015).

1.3.1.2 WikiData Graph

Other lines of work in the literature rely on the WikiData knowledge database¹, where researchers extract known types of company relationships from one-hop and two-hop connections of WikiData entries by applying hand-coded parsing rules (Feng et al., 2019; Kim et al., 2019; Sawhney et al., 2020). With this approach, researchers can identify certain relationships like collaboration, supply chain, ownership, etc.

However, due to the complex and dynamic nature of company relationships, only limited relational types can be manually captured (Xiang et al., 2022). Also, since the WikiData itself is subject to public editing, the issue of data asymmetry bias occurs in smaller companies, which questions the overall reliability of the data as evidenced by the low relation ratio in Feng et al. (2019). Furthermore, the connections between companies in this type of graph are represented with binary weights, indicating only the presence of relationships and lacking continuous measures of connection strength.

Related variants of the WikiData graph include one that uses event extraction methods to identify specific types of connections from company announcements (Xu et al., 2021) and another based on the S&P Capital IQ database (Cheng and Li, 2021). These approaches also face the issue of limited relationship types; for example, Xu et al.

¹<https://www.wikidata.org>

(2021) identifies only four types, while Cheng and Li (2021) identifies five. These relationships include industry affiliation, shareholder connections, and supply chain links. Additionally, these variants similarly use binary edge weights, limiting the effective representation of connection strengths between companies.

In comparison, our method Using news text as the direct relational information source, in comparison, can

1.3.1.3 Vector Distance Graph

The third type of company graphs utilizes vector distance-based methods. In this approach, companies are first represented as vector embeddings, and their interrelationships are quantified by calculating the vector distances. These distances are then used as edge weights in the company graph. Although distance-based methods are common practice in recommendation system networks (Lee et al., 2008; Melville and Sindhvani, 2010; Sun et al., 2015), their application in financial graphs is comparatively rare.

A notable example is Hou et al. (2021), who employ a variational autoencoder (VAE) to derive lower-dimensional vector representations of companies from a comprehensive set of fundamental features. These features include total assets, book value per share, total liabilities, retained earnings, total dividends, and others. The VAE encodes these high-dimensional fundamentals into a latent space, where Euclidean distances between companies are calculated to represent the strengths of their relationships.

The rationale behind this approach is rooted in the economic principle that companies with similar fundamentals often exhibit correlated price movements due to shared external influences. For instance, firms within the same sector or value chain are similarly affected by macroeconomic shocks or industry-specific developments. As highlighted by the authors, the COVID-19 pandemic significantly impacted the travel sector while leaving technology firms relatively unscathed. These shared dynamics, inferred through fundamental data, form the basis of the VAE graph's structure.

However, like correlation graphs, their method also models spatial relationships indirectly, basing relationship sources on company fundamentals rather than directly reflecting investors' perceptions about the companies (Hou et al., 2021). This indirect approach could potentially limit the model's ability to capture the dynamic and subjective nature of market relationships.

1.3.2 The Semantic Company Relationship Graph

To tackle the above challenges of 1) indirect relational information sources, 2) limited relational types, and 3) noncontinuous edge weights in existing company graphs, this thesis starts with the research problem of Company Relationship Modeling, with the aim to dig out the complex company interactive relationships through various business events in the business network. In this thesis, we answer this research problem by analyzing the company name co-occurrence statistics. By scanning a comprehensive financial news corpus, we count the co-occurrence pattern between company names in news reports, with the heuristic that frequent co-occurrence of two company names in news articles, especially within a close word distance (Lund and Kevin, 1997), suggests a higher likelihood of shared involvement in business activities, thereby signifying strong relational ties.

This heuristic aligns with existing semantic behavior research that the general perception of entities - companies, in our context - is cumulatively shaped through the associations they keep (Lund and Burgess, 1996; Pantel, 2005). Additionally, the numerical relationship measures within the co-occurrence pattern can be decoded using vector embedding schemes (Pennington et al., 2014), which represent companies in a unified semantic vector space, allowing for the calculation of similarity between vector embeddings that can effectively represent the semantic relationships between companies (Lund and Burgess, 1996; Turney, 2006; Pennington et al., 2014). We name such relationship modeling scheme the Semantic Company Relationship Graph (SCRG).

Compared to existing company graphs, the Semantic Company Relationship Graph (SCRG) demonstrates notable advantages in terms of information source and continuous edge weights. Unlike indirect relationship sources such as stock price correlations (Kim et al., 2019) or company fundamentals (Hou et al., 2021), SCRG directly captures real-world business interactions through co-occurrence statistics derived from news reports. This approach also eliminates the constraints of manually defined rules in WikiData graphs, which limit the types of relationships that can be captured. Furthermore, the frequency of company co-occurrence in SCRG reflects the strength of intercompany relationships, which are translated into continuous edge weights. These continuous weights enable more nuanced and effective feature propagation for trading predictions, an advantage not offered by VAE-based or WikiData graphs.

1.4 GNN Applications for Financial Market Forecasting

With prior relational knowledge between companies provided by the Semantic Company Relationship Graph (SCRG), this thesis then dives into the research of innovative applications of financial market forecasting using Graph Neural Networks (GNNs) using two distinct types of techniques: Spatial-Temporal GNNs and Graph Pooling. Spatial-Temporal GNNs are used for the two research questions, Spatial-Temporal Stock Movement Prediction and Portfolio Selection (Chapter 3), and How to Assess Investor Sentiment on Media Silent Days (Chapter 4). Graph Pooling is used for Hierarchical Graph Pooling for Intraday Market Index Prediction (Chapter 5). The SCRG works as the cornerstone of the types of two techniques, providing relational connections between companies that GNNs can work on top of it.

An overview of the relationships between SCRG and the three research questions of the thesis are shown in Figure 1.1.

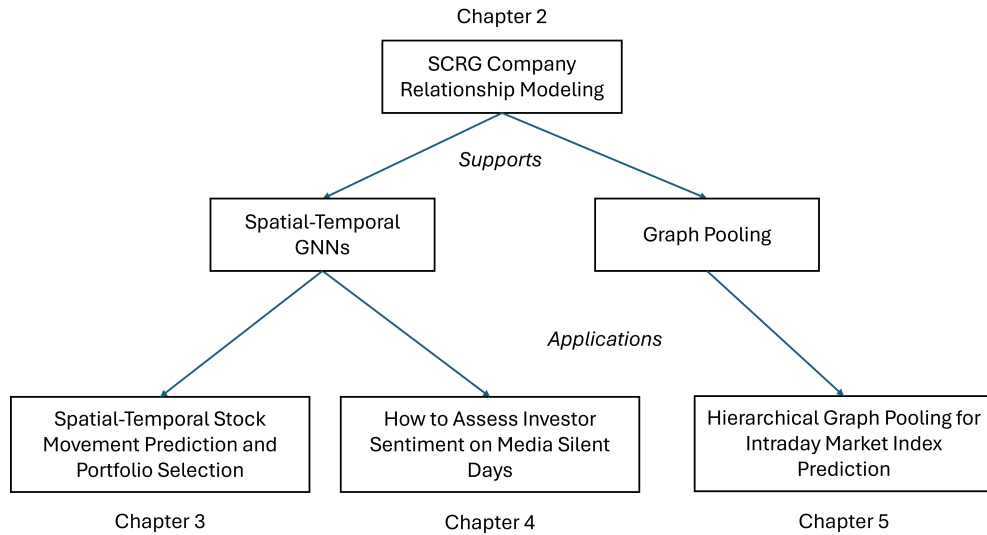


Figure 1.1: Relationship between company relationship modeling (Chapter 2) and the three applications of Graph Neural Networks (Chapter 3 to 5).

1.4.1 Spatial-Temporal GNN

Spatial-Temporal GNNs (ST-GNNs) focus on capturing both the spatial relationships among companies and the temporal dynamics of their interactions over time (Seo et al., 2018). This approach allows for the modeling of complex dependencies and patterns that evolve, providing a dynamic and comprehensive view of the financial market. By

integrating temporal features with the spatial structure provided by the SCRG, Spatial-Temporal GNNs are adept at forecasting future states of the market, taking into account how past events and neighboring related information may influence upcoming market movements.

Unlike traditional GNNs, which primarily focus on the spatial aspect by modeling relationships between nodes at a single time step, ST-GNNs extend this by incorporating changes over time, thus offering a more dynamic view of the graph (Yu et al., 2018; Zhang et al., 2020a). Given a graph, like SCRG, $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with node features $X \in \mathbb{R}^{N \times d}$, the ST-GNN operates over sequences of such graphs at different time steps $\{X_t\}_{t=1}^T$ to predict future states. The ST-GNN model can be formally described as follows.

Spatial Feature Propagation. At each time step t , graph convolution is applied to capture the spatial structure. Take the aforementioned GCN for instance,

$$H_t = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} X_t W \right) \quad (1.13)$$

where H_t is the hidden features of all company nodes after message propagation, $\tilde{A} = A + I_N$ is the adjacency matrix A with added self-connections, $\tilde{D}$ is the degree matrix and W is the weight matrix. This step ensures that the node features are updated based on both their own features and the features of their neighbors, capturing the spatial relationships within the graph.

Temporal Modeling. The sequence of spatially updated node features $\{H_t\}_{t=1}^T$ is then fed into an RNN structure, commonly a specifically a Long Short-Term Memory (LSTM) (Hochreiter and Schmidhuber, 1997) or Gated Recurrent Unit (GRU) (Chung et al., 2014), to model temporal dependencies:

$$O_t = \text{RNN}(H_t, O_{t-1}) \quad (1.14)$$

where O_t represents the output at time t , encapsulating both spatial and temporal information up to that point. The final output O_T is then used for market forecasting tasks.

In the classical ST-GNN structure, an inherent assumption is the Independent and Identical Distribution of stock time-series across different companies (temporal IID-ness), which is commonly adopted in previous research (Li et al., 2021; Xiang et al., 2022). This assumption arises due to the weight-sharing mechanism inherent in recurrent neural networks (RNNs) used across different nodes, a design that has proven effective in many general domains (Seo et al., 2018). However, this approach does not adequately capture the complexities of the financial market.

Research has demonstrated that stocks in the financial market exhibit interdependencies (Li et al., 2021) and are characterized by significant non-stationarity and randomness (Cheng et al., 2022; Lai et al., 2023). Furthermore, while stock movements exhibit random-like behavior, they do not adhere strictly to random walk models (Lo and MacKinlay, 1988; Li et al., 2015). These findings highlight the fundamental issue that stock time-series from different companies are neither independent nor identically distributed (a phenomenon we term temporal non-IIDness). This non-IIDness arises because each company’s stock movement is uniquely influenced by its operational patterns and business-specific events (Graham et al., 1962). Factors such as industry characteristics, product offerings, and management structures contribute to significant heterogeneity in the underlying processes driving stock price movements across companies.

To tackle such challenge, this thesis proposes a Non-IID Spatial-Temporal Graph Neural Network (NIST-GNN) as a Non-IID extension to the classical ST-GNN (Seo et al., 2018). NIST-GNN allows for a more nuanced understanding and processing of temporal data, setting it apart from traditional spatial-temporal approaches.

In this thesis, the NIST-GNN is first applied to stock movement prediction and portfolio selection in Chapter 3. Experimental results show that this methodology significantly outperforms benchmark models, yielding superior profitability and improving the Sharpe Ratio compared to conventional ST-GNN baselines, with statistical significance. The findings also offer valuable theoretical insights into the dynamics of information diffusion within the US market, revealing that public information from cross-correlated companies typically experiences at least a one-day lag before diffusion, challenging traditional views of market efficiency.

Additionally, a variant of the NIST-GNN, Non-IID Spatial-Temporal Chebyshev Network (NIST-Cheb), is innovatively adapted to infer absent news sentiment on ”media silent days” in chapter 4. Previous studies have highlighted news-based sentiment as a crucial proxy for investor sentiment (Tetlock, 2007; García-Laencina et al., 2010), significantly impacting stock price forecasts. However, challenges arise due to the inconsistent presence of news coverage, as media attention often shifts towards more sensational business events. Leveraging the prior knowledge from the Semantic Company Relationship Graph (SCRG), NIST-Cheb infers company-specific absent news sentiment by incorporating sentiment data from neighboring companies and the company’s own historical domestic sentiment trends. Experimental results demonstrate that asset pricing models incorporating our inferred sentiment scores consistently out-

perform traditional missing value imputation methods, doubling profits compared to models using raw sentiment data.

1.4.2 Graph Pooling

Graph pooling, on the other hand, involves the aggregation of hierarchical information across the graph to produce a condensed representation that captures the essence of the entire network. This technique is particularly useful in financial market forecasting, as graph pooling allows for the extraction of high-level features that are indicative of broader market trends, facilitating the prediction of overall market direction from the perspectives of market microstructures from the industrial sectors.

The idea of hierarchical graph pooling can trace its roots back to hierarchical learning (Hubel and Wiesel, 1962). Fundamentally, the natural of hierarchical learning enables graph pooling to abstract and learn feature representations in a layer-wise manner. By combining simpler, low-level features and progressively assembling them into more complex, high-level representations, graph learning techniques can facilitate the discernment and modeling of intricate patterns within the graph (LeCun et al., 2015; Hamilton, 2020). On the other hand, the graph neural networks (GNNs) provide graph pooling the pathways for understanding, interpreting, and manipulating spatial relational information encapsulated within graph structures (Ying et al., 2018a; Xiong et al., 2019; Zhang et al., 2020b; Ma et al., 2021), by using a message passing mechanism to iteratively integrate and propagate feature information from nodes in the graph (Gilmer et al., 2017).

Various graph pooling techniques have led to an extensive range of successful applications including molecular property prediction (Wang et al., 2020), drug discovery (Gaudeflet et al., 2021), recommendation systems (Chang et al., 2021), social network analysis (Ying et al., 2018b). Yet, their applications in the financial domain remains relatively missing.

This thesis explores the innovative use of graph pooling for intraday market index movement prediction, modeling the task as graph classification. Such idea is inspired by the inherent hierarchical structures within financial markets, where companies are classified into sectors that reflect their primary business activities. For instance, the Global Industry Classification Standard (GICS) ² provides a detailed multi-layered framework, categorizing companies into sectors, industry groups, industries, and sub-

²<https://www.msci.com/our-solutions/indexes/gics>

sectors (Bhojraj et al., 2003; Schumaker and Chen, 2009). This sector-based classification has inspired numerous trading strategies, like sector rotation (Cavaglia et al., 2001; Conover et al., 2008), and has served as a cornerstone for various asset price prediction studies (Schumaker and Chen, 2009; Seong and Nam, 2021; Liu et al., 2015).

To exploit this hierarchical structuring, we introduce the FinPool graph pooling operator, an innovation inspired by and extending the DiffPool operator with tailored modifications to suit the financial domain. The FinPool operator is designed to integrate features across the GICS hierarchy, capturing the intricate interactions within and between sectors and their subdivisions. Supported by the Semantic Company Relationship Graph (SCRG), FinPool enables a sophisticated aggregation of market data, reflecting the complex influences of sector dynamics on stock movements.

To maximize the efficacy of FinPool in predicting index movements, we propose two frameworks, Stacked FinPool and Multi-tier Attention FinPool, to enhance the integration of hierarchical features extracted via FinPool, ensuring a comprehensive aggregation of sector insights. Our empirical results showcase a noteworthy improvement in trading profits and risk-adjusted returns. This performance not only signifies a substantial advancement over existing benchmark models but also validates the efficacy of leveraging intraday constituent information for index prediction. By focusing exclusively on index components and sidestepping domestic index features, our study challenges the Efficient Market Hypothesis, demonstrating the practical value of detailed, hierarchical market analysis in financial forecasting.

1.5 Thesis Outline

The remainder of this thesis is organized as follows, with each chapter presenting its main contributions:

- Chapter 2 introduces the Semantic Company Relationship Graph (SCRG), which models company relationships using an embedding scheme based on company name co-occurrence statistics from a large-scale financial corpus. The SCRG serves as a relational prior for the market forecasting applications discussed in Chapters 3, 4, and 5.
- Chapter 3 tackles the challenges of stock movement prediction and optimal portfolio selection. We introduce the Non-Independent and Identically Distributed Spatial-Temporal Graph Neural Network (NIST-GNN), a novel extension of the

Graph Convolutional Recurrent Network (GCRN) (Seo et al., 2018). This model is designed to handle individual temporal dependencies within stock sequences, enhancing predictive accuracy. Experimental results show that integrating NIST-GNN with the Semantic Company Relationship Graph (SCRG) yields promising risk-adjusted returns compared to baseline models, providing insights into the dynamics of public information diffusion in the US financial market.

- Chapter 4 introduces a new research question: how to assess investor sentiment on days without media coverage, given that news sentiment is a crucial but inconsistently available proxy for investor sentiment. To tackle this issue, we explore the use of the Non-IID Spatial-Temporal Chebyshev Network (NIST-Cheb), an adaptation of the NIST-GNN, for inferring absent news sentiment. Traditional missing value imputation techniques are inadequate for this challenge due to the complex spatial and temporal dependencies among companies. The proposed NIST-Cheb leverages these dependencies effectively. Its efficacy has been confirmed through error-based metrics and real-world trading evaluations, with a focus on risk-adjusted profits.
- Chapter 5 presents a novel approach to predicting intraday market index movements, focusing exclusively on market microstructural information derived from the hierarchical representations of industrial clusters. To facilitate this, we first introduce the FinPool graph pooling operator for integrating hierarchical features of industrial clusters. We then detail the development of the Stacked FinPool and Multi-tier FinPool frameworks to deploy this operator effectively for market index movement predictions. Empirical analysis has demonstrated significant improvements in trading profits and risk-adjusted returns, marking a substantial advancement beyond existing benchmark models.
- Chapter 6 outlines the overall contributions and implications of the thesis, while Chapter 7 provides the conclusion and highlights potential directions for future research.

1.6 Notations

The major notations used in this thesis are listed in the below Table 1.2 for clearness and better organization:

Notation	Meaning
$\mathcal{G}$	Company graph with node set $\mathcal{N}$ and edge set $\mathcal{E}$.
X	Input feature matrix of all nodes.
H	Hidden feature matrix of all nodes.
A	Adjacency matrix of a graph.
W	Weight matrix of the graph neural networks.
$\mathcal{L}$	Loss function of the models.
$d(v)$	Degree of a node v .
$\tilde{D}$	Degree matrix of a graph.
$\tilde{L}$	Laplacian matrix of a graph.
C	Co-occurrence matrix of SCRG; C_{ij} is the co-occurrence count between company name or context word i and j .
$C_D(v)$	Degree centrality of a given node v .
$C_B(v)$	Betweenness centrality of a given node v .
V	The total length of the GloVe vocabulary for SCRG.
D	Density of a graph.
Q	Modularity of a graph.
P_t^i	Price of stock i at timestamp t .
r_t^i	Return of stock i at timestamp t .
σ_i	Standard deviation of stock i .
$Z^{(l)}$	Node embedding matrix at the l -th layer of market hierarchy.
$S^{(l)}$	Cluster assignment matrix at the l -th layer of market hierarchy.

Table 1.2: Table of the major notations used in this thesis.

Chapter 2

Semantic Company Relationship Modeling

This chapter introduces our innovative Semantic Company Relationship Graph (SCRG) for modeling company relationships, which lays the foundation for the subsequent chapters on financial market forecasting applications.

Our Semantic Company Relationship Graph is designed to address the identified gaps in the above discussed company graphs. Unlike approaches of the stock correlation graph (Li et al., 2020a; Keskin et al., 2021) and the VAE graph (Hou et al., 2021), our proposed SCRG involves direct modeling of the investors’ semantic perception of company relationships from their co-occurrence patterns. We represent companies within a unified vector space and assess their semantic relationships via vector similarities.

This approach yields continuous relationship weights between companies, in contrast to the discrete ones derived from WikiData (Feng et al., 2019; Kim et al., 2019; Sawhney et al., 2020). These continuous weights are integral in enhancing the spatial feature aggregation capabilities of GNNs for market forecasting, as they allow for the application of dedicated attentions to neighboring nodes.

It is known that stock market movements are primarily driven by the investment activities of investors, which are, in turn, heavily influenced by the information they consume (Li et al., 2015). Financial news plays a crucial role in shaping these activities due to its timeliness (Li et al., 2021), offering insights not only into focal companies but also into their related entities. However, investors’ interpretations of this information often fail to fully account for the complex interrelationships between companies, leading to inefficiencies in price formation. These inefficiencies arise from the limited

attention of investors (Hirshleifer and Teoh, 2003) and their frequent failure to fully consider the economic links between firms (Cohen and Frazzini, 2008), resulting in underreaction to news and subsequent momentum effects in markets (Hong and Stein, 1999).

Our SCRG seeks to address these challenges by capturing a comprehensive view of intercompany relationships as reflected in real-world business events reported in financial news. By modeling the overall perceptions and connections among companies, SCRG provides a relational framework for downstream Graph Neural Networks to identify systematic gaps in information processing, thereby uncovering opportunities to better understand and predict market dynamics.

2.1 Motivation of SCRG

Previous research on financial news suggests that the co-occurrence of company names within the same news article indicates a degree of relatedness between them (Yaros and Imieliński, 2015; Rönnqvist and Sarlin, 2015). This line of research posits that financial news encompasses information pertaining to the developments of inter-company relationships, as well as the perception and interpretation of such developments by market participants (Ma et al., 2009a; Yaros and Imieliński, 2015; Rönnqvist and Sarlin, 2015).

Specifically, Forss and Sarlin (2016) propose building networks that reflect company relationships by using the number of times two companies are mentioned together in news articles during a specific quarter. They argue that such co-occurrence networks can be valuable for understanding the influence of companies in financial markets and for tracking how their relationships evolve based on news flow. Similarly, Schwenkler and Zheng (2019) calculate the logarithm of co-occurrence counts between company names, suggesting that proximity in text signals a potential business relationship. Tang et al. (2019) also construct company networks based on co-occurrence frequency, demonstrating that these networks can effectively represent company relationships, capturing both expected economic linkages and uncovering unexpected connections revealed through news coverage.

Previous studies highlight the value of co-occurrence derived from financial news as a measure for capturing and analyzing inter-company relationships. This inspires us to go beyond the surface of simple co-occurrence statistics to reveal deeper insights into company relationships.

2.2 The Principle of SCRG

Building on the above previous research, our SCRG distinguishes itself by incorporating two new concepts — word distance and context words — into the analysis of company co-occurrence information.

We first extend the concept of co-occurrence through the introduction of word distance. Word distance s_{ij} refers to the literal number of words separating two terms i and j within a news sentence. Drawing on the principles of statistical semantics in linguistic theory (Weaver, 1952; Firth, 1957; Lund and Kevin, 1997), which assert that “a word is characterized by the company it keeps”, and the empirical finding that “distant word pairs are expected to contain less relevant information about the words’ relationship to one another” (Pennington et al., 2014), we posit that frequent co-occurrence of two company names, especially at shorter word distances, indicates a stronger relational tie between them. To account for this, when calculating co-occurrence counts C_{ij} , we adjust the raw figures by dividing them by the word distance. This adjustment not only captures co-occurrence information but also balances the degree of relationship strength based on proximity.

In addition to focusing on co-occurrences between company names, we also consider frequently appearing non-company words, which we refer to as context words. Since company relationships are inherently multi-dimensional (Tang et al., 2019), the co-occurrence between context words and company names contributes to the understanding of deeper and implicit non-typical relationships between companies. This idea aligns with the distributional hypothesis in linguistics (Lund and Burgess, 1996; Sahlgren, 2008), which suggests that the mathematical space of semantic relationships is often quantitatively high-dimensional. While analyzing direct co-occurrences between company names captures certain obvious relational dimensions, incorporating co-occurrence information from context words broadens this scope, uncovering additional semantic dimensions that reveal implicit and indirect connections between companies.

To better illustrate these two concepts, consider the following news snippets:

[Apple] and [Samsung] just ended their epic seven-year legal patent infringement fight. The two companies agreed to a settlement in the case, according to court documents filed Wednesday, but did not disclose the terms.¹

¹<https://money.cnn.com/2018/06/27/technology/apple-samsung-patent-infringement-settlement/>

The Supreme Court ruled for Alphabet Inc.’s [Google] in a multibillion dollar battle with [Oracle] Corp. over elements of Google’s Android smartphone-operating system.²

In the first snippet, Apple and Samsung have a short word distance of 2, indicating a strong connection between the two companies. Additionally, the context word ”patent,” which appears 9 words away from Apple and 7 words away from Samsung, highlights a specific aspect of their relationship—patent competition.

In the second example, Google and Oracle are separated by a word distance of 7, which also indicates a meaningful relationship. Context words such as ”battle” and ”Android” further enhance the understanding of their relationship, expanding it to the dimensions of legal conflict and technological rivalry.

All these proximities within news articles suggest more than simple co-mentions, implying significant connections rooted in specific business activities.

In practice, to comprehensively compile the co-occurrence statistics between companies, we utilize an extensive financial news corpus sourced from Dow Jones Factiva³, encompassing news articles from established news publishers such as *The Wall Street Journal* and *The Financial Times* (Capelle-Blancard and Petit, 2019). Factiva has been previously employed to explore the synthesis and dissemination of information across financial markets (Ferris et al., 2013; Capelle-Blancard and Petit, 2019). Our complete Factiva news dataset spans from early 2000 to the end of 2020 and contains 1.33 million news articles related to S&P 500 companies, comprising a total of 848.8 million words. The large volume of the collected corpus allows for the effective filtering of random noise, such as irrelevant company mentions within the same context, ensuring that our analysis focuses on consistent co-occurrence patterns that truly represent company relationships.

When calculating co-occurrence statistics, we apply a sliding window with a word distance of 10 to scan the financial news corpus, following the recommendations of Lund and Kevin (1997) and Pennington et al. (2014). This process produces a co-occurrence matrix C , where each entry C_{ij} represents the co-occurrence count between words i and j , which can be either company names or context words. The update rule for C_{ij} is as follows:

$$C_{ij} = C'_{ij} + \frac{1}{s_{ij}} \quad (2.1)$$

²<https://www.wsj.com/articles/supreme-court-rules-for-google-in-multibillion-dollar-copyright-battle-with-oracle-11617632233>

³<https://www.dowjones.com/professional/factiva>

where C'_{ij} is the previous count, and the digit 1 denoting the presence of co-occurrence between i and j with s_{ij} adjusting for the word distance s_{ij} in the sliding window. To avoid oversize of the co-occurrence matrix, we limit the vocabulary V to the 100,000 most frequent words in the corpus, resulting in a matrix C with dimensions $100,000 \times 100,000$.

2.3 The GloVe Embedding Scheme

The GloVe word vector embedding technique (Global Vectors for Word Representation) provides a powerful method for numerically decoding the hidden relationship patterns between companies based on the co-occurrence matrix C (Pennington et al., 2014). This technique converts co-occurrence statistics into high-dimensional vector embeddings, encapsulating the complex semantic relationships between companies and context words within a unified semantic space. As a result, the generated embeddings for company names incorporate information from frequently appearing context words. These embeddings encode valuable background information about companies, such as relevant products, industries, terminology, and operational patterns, which can be instrumental in uncovering implicit relationships between companies.

Specifically, the GloVe vector embedding scheme is defined by a weighted bilinear regression model expressed as follows:

$$J = \sum_{i,j=1}^V f(C_{ij}) \left(v_i^T v_j + b_i + b_j - \log C_{ij} \right)^2, \quad (2.2)$$

$$f(x) = \begin{cases} (x/x_{max})^\alpha, & \text{if } x < x_{max} \\ 1, & \text{otherwise} \end{cases} \quad (2.3)$$

where v_i and v_j are the vector embeddings of word i and j , which can be either company names or context words, and C_{ij} denotes the corresponding entry in the co-occurrence matrix C . The total number length of the vocabulary is represented by V . The function f acts as a weighting mechanism to balance the influence of over-frequent co-occurrences. We set x_{max} to 10,000 based on the distribution of entries in C and maintain α the default value of 0.75 in GloVe. The bias terms, b_i and b_j , ensure the symmetry of the embeddings.

We conduct 100 iterations of optimization to derive the vector embeddings, a number chosen based on research suggesting that this amount leads to the best word analogy evaluation accuracy, an essential measure of the quality of vector embeddings

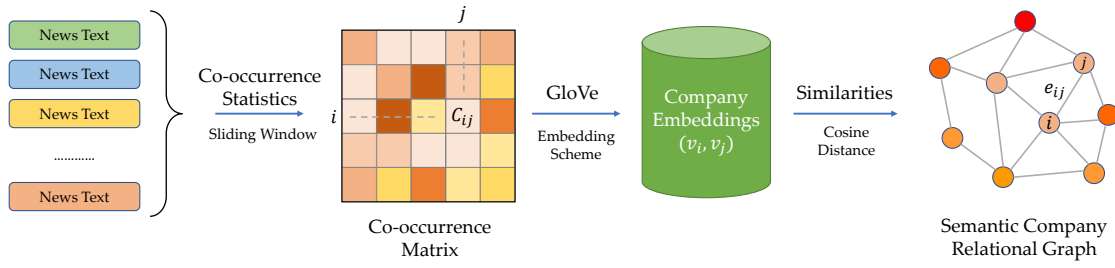


Figure 2.1: The generating process of the Semantic Company Relationship Graph.

(Mikolov et al., 2013a; Pennington et al., 2014).

2.4 The Construction of SCRG

The Semantic Company Relationship Graph (SCRG) is denoted as $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, where $\mathcal{N}$ signifies a set of nodes, each corresponding to a distinct company, and $\mathcal{E}$ comprises weighted edges that quantify the connection strengths between companies.

To build the SCRG, we only utilize the learned embeddings for company names. However, the embeddings for context words are not wasted, as the GloVe learning process of these words shape the company embeddings with richer and broader semantic dimensions.

With the derived company embeddings, the numerical relationship coefficients between company i and company j are computed through the cosine similarity between v_i and v_j , as this measure reflects the relational similarity between the vector embeddings (Lund and Burgess, 1996; Turney, 2006; Pennington et al., 2014):

$$e_{ij} = \frac{v_i \cdot v_j}{\|v_i\| \|v_j\|}, \quad e_{ij} \in \mathcal{E}. \quad (2.4)$$

By representing companies as nodes and their relationship coefficients as edges, we construct the Semantic Company Relationship Graph (SCRG).

To control edge formation within the SCRG, we implement a threshold mechanism following Hou et al. (2021) and Xiang et al. (2022). Edges with weights below a predetermined threshold are removed to maintain an optimal sparse graph structure (details on determining the optimal threshold can be found in Section 2.5.3). Additionally, nodes that are left without any edges are removed, as they do not contribute to the main connected component. The entire process of constructing the SCRG is illustrated in Figure 2.1.

2.5 Exploratory Data Analysis for SCRG

To better understand the SCRG, we conduct Exploratory Data Analysis (EDA) for it following the guide of Chatfield (1986) and Data et al. (2016).

It is important to note that, to prevent the inclusion of future information, the Factiva dataset used for SCRG construction in Chapters 3 to 5 is truncated based on the respective in-sample periods rather than using the complete dataset. We use the SCRG generated in Chapter 3 for EDA, with the dataset truncated as of 2017-12-31. Detailed information about the evaluation setting of Chapter 3 can be found in Section 3.4.

2.5.1 Global Network Measures

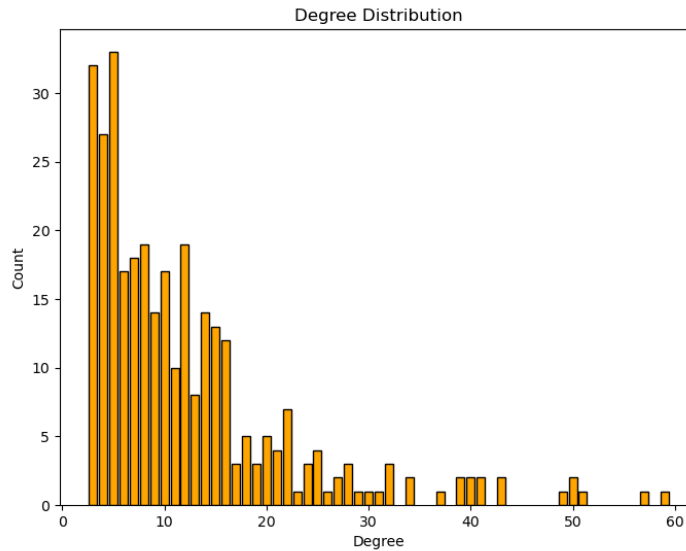


Figure 2.2: Degree distribution of SCRG

The degree distribution of the SCRG is shown in Figure 2.2. The average node degree is 12.30, with a standard deviation of 10.16. The degrees range from a minimum of 3 to a maximum of 59, with 75% of companies having a degree of 15 or less. Additionally, the average shortest path length in the SCRG is 3.79 nodes; when considering the edge weights, the average shortest path distance between nodes is 1.88. Diameter is the longest shortest path between any two nodes. SCRG has a diameter of 9 nodes, suggesting a relatively large spread across the graph.

Figure 2.3a and Figure 2.3b present the top-10 companies with highest degree centrality and betweenness centrality. Figure 2.3a shows the node D (Dominion Energy)

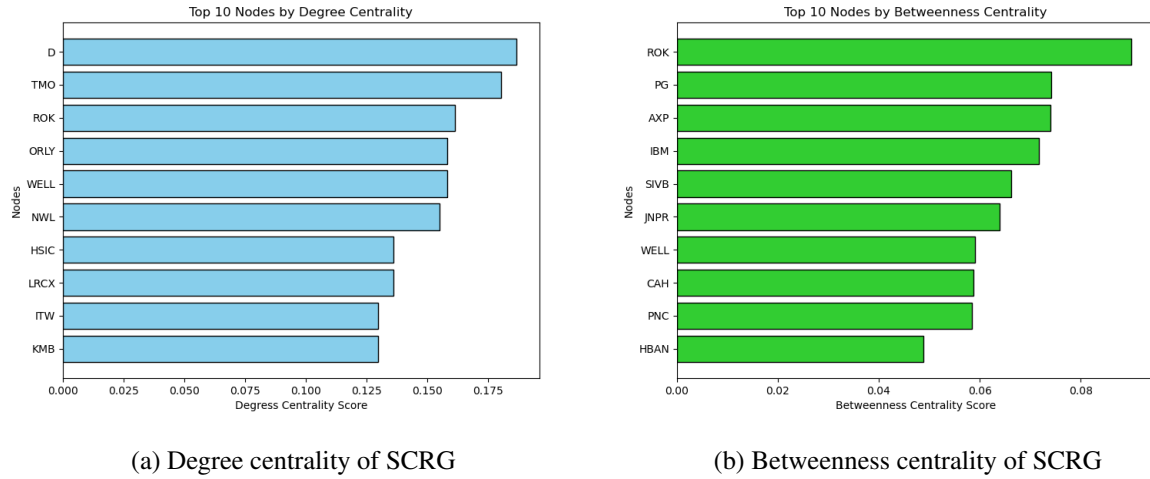


Figure 2.3: Top 10 companies by Degree centrality and betweenness centrality of SCRG.

has the highest degree centrality score, indicating that it has the most direct connections to other nodes in the network. Other companies like TMO (Thermo Fisher Scientific), ROK (Rockwell Automation), and ORLY (O'Reilly Automotive) also have high degree centrality scores, showing their prominence in terms of direct connections. This makes sense as these manufacturing companies work with lots of upstream and downstream companies with long supplier chains.

Figure 2.3b shows that the node ROK (Rockwell Automation) has the highest betweenness centrality score among all companies in the top 10, indicating that it plays a crucial role as a connector or intermediary within the network. Companies like PG (Procter & Gamble), AXP (American Express), and IBM also exhibit high betweenness centrality scores, suggesting that they are important hubs for communication or bridging different parts of the network.

It is also interesting to find that ROK appears in both figures, suggesting that it is not only a well-connected company but also a critical bridge in the S&P 500 company network.

2.5.2 Community Structures

To gain a comprehensive view of the community structures of SCRG, we first conduct community detection algorithm to partition the SCRG into communities. We adopt the widely used Louvain Community Detection Algorithm (Blondel et al., 2008) which

is a heuristic method designed to maximize modularity, a measure that quantifies the quality of a network partition by evaluating the density of edges within communities compared to between communities.

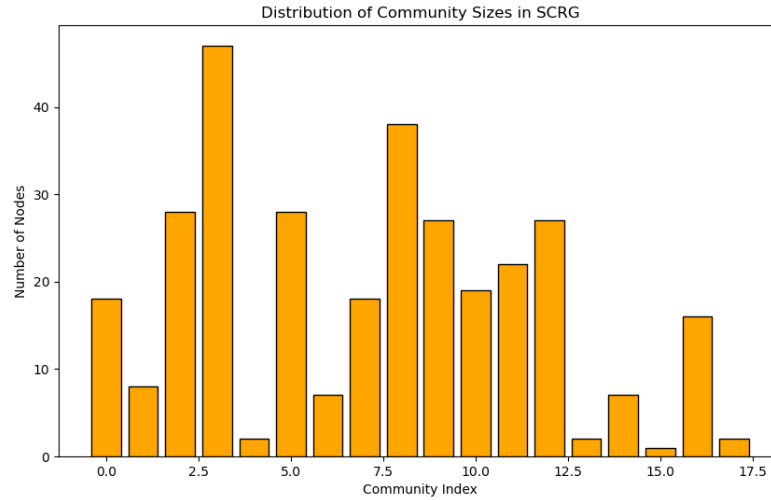


Figure 2.4: Distribution of community sizes in SCRG.

The Louvain algorithm applied to the SCRG detected 18 communities, a number between 11 sectors and 25 industry groups in GICS classification, a broadly recognized industry classification system (Bhojraj et al., 2003). The distribution of community sizes is presented in Figure 2.4. The sizes of these communities vary, ranging from as small as 1 node (an isolated company) to as large as 47 nodes, with an average community size of approximately 17.61 nodes. The standard deviation of 12.95 indicates a moderate level of variability in the sizes of the communities.

The modularity score of 0.6632 indicates a strong community structure in the network, with communities that are well-separated from each other. Modularity values closer to 1 reflect a network where nodes within the same community are densely connected, and there are relatively few connections between nodes of different communities. This score suggests that the algorithm has effectively grouped companies into communities where they share stronger internal connections, likely pointing to structural similarities or industry-based clustering.

The average intra-community density is approximately 0.77, indicating that, on average, the communities have a moderate to high level of internal connectivity. Some smaller communities exhibit very high densities (up to 3.0), which suggests that these groups are tightly interconnected, potentially representing companies that are closely

related or operate within the same sector. Larger communities tend to have lower densities, which is typical as they may encompass a broader range of companies with varying levels of interconnectedness.

2.5.3 Optimal Threshold Analysis

The threshold (Section 2.4) used to build the SCRG in this Section is 0.45, chosen as the comprehensive balance based on various network analysis measures.

Threshold	0.3	0.35	0.4	0.45	0.5	0.55	0.6
Num of Companies	368	358	342	317	266	192	101

Table 2.1: Number of companies at each threshold level.

We analyze the optimal threshold by using incremental steps of 0.05. Thresholds of 0.5 or higher result in too few remaining nodes in the SCRG, as many companies are filtered out to become outliers, creating multiple local clusters rather than a cohesive, interconnected graph. Table 2.1 shows the number of companies at each threshold level. As discussed in Chapter 3’s Section 3.4, the total number of companies with complete stock trading records is 411. However, with a threshold of 0.5 or higher, fewer than 266 companies remain, which makes them inappropriate to be the optimal threshold.

Thresholds of 0.35 or lower result in an excessive number of connections, leading to over-smoothing in spatial feature propagation and ineffective message passing. For example, with a threshold of 0.35, the degree distribution shown in Figure 2.5 reveals an average of 29.52 connections per node, with a standard deviation of 21.85. The node with the highest number of connections has 111 neighbors. Additionally, the low modularity score of 0.4497 confirms a relatively weak community structure, further indicating that the excess connections lead to over-smoothing in spatial feature propagation.

The remaining thresholds of 0.4 and 0.45 offer a good balance between the number of nodes and the connections between them. Compared to 0.4, the threshold of 0.45 shows a slight advantage in terms of modularity. The Louvain algorithm detects a similar number of communities for both thresholds, with 17 communities for 0.4 and 18 for 0.45. The distribution of community sizes of threshold 0.4 is shown in Figure 2.6. However, the modularity score for the 0.4 threshold is lower, at 0.5498, compared

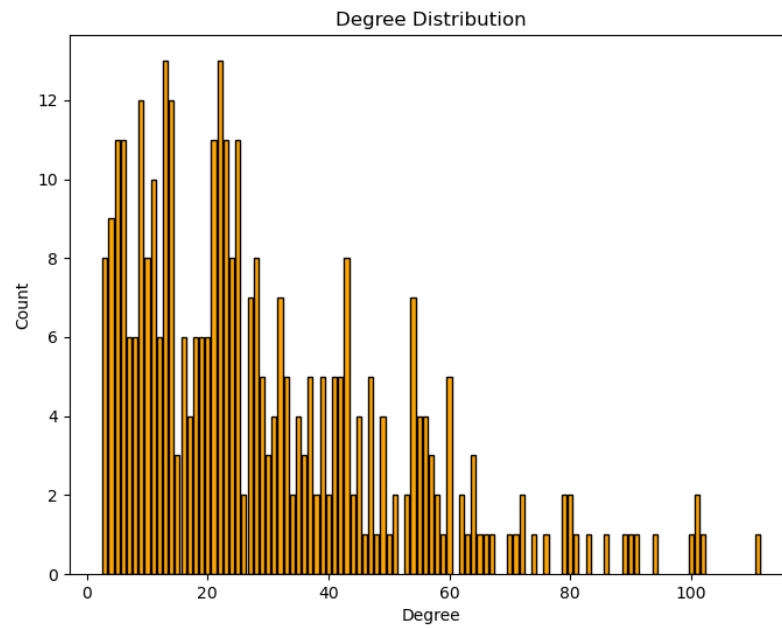


Figure 2.5: Degree distribution of SCRG with threshold 0.35.

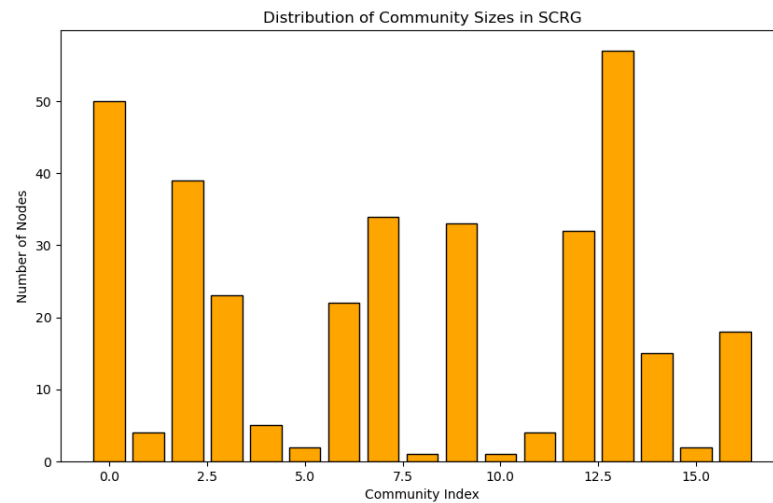


Figure 2.6: Distribution of community sizes of SCRG in threshold 0.4.

to 0.6632 for the 0.45 threshold. This indicates that a threshold of 0.45 results in a better SCRG in terms of community division. This is significant because companies naturally group into industries or sectors based on business connections, and a clearer community structure can aid information propagation among nodes in GNNs.

Chapter 3

Spatial-Temporal Stock Movement Prediction and Portfolio Selection

In this chapter, we approach stock price movements as a spatial-temporal prediction task, advancing beyond the traditional view of stocks as standalone entities, based on the Semantic Company Relationship Graph (SCRG).

We introduce a novel Non-Independent and Identically Distributed Spatial-Temporal Graph Neural Network (NIST-GNN). It is specifically designed to propagate features from both neighboring companies and internal historical data while effectively handling the inherent temporal non-IIDness in stock sequences. This innovative aspect of our NIST-GNN allows for a more nuanced understanding and processing of temporal data, setting it apart from traditional spatial-temporal approaches.

By integrating the NIST-GNN with the SCRG, we establish our NGNN-SCRG framework for predicting stock movements and selecting portfolios. Our experimental results demonstrate that this framework significantly outperforms benchmark models, yielding superior profitability and enhancing the Sharpe Ratio by 0.40 compared to the best-performing baseline, with statistical significance. Our findings also provide valuable theoretical insights into the effect of information diffusion within the US market, revealing that public information from cross-correlated companies typically experiences a minimum one-day lag before diffusion, challenging conventional perceptions of market efficiency.

* This work has been accepted and presented on the ICAIF 2022 Workshop on NLP and Network Analysis in Financial Applications.

3.1 Introduction

Financial time series prediction, especially in the context of stock price movements, presents a notably difficult yet lucrative challenge that has attracted significant interest from both academic researchers and industry practitioners (Oztekin et al., 2016; Flori and Regoli, 2021; Kim et al., 2022). Amidst the increasingly complex and volatile financial markets, the widely recognized *Efficient Market Hypothesis (EMH)* (Fama, 1970) suggests that stock prices have fully reflected all available information (Zhang et al., 2018), casting doubt on the utility of public information for predicting stock movements. Yet, studies on information diffusion in financial markets (Lo and MacKinlay, 1990; Cohen and Frazzini, 2008) offers perspectives that contrast this notion of market efficiency. These studies prove that economically linked companies, such as those involved in partnerships of supply chains, exhibit delays in information diffusion (Cohen and Frazzini, 2008). By capturing this delayed information from interconnected companies to assist, inform and enhance the decision-making process, investors can achieve positive and significant excess returns over the market (alphas) over long-time horizons (Cohen and Lou, 2012; Huang, 2015). Moreover, the well-established trading phenomenon, cross impact, also provides support to this idea: trading pressure from one stock can also move the price of another (Tomas et al., 2022; Le Coz et al., 2024).

The burgeoning research field of Graph Neural Networks (GNNs) has demonstrated potent capabilities in diverse areas, including traffic prediction (Yu et al., 2018; Guo et al., 2019), biomedical engineering (Rhee et al., 2018; Zitnik et al., 2018), recommendation systems (Fan et al., 2019), among others. Particularly, GNNs excel in integrating feature information from spatially-correlated entities. This attribute is highly pertinent to the stock market, where companies exhibit complex interrelations through competition, collaboration, outsourcing, joint ventures, and supply chains, etc (Langfield-Smith and Smith, 2003; Yusuf et al., 2004; Samaddar and Kadiyala, 2006; Gurnani et al., 2007; Lozano et al., 2013). The inherent spatial correlations within the stock market thus present a compelling case for applying GNNs to model and propagate information for enhanced stock movement predictions.

However, this idea faces two primary challenges. First, unlike more straightforward graph structures such as traffic networks or protein structures, the company relationship graph is complex and lacks a unified representation (Kim et al., 2019). To adapt GNNs to the task of stock price prediction, one needs to have a numerical relationship

scheme that effectively captures the hidden company relationships as weighted graph edges, representing the connection strengths between companies. Second, in addition to the spatial dependency across neighbors, stock price sequences themselves are temporally dependent. Since primitive GNNs cannot handle this temporal dependency, one needs to add a new temporal dimension to the GNNs architecture to accommodate both spatial and temporal factors.

Main contribution. This chapter addresses the aforementioned challenges in adapting GNNs for stock movement prediction. Our contributions are as follows:

- Introducing the *Non-Independent and Identically Distributed Spatial-Temporal Graph Neural Network* (NIST-GNN), which extends the Graph Convolutional Recurrent Network (GCRN) (Seo et al., 2018) to handle individual temporal dependencies in stock sequences. We integrate NIST-GNN with Semantic Company Relationship Graph (chapter 2) to form the NGNN-SCRG, our comprehensive framework for predicting stock price movements;
- Offering insights into the dynamics of public information diffusion in the US financial market. Our findings provide counter evidence to the strong and semi-strong forms of the Efficient Market Hypothesis (EMH) by showing that public information typically diffuses with a minimum one-day lag. These findings complement the body of research on the under-reaction of a firm's return to information about its neighboring companies in the network (Cohen and Lou, 2012; Cao et al., 2016).

3.2 Literature Review

Researchers from both the econometrics and machine learning communities have developed various methods to predict stock price movements. In this literature review, we categorize these distinct approaches into temporal and spatial-temporal methods.

Temporal methods. These methods extract information from the time-series data of individual stocks for financial predictions, encompassing empirical asset pricing based on statistical time series analysis and sequential machine learning techniques.

Dominant statistical time series analysis methods include linear regression (French, 1980; Banz, 1981; Fama and French, 1993; Campbell and Thompson, 2008; Fama and French, 2012), autoregressive moving average model (Adebisi et al., 2014), vector autoregression (Tetlock, 2007), and generalized autoregressive conditional het-

eroskedasticity models (Chou, 1988; Drost and Nijman, 1993). These models are valued for their linearity and interpretability, revealing underlying mechanisms between financial factors and dependent observations. As linear models have statistical foundations, experimental results can be evaluated by statistical tests. However, their merits also limit their ability to represent nonlinear factor relationships (Fischer and Krauss, 2018), which are prevalent in modern financial data sets.

Sequential machine learning methods, such as recurrent neural networks (Hochreiter and Schmidhuber, 1997; Chung et al., 2014) and Transformers (Zhang et al., 2022; Wang et al., 2022) treat stock prices as temporally connected sequences rather than a flat feature vector with no temporal hierarchy. Researchers incorporate multi-modal features to enhance the prediction performance of sequential machine learning models. For instance, Bao et al. (2017) explores the performance of widely recognized technical and fundamental factors. Wu et al. (2019b) states that news embeddings can boost the prediction power of LSTM based sequential models.

Although effective in capturing temporal dynamics, these methods overlook the spatial relationships within company networks, which is proved to be effective for stock movement predictions (Wang et al., 2021b). Recognizing this gap, our study introduces a spatial graph filter integrated with an LSTM structure, aiming to better concatenate features by incorporating spatially-related information from neighboring companies.

Spatial-temporal methods. Recognizing the limitations inherent in temporal methods, researchers have turned towards spatial-temporal methods. These approaches leverage Spatial-Temporal Graph Neural Networks (ST-GNNs) to concurrently process both spatial relationships from neighboring companies and temporal dependencies of individual stock price sequences.

ST-GNNs achieve this integration by combining spatial graph neural network blocks with temporal deep learning blocks, thus operating on temporal graph sequences and facilitating message passing at every time step (Seo et al., 2018; Rozemberczki et al., 2021). Typical application scenarios include traffic flow forecasting (Yu et al., 2018; Guo et al., 2019; Song et al., 2020), 3D action recognition (Yan et al., 2018; Cai et al., 2019) and time series forecasting (Wu et al., 2020b).

While ST-GNNs have not been widely applied in stock prediction, they hold inherent suitability for this domain. Hou et al. (2021) is among the pioneers in this area, applying an ST-GNN on the aforementioned VAE graph (section 1.3.1.3) to predict minute-level binary price movements using minute-level stock sequences and auxil-

iary fundamental variables. However, their focus is primarily on model predictability metrics, without delving into portfolio selection or trading simulations, which are crucial for practical financial applications.

A major hurdle in the broader application of ST-GNNs in finance is the temporal non-IIDness in stock sequences, which traditional ST-GNN models struggle to address (Seo et al., 2018). Our research contributes to this area by relaxing the temporal IIDness assumption inherent in conventional ST-GNNs. We assign unique LSTM weights to each company node, enabling more effective handling of the temporal complexities in stock sequence data.

3.3 The Non-IID Spatial-Temporal Graph Neural Network

We introduce the Non-IID Spatial-Temporal Graph Neural Network (NIST-GNN) as an innovative advancement beyond the Graph Convolutional Recurrent Network (GCRN) (Seo et al., 2018) for the context of stock movement predictions. Diverging from the conventional GCRN, our NIST-GNN is specifically designed to 1) more effectively leverage the feature information from spatially-connected neighboring companies in the SCRG, and 2) address the limitations of the temporal independence and identical distribution (IID) assumption foundational to the GCRN, which are unsuitable for the fluctuating dynamics of financial markets.

To provide a foundation of NIST-GNN, we first introduce the architecture of GCRN and discuss its limitations in the financial context. The original GCRN incorporates a Chebyshev spatial graph filter (Defferrard et al., 2016) integrated with an FC-LSTM temporal filter (Graves, 2013) to simultaneously process spatial-temporal features at different timestamps. The following equation set defines the GCRN

$$\begin{aligned}
 x_t^{\text{Cheb}} &= \text{Chebyshev}_{\mathcal{G}}(x_t) \\
 i_t &= \sigma \left(W_{xi} x_t^{\text{Cheb}} + W_{hi} h_{t-1} + w_{ci} \odot c_{t-1} + b_i \right) \\
 f_t &= \sigma \left(W_{xf} x_t^{\text{Cheb}} + W_{hf} h_{t-1} + w_{cf} \odot c_{t-1} + b_f \right) \\
 c_t &= f_t \odot c_{t-1} + i_t \odot \tanh \left(W_{xc} x_t^{\text{Cheb}} + W_{hc} h_{t-1} + b_c \right) \\
 o_t &= \sigma \left(W_{xo} x_t^{\text{Cheb}} + W_{ho} h_{t-1} + w_{co} \odot c_t + b_o \right) \\
 h_t &= o_t \odot \tanh(c_t).
 \end{aligned} \tag{3.1}$$

In the context of stock movement prediction, $x_t \in \mathbb{R}^{d_x}$ denotes the input feature of the target company at timestamp t . $\text{Chebyshev}_{\mathcal{G}}$ denotes the Chebyshev spatial filter to

aggregate and integrate the features of neighboring companies at the same timestamp, generating a spatially propagated input feature, $x_t^{Cheb} \in \mathbb{R}^{d_x}$ Defferrard et al. (2016). Additionally, the LSTM components – i_t (input gate), f_t (forget gate), c_t (cell state), o_t (output gate), and h_t (hidden state) – form a temporal filter, crucial for processing the long-term historical influences on the stock price at timestamp t , based on previous spatially propagated features. In this LSTM structure, the input gate, i_t , is pivotal in determining the proportion of the input feature, x_t^{Cheb} , and the previous hidden state, $h_{t-1} \in \mathbb{R}^{d_h}$, that will be incorporated into the current cell state, c_t . Conversely, the forget gate, f_t , regulates how much of the previous cell state, c_{t-1} , is preserved in c_t . The output gate, o_t , then decides the extent to which the current cell state contributes to the cell's output - hidden state h_t . The learning process within these gates is governed by trainable weights: $W_x \in \mathbb{R}^{d_h \times d_x}$ for the input and $W_h \in \mathbb{R}^{d_h \times d_h}$ for the hidden states. The LSTM's peephole connections, denoted by $w_c \odot c_{t-1}$ and controlled by $w_c \in \mathbb{R}^{d_h}$ as per Gers and Schmidhuber (2000), play a crucial role in linking the cell state to the gates. Finally, the biases in the LSTM, represented by $b \in \mathbb{R}^{d_h}$, further modulate the gate functions.

While the GCRN is applicable to the task of financial predictions, it exhibits two significant limitations that impact its effectiveness in this domain.

Restricted spatial filtering performance on graphs with community structures.

The Chebyshev spatial filter employs pre-defined Chebyshev polynomials to extract features from spatially-connected neighboring companies of the target node (Defferrard et al., 2016). It is formalized as follows:

$$Y = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{L}) X_t, \quad (3.2)$$

where $\tilde{L} = 2L/\lambda_{max} - I$ is the scaled Laplacian of the Semantic Company Relationship Graph (SCRG), λ_{max} is the largest eigenvalue of L , θ_k denotes the learnable weights, X_t is the the input feature matrix of all SCRG companies, and K is a hyper-parameter determining the hops of local neighborhood integration in the convolution. This spatial filtering rule is derived from the K -th order approximation of spectral graph convolution in spectral graph theory (Chung, 1997; Chen et al., 2020). The Chebyshev polynomials in equation (3.2) are recursively defined as,

$$T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x), \quad (3.3)$$

with $T_0(x) = 1$ and $T_1(x) = x$. A key disadvantage of the Chebyshev filter is its reliance on polynomials, which complicates the production of narrow-band graph fil-

ters and therefore limits its effectiveness on graphs with community structures (Levie et al., 2018), such as the Semantic Company Relationship Graph (SCRG) utilized in this study. This limitation is particularly relevant because companies naturally exhibit hierarchical cluster structures, like industries and sectors, based on their interactive business operations (Bell, 2005).

Inherent temporal IIDness among company nodes. In the GCRN architecture (eq. (3.1)), the trainable weights and the state representations of the FC-LSTM structure are uniformly shared across all company nodes. This approach, embodying the assumption of temporal IIDness, implies a consistent pattern in the temporal dynamics of various nodes. While such a weight-sharing mechanism is effective in various general node classification tasks (Seo et al., 2018), it falls short in the specific context of stock movement predictions. In fact, each company’s stock price is distinctively influenced by its specific operation patterns and business events (Graham et al., 1962), with impacting factors varying significantly across different industries, product lines and management structures. This diversity renders the assumption of temporal IIDness among stocks unrealistic, making it inappropriate to apply the same LSTM weights universally to all stocks. Therefore, it is improper to share the same set of LSTM weights across the graph.

To address the above limitations of GCRN, we propose our NIST-GNN which is defined by the following architecture:

$$\begin{aligned}
\tilde{X}_t &= \text{GCN}_{\mathcal{G}}(X_t) \\
I_t &= \sigma \left(\mathbf{W}_{XI} \otimes \tilde{X}_t + \mathbf{W}_{HI} \otimes H_{t-1} + W_{CI} \odot C_{t-1} + B_I \right) \\
F_t &= \sigma \left(\mathbf{W}_{XF} \otimes \tilde{X}_t + \mathbf{W}_{HF} \otimes H_{t-1} + W_{CF} \odot C_{t-1} + B_F \right) \\
C_t &= F_t \odot C_{t-1} + I_t \odot \tanh \left(\mathbf{W}_{XC} \otimes \tilde{X}_t + \mathbf{W}_{HC} \otimes H_{t-1} + B_C \right) \\
O_t &= \sigma \left(\mathbf{W}_{XO} \otimes \tilde{X}_t + \mathbf{W}_{HO} \otimes H_{t-1} + W_{CO} \odot C_t + B_O \right) \\
H_t &= O_t \odot \tanh(C_t),
\end{aligned} \tag{3.4}$$

where $X_t \in \mathbb{R}^{N \times d_x}$ represents the raw input feature matrix encompassing all companies, while $\tilde{X}_t \in \mathbb{R}^{N \times d_x}$ is the processed feature matrix after integrating spatial information from neighboring companies. The gates I_t , F_t , and $O_t \in \mathbb{R}^{N \times d_h}$ – input, forget, and output respectively – function similarly to those in the GCRN, serving as temporal filters for feature propagation from preceding timestamps. $H_t \in \mathbb{R}^{N \times d_h}$ denotes the hidden state matrix of all companies.

GCN spatial filtering. To tackle the problem of restricted performance on spatial

filtering, our NIST-GNN replaces the Chebyshev filter in GCRN (eq. (3.1)) with the Graph Convolution Network (GCN) filter proposed by Kipf and Welling (2017) with two reasons. First, the GCN has been shown to surpass the Chebyshev filter in various node classification tasks due to its effective localized first-order approximation of spectral graph convolutions, which simplifies the computation while preserving the essence of graph convolutions (Kipf and Welling, 2017; Chen et al., 2020). Second, the GCN filter is particularly adept at effectively leveraging the weighted relationship information inherent in the SCRG thanks to the symmetric normalization, thereby enhancing stock movement prediction accuracy. The GCN filter employs the following spatial feature aggregation pattern

$$\tilde{X}_t = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} X_t W_g \right), \quad (3.5)$$

where $\tilde{A} \in \mathbb{R}^{N \times N}$ represents the weighted adjacency matrix of $\mathcal{G}$ with self-loops added to include each node's own features, and $\tilde{D} \in \mathbb{R}^{N \times N}$ is the diagonal degree matrix corresponding to $\tilde{A}$, with diagonal entries $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Unlike simple averaging, the symmetric normalization operation, $\hat{A} = \tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}$, differentially weights the features from neighboring companies based on the strength of their connections and the node degrees, as defined by

$$\alpha_{u,v} = \frac{e_{u,v}}{\sqrt{\hat{d}_u \hat{d}_v}}. \quad (3.6)$$

Here, $\alpha_{u,v}$ is the element in $\hat{A} \in \mathbb{R}^{N \times N}$ that represents the normalized weight between company u and company v ; $\hat{d}_u$ and $\hat{d}_v$ represent the degrees of nodes u and v , respectively, and $e_{u,v}$ is the edge weight between them in $\mathcal{G}$. This normalization deals with the problem of scale disparity among different node degrees, which can lead to numerical instability and poor convergence during training (Kipf and Welling, 2017). Such attribute is particularly important as SCRG features highly varying node degrees. The $W_g \in \mathbb{R}^{d_x \times d_x}$ in equation (3.5) is the learnable weight matrix that transforms features after the symmetrically normalized spatial feature propagation.

Unique temporal weight learning. To tackle the problem of temporal IIDness among company nodes, in our NIST-GNN, a new dimension, N , is added to the weight matrices in GCRN, $W_x \in \mathbb{R}^{d_h \times d_x}$ and $W_h \in \mathbb{R}^{d_h \times d_h}$, to form weight tensors, $\mathbf{W}_X \in \mathbb{R}^{N \times d_h \times d_x}$ and $\mathbf{W}_H \in \mathbb{R}^{N \times d_h \times d_h}$. This dimension expansion ensures that each node in our model has its own dedicated FC-LSTM weights and hidden states, effectively accommodating the unique temporal characteristics of each stock, thus significantly enhancing the model's applicability and effectiveness in predicting stock movements.

An immediate issue comes with the dimension expansion is the tensor-matrix multiplication in equation set (3.4), such as the multiplication between $\mathbf{W}_{HI}$ and H_t , because tensors and matrices cannot be directly multiplied. To tackle this issue, we clearly define a node multiplication operator $\otimes$ to assign a rule of multiplication, which works as follows

$$\begin{aligned} z_t^k &= W^k h_t^k, \\ Z_t &= z_t^k \Big|_N, \quad \text{for all } k \in \mathcal{N}. \end{aligned} \quad (3.7)$$

In this operation, h_t^k represents company k 's hidden state vector, i.e. the k -th row of hidden state matrix H_t , and W^k is its corresponding matrix in the weight tensor $\mathbf{W}_{HI}$. The intermediate result vector z_t^k is then stacked along the N axis to form the result matrix $Z_t \in \mathbb{R}^{N \times d_h}$.

For predicting the stock movement of a specific company k , our model processes k 's final hidden state, $h_t^k \in H_t$, through a fully-connected layer with a softmax activation function, generating a binary prediction for the next trading day

$$\hat{y}_{t+1}^k = \begin{cases} +1 \\ -1 \end{cases}, \quad (3.8)$$

where $+1$ signifies an expected increase, and -1 represents a decrease in company k 's closing price from time t to $t+1$. The whole architecture of NIST-GNN illustrating this process, is depicted in Figure 3.1.

This prediction task allows the model to learn stock movement dynamics for company k under various market conditions, represented by the node features X_t . Following (Ding et al., 2015; Chen et al., 2018), this is achieved by minimizing the Binary Cross Entropy (BCE) loss $\mathcal{L}_k$, defined as

$$\mathcal{L}_k = -\frac{1}{T} \sum_t (y_t^k \log(\hat{y}_t^k) + (1 - y_t^k) \log(1 - \hat{y}_t^k)), \quad (3.9)$$

where T is the total length of the stock movement sequence for company k , y_t^k is the true label for k 's movement at timestamp t , and $\hat{y}_t^k$ is the predicted probability of the stock moving up (between 0 and 1).

When implementing the model, we compute the BCE loss for all companies in the SCRG by vectorizing the calculation and averaging the individual losses $\mathcal{L}_k$ to obtain the overall loss

$$\mathcal{L}_{\text{BCE}} = \frac{1}{N} \sum_k \mathcal{L}_k, \quad (3.10)$$

where N represents the total number of companies in the graph.

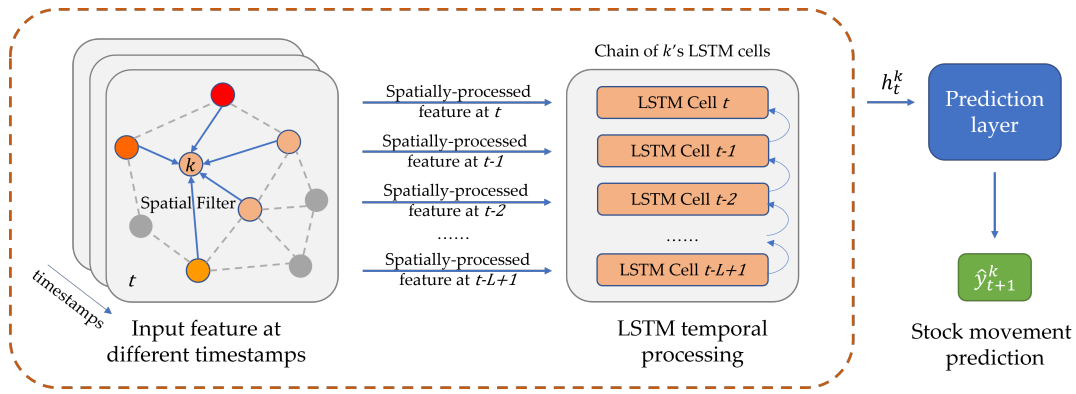


Figure 3.1: Visualization of our NIST-GNN architecture from the perspective of company node k .

In summary, our NIST-GNN significantly extends the GCRN framework to be aligned with the specific requirements of stock movement prediction. Key to this adaptation is 1) the replacement of the Chebyshev filter with the GCN filter, a change that allows for effective spatial aggregation of the weighted feature information from neighboring companies in the SCRG; and 2) the modification of the LSTM weight-sharing mechanism, enabling each stock to have its own dedicated LSTM weights and state representations. This adjustment is crucial in addressing the temporal non-IIDness characteristic of different stocks, arising from their distinct market behaviors and influences. We believe this comprehensive approach sets a new standard in the predictive analysis of stock price movements, offering potential for significant improvements in many other financial forecasting tasks.

3.4 Evaluation Settings

In our evaluation, we use S&P 500 components as the target companies for stock movement prediction and portfolio selection. The research period spans from 2005-01-01 to 2020-12-31, divided into an in-sample period (2005-01-01 to 2017-12-31) and an out-of-sample period (2018-01-01 to 2020-12-31). We specifically designate the trading days between 2015-01-01 and 2017-12-31 as the validation set serving as the basis for portfolio selection. Our study operates on a daily interval, targeting the binary close price movement of the next trading day. The hidden size hyperparameter of the model, d_h , is determined through grid search based on the validation set loss (Feng et al., 2019), with values selected from $\{16, 24, 32, 48, 64\}$, and 32 is used as the optimal choice. The number of training epochs is set to 200, based on the model's performance

on the validation set.

We include 13 technical and numerical stock features in our analysis, including open, close, high, low, volume, VWAP (volume-weighted average price), ADX (average directional index), EMA (exponential moving average), RSI (relative strength index), SMA (simple moving average), standard deviation, Williams, and close change percent. All data is sourced from Financial Modeling Prep (FMP) ¹, a data service providing historical stock data, company financial statements, and other financial information, which has been utilized in previous studies (Liu et al., 2023; Zhang et al., 2024).

These variables are standard and widely used in financial empirical research, and their selection is grounded in the relevant literature (Campbell and Thompson, 2008; Fama and French, 2012; Chen et al., 2015; Bao et al., 2017; Vargas et al., 2018; Matsunaga et al., 2019; Wang and Ma, 2024). For instance, Wang and Ma (2024) employ open, close, high, low, and volume for stock prediction tasks. Matsunaga et al. (2019) include features such as moving averages of stock prices and 1-day return ratio (close change percent). Similarly, Vargas et al. (2018) utilize features like Williams, EMA, and RSI.

Specifically, the VWAP calculates the average price of a stock, weighted by volume over the sliding window (see Section 3.4.2).

$$VWAP_t = \frac{\sum_{i=1}^n TP_i \cdot V_i}{\sum_{i=1}^n V_i} \quad (3.11)$$

where $TP_i = \frac{(High_i + Low_i + Close_i)}{3}$ stands for the typical price and V_i is the volume and n is the number of periods in the sliding window. The SMA is the unweighted mean of prices over the sliding window:

$$SMA_t = \frac{\sum_{i=1}^N P_i}{N} \quad (3.12)$$

where P_i is the stock price. The EMA places more weight on recent prices:

$$EMA_t = P_t \cdot \frac{2}{n+1} + EMA_{t-1} \cdot \left(1 - \frac{2}{n+1}\right) \quad (3.13)$$

The Williams is a momentum indicator that measures the closing price relative to the highest high and lowest low over a specified period of time. It identifies overbought and oversold levels:

$$Williams = \frac{High_n - Close_t}{High_n - Low_n} \times (-100) \quad (3.14)$$

¹<https://site.financialmodelingprep.com/>

	open	high	low	close	volume	vwap	adx	ema	rsi	sma	std	williams	change_pct
mean	72.40	73.20	71.58	72.41	6395379.49	72.44	12.89	71.66	52.09	71.65	3.08	-42.01	0.00
std	134.86	136.47	133.24	134.89	32588648.82	134.86	5.16	132.40	6.31	132.48	7.33	29.68	0.02
min	0.70	0.72	0.67	0.70	600.00	0.70	0.00	0.66	0.00	0.65	0.00	-100.00	-0.92
25%	26.89	27.23	26.53	26.90	1021600.00	26.95	9.11	26.82	47.84	26.79	0.95	-67.48	-0.01
50%	46.41	46.92	45.88	46.42	2230900.00	46.46	11.84	46.15	52.17	46.15	1.69	-37.33	0.00
75%	78.45	79.25	77.64	78.47	5101000.00	78.49	15.60	77.82	56.42	77.84	3.11	-15.05	0.01
max	4450.10	4530.00	4375.00	4446.88	3372969600.00	4432.29	45.60	4139.99	81.27	4168.66	655.05	0.00	1.02

Table 3.1: Descriptive statistics of the stock market dataset.

Where $High_n$ is the highest price over the sliding window; Low_n is the lowest price over the sliding window and $close_t$ is the most recent closing price. The calculations of ADX and RSI are rather complex. We refer the reader to the FMP documentation² for more details.

We apply z-score normalization to standardize the features following (Li et al., 2020b), as research has shown that data normalization improves classification performance (Singh and Singh, 2020). Companies that do not span the entire research period are excluded, resulting in a dataset of 411 companies with a total of 1,655,508 data entries. A descriptive statistics of our dataset can be found in Table 3.1.

3.4.1 Benchmark models

To evaluate the effectiveness of our NIST-GNN-SCRG (NGNN-SCRG) framework, we have adopted a comprehensive set of benchmark models, aligning with the recommendations provided by Fischer and Krauss (2018). The benchmark models are categorized into two groups to ensure a broad spectrum of comparisons.

The first group comprises deep learning models, each chosen to test a specific aspect of our NGNN-SCRG framework. This includes the NIST-GNN model coupled with a baseline correlation graph, NGNN-Corr. The correlation graph is constructed using a Pearson correlation matrix over the entire in-sample period, following the methodology of Li et al. (2020a). To ensure comparability with the SCRG, the correlation graph's threshold is calibrated such that the average neighbor count matches that of the SCRG. The NGNN-Corr aims at assessing the comparative performance of our SCRG against traditional correlation-based methods. This comparison is critical in validating the SCRG's effectiveness in capturing nuanced market relationships. Additionally, the original GCRN integrated with our SCRG is included to evaluate the

²<https://site.financialmodelingprep.com/developer/docs>

Model	Number of parameters
NGNN-SCRG	1,897,672
NGNN-Corr	2,352,456
GCRN-SCRG	6,728
LSTM	5,888 (per node)
MLP	562 (per node)
RF	120,548 (per node)
LR	781 (per node)

Table 3.2: The number of parameters of each benchmark models.

enhancements brought about by our modifications, specifically in terms of the temporal weight-sharing mechanism and the adaption of GCN filter. A standard LSTM model is also part of this group, serving as a baseline to highlight the added benefits of spatial information propagation, a notable feature of our NGNN-SCRG. For consistence, the LSTM baseline uses the same hidden size of $d_h = 32$.

The second group consists of traditional machine learning models, including Logistic Regression (LR), Random Forest (RF), and Multi-Layer Perceptron (MLP). These models were selected due to their established robustness and high performance in various baseline modeling scenarios, as documented in the extant literature (Krauss et al., 2017; Fischer and Krauss, 2018). Their inclusion offers a comprehensive benchmark against advanced deep learning techniques. We utilize the standard scikit-learn implementation for the baseline models (Pedregosa et al., 2011). For the Random Forest (RF), we follow the configuration used by Fischer and Krauss (2018), employing 1,000 trees with a maximum depth of 20. For Logistic Regression (LR), we use the default settings provided by the library. The Multi-Layer Perceptron (MLP) employs a two-layer architecture with hidden layer sizes of 32 and 16, consistent with the configurations of the LSTM and NIST-GNN models. To ensure consistency and comparability, both the machine learning models and the LSTM model are renewed and trained individually for each company, in line with the needs of temporal non-IIDness.

The number of parameters for each model is listed in Table 3.2. Both NGNN-SCRG and NGNN-Corr models have over 1 million parameters. This is primarily due to the unique LSTM weights assigned to each company node, enabling them to capture company-specific temporal dependencies. The NGNN-Corr model has slightly more parameters than NGNN-SCRG because the correlation graph includes more compa-

nies, as SCRG excludes companies not present in the GloVe vocabulary. In contrast, the GCRN-SCRG model has significantly fewer parameters, as it employs shared LSTM weights across all nodes, reducing the overall parameter count. For the remaining benchmark models, the parameter counts are reported per company, as these models are trained individually for each company. Notably, the RF model exhibits a much larger parameter count per node compared to the MLP and LR models, owing to the high number of decision trees utilized in the ensemble.

3.4.2 Trading simulations

The primary objective of stock trading is to maximize profits while managing the associated risks (Sawhney et al., 2021). It is important to note that, although the benchmark models are trained to minimize the BCE loss, high predictive accuracy does not always translate into the most profitable stock selections (Sawhney et al., 2021). Therefore, to comprehensively evaluate model performance, we prioritize profitability by conducting trading simulations based on online portfolio selection strategies.

Our trading simulation adopts a long-short portfolio approach, as detailed in studies by Hsu et al. (2016) and Wu et al. (2021). A portfolio is a list of stocks (components) for collective investment. For stocks predicted to rise (+1) on day $t + 1$, we simulate a buy trade at time t and a subsequent sell trade at $t + 1$. Conversely, for stocks predicted to fall (-1), we simulate a short position at t , followed by a repurchase at $t + 1$. In line with Feng et al. (2019), Xiang et al. (2022) and Vuletić et al. (2024), we exclude transaction costs from our simulation, based on the relatively low cost of trading US stocks through brokers, whether calculated per transaction or on a stock-by-stock basis. In addition, we report our PnL results also in basis points, which can be placed by the end user in the context of their own trading costs, which are typically also quoted in basis points.

We also employ a step-by-step training strategy incorporating time series cross-validation (Bergmeir and Benítez, 2012). Each training step spans 5 trading days, balancing training time with data efficiency. For each day t in the validation and out-of-sample periods, benchmark models are trained using data up to and including day t , with predictions then made for the next 5 days ($t + 1$ to $t + 5$). For spatial-temporal models and the LSTM model, we set a relatively long 60-day LSTM sliding window, following Fischer and Krauss (2018)’s recommendation of a longer horizon, to capture a comprehensive range of temporal dynamics.

Our evaluation involves implementing an online portfolio selection informed by the stock movement predictions generated by our models (Li et al., 2018a; Guo et al., 2023). We construct daily dynamic portfolios, which are adapted every trading day to reflect changes in the market. The composition of these portfolios is determined based on the historical prediction accuracy scores of the past three years from the current timestamp t . This strategy ensures that the portfolios are continually optimized based on the most recent and relevant predictive performance, thereby mirroring the adaptive decision-making process typical in actual trading environments. We set accuracy thresholds at 0.51, 0.52, 0.53, and 0.54. On each trading day, each benchmark model selects stocks exceeding these historical prediction accuracy thresholds. Thresholds above 0.54 were not considered due to their stringency, which limited the number of qualifying stocks and resulted in unstable portfolio performance. Similarly, thresholds below 0.51 were excluded to avoid stocks whose accuracy nears 0.50, tantamount to random guesses.

3.4.3 Evaluation metrics

To assess the profitability performance of portfolios created by our benchmark models, we utilize Profit and Loss (PnL) and Sharpe Ratio (SR) as our key metrics following previous studies (Hsu et al., 2016; Matsunaga et al., 2019; Wu et al., 2021). PnL, a standard measure in trading simulations, reflects the total profit or loss of a portfolio over a specified period and is calculated as follows:

$$\text{PnL} = \sum_t \left(\frac{1}{n_t} \sum_{i=1}^{n_t} \text{sign}_t^i * r_t^i \right), \quad (3.15)$$

where t denotes a timestamp in the out-of-sample period, i represents a stock component in the daily dynamic portfolio and n_t is the number of components at day t . The sign_t^i represents the model's prediction for stock i , while r_t^i denotes the daily return of stock i , calculated as $r_t^i = \frac{P_t^i - P_{t-1}^i}{P_{t-1}^i}$. The PnL is also converted into daily basis points, PnL_{bps} , by multiplying by 10,000 and dividing by the total number of trading days (755 days) in the out-of-sample period.

The Sharpe Ratio (SR) offers an adjusted measure of profitability that accounts for risk. We calculate the annualized Sharpe Ratio using the formula:

$$\text{SR} = \frac{\overline{\text{PnL}} - r_f}{\sigma_{pf}} * \sqrt{252}, \quad (3.16)$$

where $\overline{\text{PnL}}$ is the mean daily PnL of the portfolios, r_f is the risk-free rate, and σ_{pf} is the standard deviation of the risk-free excess return series. In this analysis, the US 10-year Treasury rate is used as the risk-free rate (Pratt and Grabowski, 2008), aligning with standard practices in financial evaluations.

For statistical predictability performance metrics, we follow prior research in using Accuracy (Acc), Precision, Recall and Matthews Correlation Coefficient (MCC) (Sawhney et al., 2020; Hsu et al., 2021; Lin et al., 2021). These metrics are calculated based on the confusion matrix $\begin{pmatrix} TP & FN \\ FP & TN \end{pmatrix}$:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.17)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.18)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3.19)$$

$$\text{MCC} = \frac{(TP \cdot TN) - (FP \cdot FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (3.20)$$

where TP denotes True Positives, TN is True Negatives, FP stands for False Positives, and FN indicates False Negatives. We use MCC because, unlike the F1 score, it mitigates data skew bias by being independent of the positive class choice and accounting for True Negatives.

In this study, we prioritize profitability metrics over predictability metrics, with a particular focus on the Sharpe Ratio (SR) due to its ability to balance profitability with risk. This focus reflects the core objective of stock trading: maximizing profits while managing risk effectively (Sawhney et al., 2021).

3.5 Results

In our results section, we detail the outcomes of our portfolio-based trading evaluation scheme, which is essential for understanding the practical applicability of our model in real-world trading scenarios and assessing its economic benefits.

Model	Portfolios (acc > 0.51)				Portfolios (acc > 0.52)				Portfolios (acc > 0.53)				Portfolios (acc > 0.54)			
	PnL	PnL _{bps}	SR	Mean acc	PnL	PnL _{bps}	SR	Mean acc	PnL	PnL _{bps}	SR	Mean acc	PnL	PnL _{bps}	SR	Mean acc
NGNN-SCRG	0.34	4.53	0.88	52.75%	0.35	4.65	0.84	53.40%	0.44	5.79	0.91	54.20%	0.46	6.04	0.79	55.02%
NGNN-Corr	0.09	1.23	0.08	52.74%	0.21	2.84	0.33	53.41%	0.31	4.17	0.51	54.22%	0.28	3.77	0.40	54.89%
GCRN-SCRG	0.20	2.67	0.24	53.21%	0.23	3.11	0.29	53.69%	0.24	3.13	0.29	54.38%	0.27	3.62	0.35	55.13%
LSTM	0.13	1.74	0.24	52.72%	0.16	2.17	0.32	53.38%	0.24	3.15	0.47	54.22%	0.26	3.44	0.44	55.02%
LR	-0.03	-0.39	-0.33	52.51%	-0.08	-1.07	-0.50	53.23%	-0.01	-0.09	-0.22	54.03%	-0.06	-0.73	-0.34	54.83%
RF	-0.10	-1.32	-0.68	52.26%	-0.09	-1.16	-0.56	52.97%	-0.10	-1.36	-0.54	53.85%	-0.14	-1.86	-0.38	54.56%
MLP	-0.02	-0.31	-0.41	52.25%	0.07	0.95	-0.52	52.95%	0.08	1.05	-0.49	53.83%	0.03	0.37	-0.40	54.70%
Market	0.41	5.40	0.50	-	0.41	5.40	0.50	-	0.41	5.40	0.50	-	0.41	5.40	0.50	-

Table 3.3: Performance characteristics of dynamic portfolios created by models with various mean accuracy thresholds (2018-01-01 to 2020-12-31).

3.5.1 Portfolio performance characteristics and portfolio selection

Table 3.3 details the performance characteristics of dynamic portfolios generated by our NGNN-SCRG and various benchmark models across the chosen accuracy thresholds. These metrics include Profit and Loss (PnL), daily PnL in basis points, Sharpe Ratio, and mean accuracy scores. A key finding is that our NGNN-SCRG consistently achieves superior results in both PnL and risk-adjusted returns (Sharpe Ratio) compared to the benchmarks. Notably, for the 0.53 and 0.54 thresholds, the NGNN-SCRG recorded its highest PnL, with daily PnLs of 5.79 and 6.04 basis points, respectively, exceeding the S&P 500 index's 5.40 bps. This performance, particularly a Sharpe Ratio peak of 0.91 at the 0.53 threshold, underscores the model's effective profitability and risk management capabilities.

An interesting aspect of the results is that despite the GCRN-SCRG showing slightly higher mean accuracy scores than the NGNN-SCRG (e.g. 54.38% vs. 54.20% with the 0.53 threshold), this marginal improvement in accuracy does not correspond to higher PnL and Sharpe Ratio. This discrepancy suggests that our NGNN-SCRG is more effective in identifying and capitalizing on trading days with higher profit opportunities.

Furthermore, classical machine learning benchmarks, including Logistic Regression, Random Forest, and MLP, predominantly show negative Sharpe Ratios. While GCRN-SCRG and LSTM achieve positive profits, they consistently fall short of the S&P 500 index in both PnL and Sharpe Ratio, highlighting the effectiveness of our NGNN-SCRG model.

We select the threshold of 0.53 for further detailed analysis of the model performance, because most benchmark models, including NGNN-SCRG, NGNN-Corr, and LSTM, achieved their best Sharpe Ratios.

	PnL				PnL _{bps}				Sharpe Ratio				<i>t</i> -Statistics (<i>p</i> -value)			
	2018	2019	2020	Overall	2018	2019	2020	Overall	2018	2019	2020	Overall	2018	2019	2020	Overall
NGNN-SCRG	0.08	0.20	0.16	0.44	3.08	7.81	6.46	5.79	0.58	2.29	0.70	0.91	0.04	0.13	0.04	0.05
NGNN-Corr	-0.04	0.18	0.17	0.31	-1.40	7.08	6.82	4.17	-0.44	1.63	0.62	0.51	0.01	0.09	0.03	0.03
GCRN-SCRG	-0.04	0.24	0.03	0.24	-1.46	9.59	1.26	3.13	-0.40	2.13	0.04	0.29	0.01	0.11	0.01	0.02
LSTM	-0.02	0.15	0.10	0.24	-0.75	6.06	4.14	3.15	-0.41	1.73	0.46	0.47	0.01	0.10	0.03	0.03
Market	-0.06	0.26	0.20	0.41	-2.31	10.38	8.10	5.40	-0.46	1.94	0.54	0.50	0.02	0.10	0.03	0.03

Table 3.4: Yearly performance characteristics of dynamic portfolios with the accuracy threshold 0.53. The last column shows the *p*-value of *t*-Statistics of the Sharpe Ratio.

3.5.2 Yearly performance breakdown

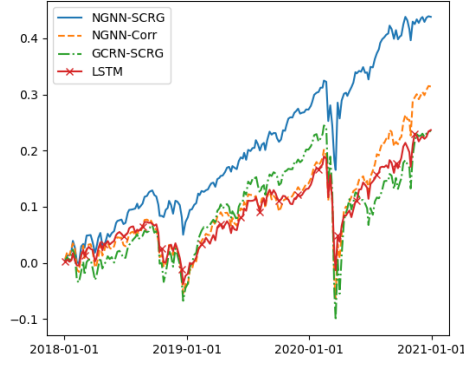
To assess the robustness of our NGNN-SCRG model, Table 3.4 provides a yearly breakdown of return performance for the out-of-sample period. Since machine learning models exhibit negative risk-adjusted returns, as indicated in Table 3.3, we exclude them and focus on comparing the deep learning based models for this robustness evaluation.

As shown in Table 3.4, in 2018, a particularly challenging year for market predictions due to adverse conditions, our NGNN-SCRG is the only model to record a positive PnL of 3.08 bps and a positive Sharpe Ratio of 0.58, outperforming both the benchmark models and the S&P 500 index, which all post negative returns. This underscores the effectiveness of the NGNN-SCRG in extracting valuable information from neighboring companies, even in challenging market conditions.

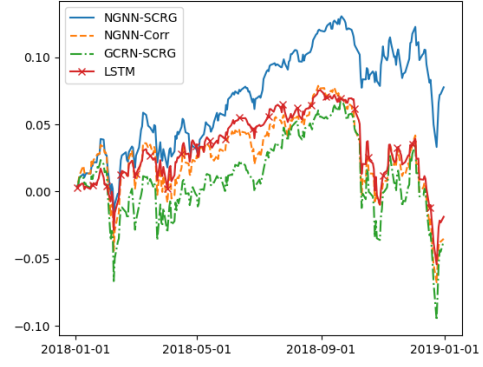
In 2019, the S&P 500 index achieves the highest PnL of 10.38 bps. However, this is accompanied by a relatively moderate Sharpe Ratio of 1.94, reflecting higher risk exposure of the market. In contrast, NGNN-SCRG achieves the highest Sharpe Ratio of 2.29 among all models, underscoring its superior risk-adjusted performance. In 2020, a year marked by extreme volatility due to the COVID-19 pandemic, NGNN-SCRG maintains strong risk-adjusted performance with a Sharpe Ratio of 0.70, outperforming the S&P 500 index (0.54) and most benchmark models. While the S&P 500 index records the highest PnL of 8.10 bps, NGNN-SCRG's ability to balance returns with reduced risk further validates its robustness.

Additionally, GCRN-SCRG surpasses the S&P 500 index in terms of Sharpe Ratio in 2019, while NGNN-Corr is the only baseline model to exceed the S&P 500 index's Sharpe Ratio in 2020. However, both models fall short of NGNN-SCRG, which consistently outperforms them in risk-adjusted returns across these periods.

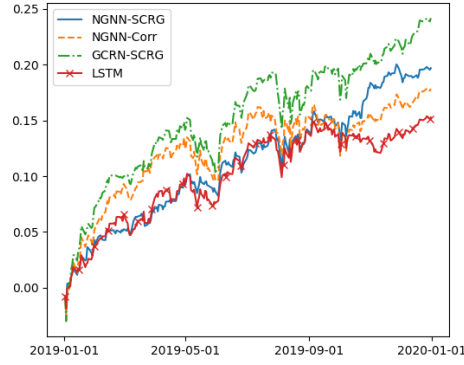
To statistically assess the significance of the Sharpe Ratio results, i.e., whether



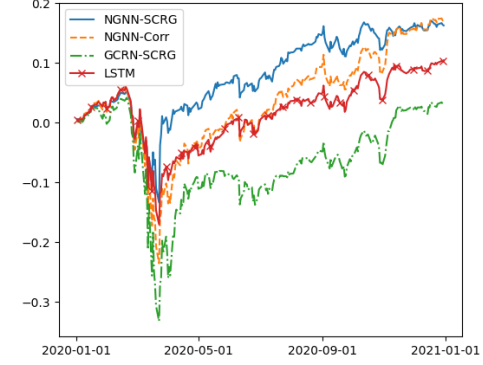
(a) Cumulative PnL across the out-of-sample period



(b) Cumulative PnL in 2018



(c) Cumulative PnL in 2019



(d) Cumulative PnL in 2020

Figure 3.2: Cumulative PnL from January 2018 to December 2020 of dynamic portfolios created by models with the accuracy threshold 0.53.

they are significantly different from zero, Bailey and López de Prado (2014) proposes a statistic that incorporates the effects of skewness and kurtosis, ensuring the test reflects the actual distribution of returns. The statistically significant Sharpe Ratio of our model, as shown in Table 3.4, underscores its robustness in consistently delivering superior results compared to competing methods.

3.5.3 Cumulative profit and loss

The trends in cumulative profit and loss (PnL) depicted in Figure 3.2 further substantiate the robustness of our NGNN-SCRG model's performance, complementing the findings in Table 3.4. Specifically, Figure 3.2a illustrates the overall cumulative PnL

across the entire out-of-sample period. Figure 3.2b through Figure 3.2d, each starting from zero at the beginning of the year, demonstrates the cumulative PnL trends for these individual years.

Figure 3.2a shows a significant divergence in the NGNN-SCRG's PnL trajectory from the other benchmark models, ultimately achieving a 41.9% higher PnL than the second-best NGNN-Corr, as detailed in Table 3.4. This growing divergence underscores the NGNN-SCRG's consistent and superior performance over time. In 2018, as demonstrated in Figure 3.2b, the NGNN-SCRG is uniquely successful in maintaining positive cumulative profits throughout the year, a distinction not shared by other models. The following year, Figure 3.2c indicates that despite GCRN-SCRG's better PnL trend, its higher risk exposure leads to a lower Sharpe Ratio compared to NGNN-SCRG, highlighting the latter's effective risk management strategies.

Lastly, Figure 3.2d illustrates the resilience of our model in the face of systematic risks during the early part of 2020, marked by the global outbreak of the Covid-19 pandemic (Mazur et al., 2021). The initial trend of our model mirrors that of the LSTM; however, a distinct upward shift in our model's performance is observed starting in April 2020. This 'U-turn' signifies the NGNN-SCRG's capability to swiftly recover from the market downturn and achieve positive profits, a resilience not as evident in other models. This ability to rebound from a black swan event like the Covid-19 pandemic is a testament to the NGNN-SCRG model's ability to consistently produce more robust forecasts.

3.5.4 Statistical performance analysis

	Precision	Recall	MCC
NGNN-SCRG	0.558	0.748	0.041
NGNN-Corr	0.559	0.797	0.023
GCRN-SCRG	0.550	0.866	0.042
LSTM	0.560	0.734	0.042

Table 3.5: Statistical predictability performance metrics of the benchmark models.

For statistical predictability performance analysis, we report the Precision, Recall and Matthews Correlation Coefficient (MCC) for the benchmark models in Table 3.5. From this table, we can tell that, NGNN-SCRG shows a balanced competitive performance with a Precision of 0.558, a Recall of 0.748, and an MCC of 0.041. While it

does not lead in the metrics, its balanced scores across the board indicate robust predictive capability. NGNN-Corr achieves a slightly higher Recall of 0.797 but has a severely lower MCC (0.023), suggesting that although it can identify more true positives, it does so with a higher rate of false positives, impacting its overall reliability.

GCRN-SCRG performs the best in terms of Recall (0.866) and MCC (0.042), indicating superior ability to capture true positives while maintaining a balanced prediction quality. This model sets the benchmark for performance. LSTM records the highest Precision (0.560) and an MCC of 0.042, very close to GCRN-SCRG. However, its lower Recall (0.7334) suggests that while it is precise in its positive predictions, it might miss a considerable number of true positive cases.

It is important to note that high predictive performance does not always translate into the most profitable stock selections (Sawhney et al., 2021). Thus, while GCRN-SCRG and NGNN-SCRG demonstrate competitive statistical predictability, profitability metrics provide a more comprehensive assessment of practical model performance in stock prediction applications, as reflected in Table 3.4.

3.5.5 Ablation study

	SR	PnL
GCN	0.14	0.14
GCN-LSTM	0.58	0.32
NGNN-SCRG	0.91	0.44
LSTM-NIID	0.47	0.24
GCRN-SCRG-NIID	0.36	0.17

Table 3.6: Model component ablation study results.

Following Sawhney et al. (2021) and Xiang et al. (2022), we conduct a model component ablation study of NGNN-SCRG to better understand what is driving the performance of our proposed approach.

We benchmark five model components. The first model component is a GCN only model (GCN) that follows the setting of GCN filter in equation 3.4. The second is GCN model with shared LSTM weights (GCN-LSTM), same as the architecture of GCRN-SCRG in equation 3.1 but replacing the Chebyshev filter with GCN filter. The third one is our proposed NGNN-SCRG, with architecture represented in equation 3.4. The fourth one is LSTM baseline that is trained individually for each stock. The

final added component benchmark is GCRN-SCRG with individual LSTM weights (GCRN-SCRG-NIID) that essentially replaces the GCN spatial filter with GCRN's Chebyshev filter in equation 3.4.

The ablation study results are presented in Table 3.6. From these results, we can observe a clear trend: performance improves progressively from GCN to GCN-LSTM and reaches its peak with NGNN-SCRG. Specifically, introducing the LSTM layer in GCN-LSTM yields a substantial boost in performance compared to the GCN-only model, with SR increasing from 0.14 to 0.58 and PnL from 0.14 to 0.32. This improvement indicates that the LSTM component contributes valuable temporal processing capabilities, allowing the model to better capture the sequential dependencies in stock data.

Our proposed NGNN-SCRG, which incorporates a Non-IID LSTM, builds on this performance gain by further enhancing the temporal dynamics in a way that accounts for the unique, non-IID characteristics of financial time series. This results in the highest performance metrics in the study, with an SR of 0.91 and a PnL of 0.44, demonstrating the effectiveness of NGNN-SCRG in harnessing both spatial and temporal information.

In contrast, while the LSTM-NIID model achieves relatively competitive results on its own, introducing the Chebyshev filter in the GCRN-SCRG-NIID configuration actually leads to a performance drop (SR of 0.36 and PnL of 0.17). This decline confirms our analysis that the Chebyshev filter is not well-suited for graphs with a community structure, such as our SCRG. Specifically, the Chebyshev filter's restricted spatial filtering capacity limits its ability to capture the community-based connections inherent in the SCRG, thereby impacting overall model performance.

3.5.6 Performance comparison against other spatial-temporal methods

Xiang et al. (2022) proposed a Temporal and Heterogeneous Graph Neural Network (THGNN) based on stock price correlation information (correlation graph) for stock movement prediction and portfolio selection. In their study, they summarized the state-of-the-art spatial-temporal models' performance on the year of 2020 and benchmarked THGNN against them. These models are TGC (Feng et al., 2019), MAN-SF (Sawhney et al., 2020), HATS (Kim et al., 2019), REST (Xu et al., 2021) and AD-GAT (Cheng and Li, 2021). Among these, TGC, MAN-SF, and HATS rely on WikiData-

	SR	ARR
TGC	1.16	0.53
MAN-SF	1.13	0.53
HATS	1.06	0.49
REST	1.08	0.50
AD-GAT	1.17	0.54
THGNN	1.42	0.67
NGNN-SCRG	1.94	0.55

Table 3.7: Profitability performance of NGNN-SCRG against the state-of-the-art models from the evaluation year 2020 with top- k portfolio selection.

based graphs, while REST uses a WikiData variant that employs event extraction to identify specific types of relations from announcements. AD-GAT, on the other hand, utilizes a variant based on the S&P Capital IQ database to extract particular relations (see Section 1.3.1.2).

To assess how our proposed model aligns with existing literature, we benchmark NGNN-SCRG against these models, reporting both the Sharpe Ratio (SR) and Annual Return Rate (ARR). ARR is calculated by compounding daily returns rather than summing them like the PnL (Lin et al., 2021). For a fair comparison, we adhere to the portfolio selection standards used in those studies, selecting portfolios based on the top- k best-performing stocks for online portfolio construction. Specifically, in the study of TGC (Feng et al., 2019), the authors examined the effect of varying k in stock selection and identified $k = 5$ as the most representative value for performance assessment. Consequently, we adopt this setting for consistency. The results are showed in Table 3.7.

From these results, we observe that NGNN-SCRG outperforms the other models in terms of Sharpe Ratio (SR), achieving an SR of 1.94, which is notably higher than that of any other model. This significant improvement in SR suggests that NGNN-SCRG effectively balances return and risk, delivering more stable performance over time compared to other methods. Regarding ARR, NGNN-SCRG achieves a competitive value of 0.55. While this is slightly lower than the THGNN model’s ARR of 0.67, NGNN-SCRG still surpasses all other models, demonstrating that it provides substantial profitability along with a high SR.

Throughout this study, we prioritize SR as the key metric, emphasizing risk-adjusted

returns. The high SR achieved by NGNN-SCRG highlights its robustness and effectiveness in dynamic market conditions, underscoring its potential as a strong alternative to existing spatial-temporal models.

3.5.7 Discussion of experiment results

The results presented in Table 3.4 are particularly noteworthy in underscoring the comparative effectiveness of our NGNN-SCRG model. Specifically, the table reveals that the NGNN-SCRG not only achieved better risk-adjusted returns but also higher profits in two out of the three out-of-sample years when compared to NGNN-Corr, notably during a time when the market faced systematic risks. This enhanced performance is rooted in the creation of the semantic space for company vector embeddings (Burgess and Cottrell, 1995; Burgess and Lund, 1995; Lund and Burgess, 1996) as discussed in Chapter 2.

Previous studies in the area of quantitative finance have proposed various approaches to identify company interrelationships based on information including historical stock prices (Li et al., 2020a; Keskin et al., 2021), WikiData (Feng et al., 2019; Kim et al., 2019; Sawhney et al., 2020), and stock fundamental information (Hou et al., 2021). However, to identify contemporaneous changes in company interrelationships, investors often need to rely on sources of information that are available promptly. In real-time, this information is typically disseminated to the market via financial news articles, which distill complex financial events and their consequences into a more comprehensible sequence, enabling investors to make informed decisions (Ma et al., 2009b; Box, 2018). Our study successfully introduces the methodology of news-corpus-based semantic company relationship representation, showcasing its advantages over traditional relationship modelling methods, such as those relying on correlation graphs (Li et al., 2020a; Keskin et al., 2021).

Downstream studies based on financial relational modeling could benefit from our contribution. For instance, in the study of business networking (Österle et al., 2001; Ford and Mouzas, 2013), scholars could not only establish the business-to-business network through the semantic embedding, but also the business-to-product network, business-to-industry network, etc, empowering the potential of exploring multi-dimensional relationships and their complex dynamics.

Furthermore, the similar enhanced performance of NGNN-SCRG over GCRN-SCRG demonstrates NIST-GNN’s capability to aggregate more pertinent feature infor-

mation from adjacent companies than the original GCRN (eq. (3.1)). This experimental finding confirms the significance of temporal non-IIDness among different stocks, and the effectiveness of our novel LSTM weight-sharing mechanism (eq. (3.4)).

Several lines of research in financial predictions can benefit from this finding. First, when exploring risk propagation in financial networks (Dungey* et al., 2005), such as banking (Kaufman, 1994) and stock markets (Masih and Masih, 1997; Zhang and Liu, 2018; Yu et al., 2021), our NIST-GNN model facilitates the conceptualization of interconnected financial entities within a risk graph and encapsulates the dynamics of risk propagation effectively.. Second, in the domain of trading strategies, researchers can embed sector relationships within a graph to support sector rotation decisions (Cavaglia et al., 2001; Conover et al., 2008). By harnessing the NIST-GNN, it becomes feasible to assess sector significance and pinpoint profitable sectors, thus offering an alternative to conventional empirical metrics.

Finally, identifying company interrelationships is of significant value for managing portfolio risk (Yaros and Imieliński, 2015; Box, 2018). Our NIST-GNN approach captures these interrelationships through the use of both historical data and contemporaneous information flows across firms, thereby resulting in portfolios that demonstrate increased resilience in periods of market downtime.

3.5.8 Implications and future directions

Our study challenges the Efficient Market Hypothesis (EMH) as articulated by Fama (1970) in the aspect of slow market information diffusion (Lo and MacKinlay, 1990; Cohen and Frazzini, 2008; Menzly and Ozbas, 2010; Cohen and Lou, 2012; Huang, 2015; Sharifkhani and Simutin, 2021; Yu et al., 2024). Through our empirical trading evidence, we demonstrate that leveraging publicly available information from related companies enables NGNN-SCRG to achieve positive alphas. This finding directly questions the EMH's assumption that market prices instantaneously and fully reflect all available information. The ability of NGNN-SCRG to capture and exploit inter-company dependencies highlights inefficiencies in how the market processes relational information, suggesting that such dependencies are underutilized by traditional approaches.

Lo and MacKinlay (1990) was among the earliest researchers examining the phenomena of slow market diffusion, showing that lagged weekly returns of large companies could contribute to the explanation of current weekly returns of smaller com-

panies, hinting at delayed market responses. Corroborating this study, our significant results demonstrate that using $t - 1$'s spatial and temporal information can effectively predict stock movements on day t , suggesting a delay in the diffusion of market information.

Later studies proposed that economic links between companies, such as supplier-customer relationships or competitors, could serve as indicators of information diffusion delays, often due to investors' limited attention and the complexity involved in analysing companies that are inter-correlated Cohen and Frazzini (2008); Sharifkhani and Simutin (2021); Yu et al. (2024). Based on this perspective, Cohen and Lou (2012) and Huang (2015) devised long-short trading strategies that yielded persistently positive alphas over extended periods, capitalizing on lagged information from economically linked companies.

Echoing these foundational insights, our experimental results showed the existence of slow information diffusion in the current US stock markets. In particular, public information from cross-correlated stocks in the US market experiences a minimum one-day lag in diffusion. Such insights have profound implications for investment strategies, market analysis, and the broader understanding of market efficiency, indicating that investors can potentially exploit these lags for profitable trading opportunities.

Our study also has limitations worth noting for future research. First, our Semantic Company Relationship Graph is static, yet company relationships are dynamic. Future work could explore modeling the SCRG dynamically to better reflect evolving company interactions. Second, the SCRG and NIST-GNN are trained separately, with the SCRG serving as a preprocessing step. An integrated, end-to-end training approach could allow for gradient propagation through both the SCRG and NIST-GNN, potentially enabling finer tuning of the SCRG with additional market information. Another limitation of our research lies in its binary stock movement predictions. While binary predictions provide valuable insights, they inherently simplify the complex nature of stock price movements. Future research could expand this scope by implementing continuous predictions of stock prices which could offer a more subtle understanding of stock market dynamics, capturing not just the direction but also the magnitude of price changes.

Addressing these limitations could potentially enhance the adaptability of the framework to continuously evolving market conditions and different economic regimes.

3.6 Conclusion

In this chapter, we introduced the NGNN-SCRG framework for stock movement prediction and portfolio selection, combining the Semantic Company Relationship Graph (SCRG) with the Non-IID Spatial-Temporal Graph Neural Network (NIST-GNN). The SCRG, leveraging a novel company vector embedding scheme, effectively captures investors' perceptions of company relationships based on real-world financial events. This approach addresses the limitations associated with the indirect relationship modeling inherent in traditional correlation graphs. Furthermore, our NIST-GNN, expanding upon the GCRN architecture, efficiently handles the non-IID nature among different stock sequences, offering an innovative solution to a complex problem in financial analytics.

Our comprehensive evaluation demonstrates that the NGNN-SCRG framework not only outperforms leading benchmark models but also significantly improves the ability to identify profitable stocks while offering superior risk-adjusted returns. These results provide compelling evidence that public information in the US stock market is subject to at least a one-day lag, presenting a challenge to the traditional Efficient Market Hypothesis (EMH). This finding has far-reaching implications, suggesting a need for a revised understanding of information diffusion within financial markets.

Chapter 4

How to Assess Investor Sentiment on Media Silent Days?

News sentiment, serving as a crucial proxy for investor sentiment, is widely used for asset pricing. Despite its importance, not all companies receive consistent media coverage, resulting in many experiencing “media silent” days without news coverage, as the media often focuses on the most sensational business events. By analyzing 14 years of comprehensive news data, we discover that even the highly visible companies lack daily news coverage on nearly half of the trading days.

This study addresses the challenge of assessing investor sentiment by inferring the absent news sentiment during these “media silent” days. Existing missing value imputation (MVI) techniques, while numerous, fall short in the financial context due to complex spatial and temporal dependencies among companies. To bridge this gap, we develop NIST-Cheb (Non-IID Spatial-Temporal Chebyshev Network) that leverages these dependencies to infer non-existing news sentiment. This approach is enhanced by a semi-supervised training scheme that optimizes the utilization of available news sentiment data.

The effectiveness of our methodology is rigorously validated using both error-based metrics and real-world trading evaluations. The results demonstrate that our inferred sentiment achieves the lowest error scores, and asset pricing models incorporating our sentiment significantly outperform baseline methods, effectively doubling profits compared to models using raw sentiment data. Moreover, our research contributes theoretical insights into the information spillover effect of news sentiment, highlighting the practicality and significance of NIST-Cheb in addressing the challenges of absent financial news sentiment inference.

* This work has been accepted and presented on the Economics of Financial Technology Conference 2023, and the ICAIF 2023 Workshop on NLP and Network Analysis in Financial Applications.

4.1 Introduction

Investor sentiment is pivotal in asset price prediction, as demonstrated by extensive prior research (Baker and Wurgler, 2007; Huang et al., 2015; Zhou, 2018). However, directly measuring investor sentiment is so challenging that proxies are commonly used (Baker and Wurgler, 2007). Research by Tetlock (2007) and García-Laencina et al. (2010) has highlighted news sentiment as a significant proxy for investor sentiment towards companies, greatly influencing stock price forecasts. Subsequent studies by Li et al. (2014a), Heston and Sinha (2017) and Obaid and Pukthuanthong (2022) have expanded this understanding by exploring the impact of news sentiment across various financial markets and types of news.

Despite the recognized proxy role of news sentiment, substantial challenges arise from the uneven frequency of news coverage, as media attention is often drawn to the most sensational business events. Our comprehensive analysis of financial news, encompassing major publishers from 2005 to 2018, uncovers a significant issue: even the most prominent S&P 500 companies lack news coverage on approximately 50.67% of trading days.

Nevertheless, the absence of news coverage does not imply a lack of investor sentiment since stocks continue to see investment decisions and trading activities even on days without media coverage. This raises an intriguing research question: how can investor sentiment be assessed on these media silent days?

Missing Value Imputation (MVI) is a related and widely recognized technique originally devised to address data incompleteness (Lin and Tsai, 2020). These methods rely on the presence of other non-missing attributes within the same data record to infer the missing one (Brás and Menezes, 2007; Bryzgalova et al., 2022). While MVI methods provide valuable insights, they typically aim to fill gaps where data is expected but missing, often due to issues in data collection or processing. These methods assume the presence of data that, for various reasons, was not captured or recorded.

Our research, however, addresses a fundamentally different challenge: we deal with days where financial news is inherently absent, not because of collection or recording failures, but due to a lack of noteworthy events. This distinction is crucial as

it shifts our focus from imputing “missing” data to inferring the implied sentiment.

In fact, even in the absence of news, investors continue to make decisions based on available sources of information, such as historical sentiment and events from related companies (Langfield-Smith and Smith, 2003; Lozano et al., 2013). Building on this insight, we propose a Non-IID Spatial-Temporal Chebyshev Network (NIST-Cheb) to infer absent news sentiment by leveraging both historical sentiment records and the sentiment of related companies.

Traditional missing value imputation (MVI) research often employs error-based evaluation metrics such as Mean Absolute Error (MAE) and Mean Relative Error (MRE) to assess model performance (Cao et al., 2018; Yang et al., 2021; Wang et al., 2023b,a). While these metrics are effective for standard MVI applications, such as air quality (Yi et al., 2016; Cao et al., 2018; Wang et al., 2023b) and taxi trajectory data (Yang et al., 2021; Zhang et al., 2017), they are insufficient in the context of financial sentiment data. This limitation arises because sentiment data serves as an intermediary factor in asset pricing tasks (Tetlock, 2007; García-Laencina et al., 2010), where its influence on financial outcomes is more pivotal than what conventional error metrics can capture. As Lin and Tsai (2020) suggests, a more appropriate evaluation of our inferred data should involve its application in practical trading scenarios, reflecting its real-world efficacy.

To systematically confirm the effectiveness of our sentiment inference approach, we employ a dual evaluation methodology. Initially, we utilize conventional error-based metrics to demonstrate our method’s capability to minimize numerical errors. Subsequently, we validate the practical applicability of our method through stock prediction-based evaluations. By conducting trading experiments using real-world financial market data, we directly connect the quality of our inferred sentiment to tangible financial outcomes.

Contributions. This study provides several key contributions:

- Our study innovatively defines the research problem of non-existing news sentiment inference on media silent days. Distinct from Missing Value Imputation that addresses data gaps by filling in missing values based on existing attributes of the same data record (Brás and Menezes, 2007; Bryzgalova et al., 2022), our study pioneers in the challenge of inferring sentiment where the entire attributes, (Negative, Neutral, Positive), of the sentiment are absent.
- To address this unique research question, we propose a Non-IID Spatial-Temporal

Chebyshev Network (NIST-Cheb). The NIST-Cheb is a specialized adaptation of the Non-IID Spatial-Temporal Graph Neural Network (NIST-GNN) designed to leverage spatial-temporal relationships among companies to model sentiment dynamics in the absence of direct news reports. A key feature of this adaptation is the strategic use of a Chebyshev filter (Defferrard et al., 2016), which enables the retrieval and sampling of relevant news sentiment from a broad spectrum of related companies for the target company.

- To effectively train NIST-Cheb with incomplete sentiment data, we propose a masked semi-supervised optimization scheme, inspired by Kipf and Welling (2017). This innovative approach involves intentionally masking known sentiment scores to simulate the absence of sentiment data, thereby treating them as ground truths. By leveraging both labeled and unlabeled data, this strategy aims to enhance the learning accuracy of the model under conditions of incomplete data availability.

4.2 Related Work

The research area most closely related to our study is Missing Value Imputation (MVI). Traditional MVI methods include statistical approaches that utilize the sample mean, median, mode, or estimates based on linear regression (Little and Rubin, 2019), as well as machine learning techniques like supervised regression and unsupervised clustering methods, including k -nearest neighbor (k -NN) and support vector machines (SVM) (García-Laencina et al., 2010). These conventional MVI approaches are designed for a general context of missing data and are based on assumptions that are often invalid in the complex environment of financial data (Cao et al., 2018; You et al., 2020), thus they are unsuitable for directly addressing our research problem of inferring non-existing financial news sentiment.

Specifically, the first mismatch arises from the complete absence of financial sentiment, wherein the three sentiment elements, (Negative, Neutral, Positive), are entirely non-existent, rather than partially missing as in traditional MVI scenarios. This complete absence of data makes prevalent machine learning imputation methods, such as k -nearest neighbor (k -NN), ineffective. These methods depend on the presence of other non-missing attributes within the same data record to infer missing values (Brás and Menezes, 2007; Bryzgalova et al., 2022), a condition not met in our research problem.

Additionally, the inherent temporal dependencies within financial data pose further challenges (Bryzgalova et al., 2022). Financial sentiment data, typically structured as time-series sequences, inherently connects to historical data points reflecting the enduring impacts of sentiments (Tetlock, 2007). This relational aspect makes it impractical to use future data for imputing current values, a significant limitation that is not commonly faced in general MVI methods.

Moreover, company news sentiment is also influenced by spatially related companies, contrasting with the isolated instances often assumed in MVI scenarios. Companies operate within an interconnected business ecosystem, where they influence and are influenced by one another (Langfield-Smith and Smith, 2003; Yusuf et al., 2004; Lozano et al., 2013). This necessitates the inclusion of sentiment data from spatially related firms to accurately infer the non-existing sentiment for a target company (Hou et al., 2021; Wu et al., 2021), further complicating the application of traditional MVI techniques in our context.

Recent advancements in Spatial-Temporal Graph Neural Networks (ST-GNNs) applied to traffic data MVI scenarios have provided valuable inspirations for addressing the aforementioned challenges. For instance, Ye et al. (2021) introduced a Graph Attention Convolutional Network (GACN) to capture spatial correlations among traffic data from adjacent sensors and employed temporal convolutions to unravel time-series relationships. Similarly, Asadi and Regan (2019) proposed a convolution bidirectional LSTM to detect spatial and temporal patterns in traffic flow data, where the convolutional layer analyzes spatial time series data with kernels and the bidirectional LSTM layer decodes temporal relations.

However, ST-GNN models used in traffic data often assume temporal independence and identically distributed (IID) nodes, facilitated by shared LSTM weights across the graph (Seo et al., 2018). This assumption does not hold in the financial context, where companies exhibit significant diversity in aspects such as industry sectors, product lines, and corporate governance, leading to non-IID temporal characteristics among entities.

Furthermore, unlike the traffic domain where well-defined traffic networks exist (Ye et al., 2021; Ren et al., 2021; Chen and Tsourakakis, 2022), explicit relationship graphs are not readily available in the financial domain (Kim et al., 2019). Capturing the spatial dependencies between companies demand an in-depth understanding of business interactions among them with domain knowledge (Zheng et al., 2021).

To overcome these challenges and apply ST-GNNs to the problem of absent news

sentiment inference, this study adopts the Semantic Company Relationship Graph and tailors a Non-IID Spatial-Temporal Graph Neural Network (NIST-GNN) with architectural modifications to model the complex company relationships and address the non-IID characteristic of the financial sector.

4.3 Methodology

Our methodology is organized into three distinct sections. Initially, we introduce the comprehensive news sentiment dataset we have compiled, accompanied by a statistical analysis of news absent rates, providing readers with a detailed understanding of the data landscape. Subsequently, we propose the Non-IID Spatial-Temporal Chebyshev Network (NIST-Cheb) by implementing specific architectural modifications to the NIST-GNN. NIST-Cheb works in conjunction with the SCRG, which quantitatively illustrates the connections among closely related companies. Finally, we outline the semi-supervised optimization strategy for NIST-Cheb, designed to make optimal use of the available limited news sentiment ground truths.

4.3.1 The Financial News Sentiment Dataset

4.3.1.1 The data source

Our collected dataset encompasses over 2.6 million financial news items with sentiment scores spanning from January 1, 2005, to December 31, 2018, sourced from Refinitiv News Analytics, a leading financial news provider. Refinitiv tracks news from established financial media including the Wall Street Journal, Reuters, Financial Times, and CNN, among others. The dataset spans more than 700 topics, including business, science, technology, and sports, with a strong emphasis on the business implications of reported events Calomiris and Mamaysky (2019). Refinitiv News Analytics data has been widely utilized in the academic literature to model and forecast financial market returns and volatilities (Bali et al., 2018; Heston and Sinha, 2017; Ito et al., 2023). For each news item, Refinitiv News Analytics provides a three-dimensional sentiment score — Negative, Neutral, Positive — attributed to the targeted company. These scores reflect the likelihood that the news content will be perceived by the investors of the company as negative, neutral, or positive, respectively, with the sum of these scores equaling one.

We focus our analysis on companies listed in the S&P 500 index as those are the

most valuable and representative firms in the capital market. The analysis adopts a daily interval, including news from 4:00 PM of the previous trading day ($t - 1$) to 3:59 PM of the current trading day (t) as the sentiment data for day t . It's important to note that t refers to trading days, not calendar days. For example, if t is January 10, 2005, $t - 1$ would be January 7, 2005. For cases where multiple news reports are available for a company on a given day, we calculate the daily sentiment score as the simple average of all sentiment scores for that day.

4.3.1.2 Exploratory data analysis

	Mean	Std	Min	25%	50%	75%	Max
Negative	0.262	0.199	0.006	0.110	0.231	0.366	0.819
Neutral	0.377	0.191	0.112	0.214	0.358	0.539	0.928
Positive	0.361	0.212	0.019	0.180	0.317	0.505	0.857

Table 4.1: Statistical description of the financial news sentiment scores.

To better understand the data properties, we first give the statistical description of the news sentiment scores in Table 4.1. This statistics shows that the average sentiment polarity of the news is neutral, which average polarity of 37.74%. The number of positive scores takes a very close percentage of 36.09%. All the three sentiment scores have the similar standard deviation of around 0.2.

To figure out the trends of overall company sentiment, we specially draw a line plot in Figure 4.1 depicting the monthly average sentiment scores for company news from 2005 to 2018, showing trends in negative, neutral, and positive sentiment over time. The data has been resampled to a monthly frequency to provide a smoother visualization of sentiment changes. We observe that notable sentiment patterns align with significant economic events. During the 2008-2009 financial crisis, there is a sharp increase in negative sentiment (blue line) and a corresponding drop in positive sentiment (green line). This period reflects heightened pessimism in news coverage likely due to economic uncertainty and market instability. In contrast, the 2016-2017 bull market is marked by a rise in positive sentiment, reflecting increased optimism and confidence in the economy. This period sees positive sentiment scores (green line) consistently trending higher, while negative sentiment scores remain subdued, suggesting that news coverage was increasingly favorable.

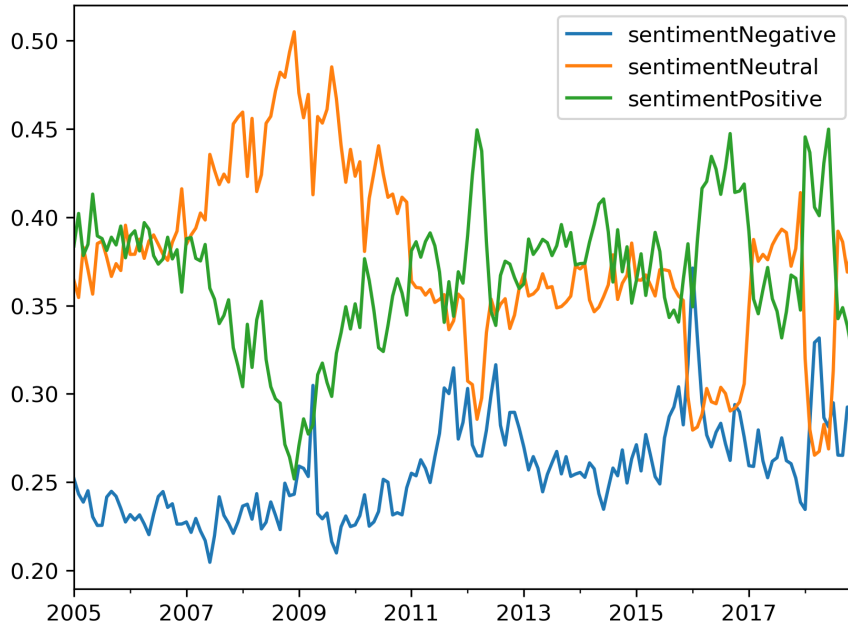


Figure 4.1: The trends of monthly average sentiment scores for all companies from 2005 to 2018.

To quantify the extent of non-existing news sentiment of the S&P 500 components, we define the news absent rate as the following equation,

$$\text{Absent Rate} = \frac{\sum day_t}{3523}, \quad (4.1)$$

where day_t is a binary variable indicating trading day t has news report ($day_t = 1$) or not ($day_t = 0$), and 3,523 is the total length of the news sentiment dataset. Across S&P 500 companies, we observe an average news absent rate of 50.67%, demonstrating that these firms lack news coverage on half of the total trading days. Detailed sector-level absent rate statistics, presented in Table 4.2, indicate that the Materials and Real Estate sectors experience absent rates above 60%, whereas the Communication Services sector exhibits a absent rate below 30%.

Additionally, Figure 4.2 presents the distribution of the length of continuous days without news coverage for all companies. From this figure, it is evident that most companies on most trading days experience continuous periods without news coverage lasting less than 3 days. This is also supported by the summary statistics in Table 4.3, where the third quartile is only 3, indicating that 75% of the periods are 3 days or fewer. The distribution exhibits significant positive skewness (8.09), with a long right tail, indicating that while most periods without news coverage are short, there are occasional extended periods. Additionally, the high kurtosis value (279.63) suggests

Sector	Company Count	Absent Rate
Communication Services	13	28.17%
Consumer Discretionary	52	51.94%
Consumer Staples	29	47.89%
Energy	20	38.13%
Financials	57	40.05%
Health Care	57	56.39%
Industrials	52	55.82%
Information Technology	52	49.97%
Materials	20	64.63%
Real Estate	31	64.67%
Utilities	27	46.75%
Total	410	50.67%

Table 4.2: Sector-level absent news statistics of the financial news sentiment dataset.

Mean	Std	Min	25%	50%	75%	Max	Skewness	Kurtosis
2.93	3.60	1	1	2	3	276	8.09	279.63

Table 4.3: Descriptive statistics of the length of continuous days without news coverage.

a leptokurtic distribution, characterized by a sharp peak and heavy tails. This implies that extreme values, such as the maximum observed length of 276 days, are much more likely than they would be in a normal distribution. These statistical measures reflect the rarity but presence of extended gaps in news coverage for certain companies.

To examine potential relationships in the timing of news absences across companies, we calculated the correlations between the binary time series representing news coverage for each company, where 0 indicates a day without news and 1 represents a day with news. The resulting correlation matrix reveals that most correlations are quite low, with an overall mean correlation of only 0.036. When examining the highest correlation value for each company, the mean of these maximum values is 0.153, with a standard deviation of 0.049. The highest observed correlation is 0.275, indicating that even the strongest relationship between companies is relatively weak. These low correlations suggest that the occurrence of days without news is largely independent across companies, with minimal shared patterns in news absence timing.

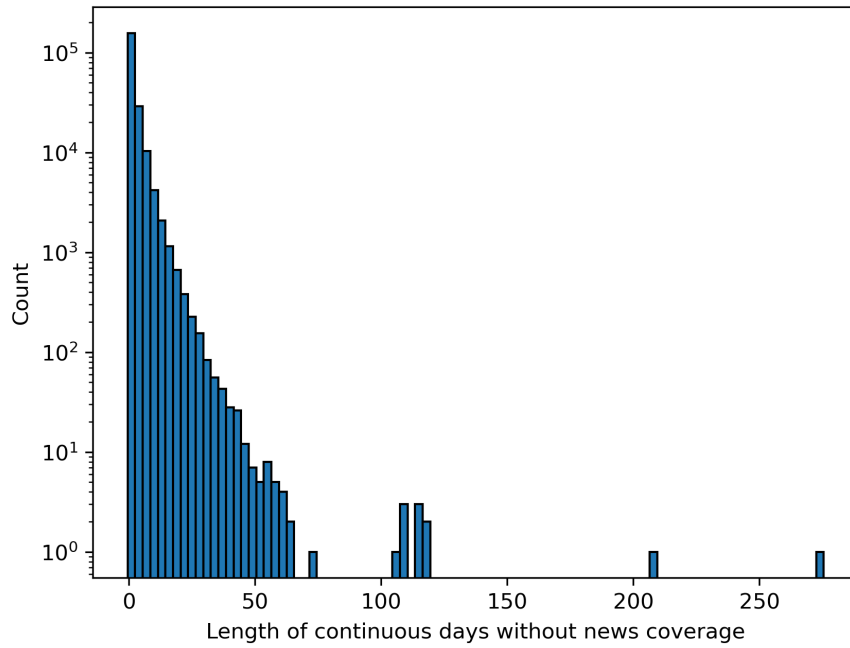


Figure 4.2: Distribution of length of continuous days without news coverage.

4.3.2 The NIST-Cheb Sentiment Inference Model

Spatial-Temporal Graph Neural Networks (ST-GNNs) are specialized GNNs crafted to process spatially-correlated time-series sequences of nodes within a graph (Seo et al., 2018; Chen et al., 2022). As discussed in section 4.2, directly applying traditional ST-GNNs to our research problem of absent news sentiment inference encounters three major challenges: the complete absence pattern of sentiment scores, the temporal IIDness of each company node, and the absence of explicit company relationship representations. The last challenge has been addressed by incorporating the Semantic Company Relationship Graph (SCRG), while the first two are tackled by our proposed NIST-Cheb model, an adaptation of NIST-GNN.

Specifically, NIST-Cheb employs a Chebyshev spatial graph filter g for aggregating spatial sentiment across all company nodes (Defferrard et al., 2016), and uses distinct temporal LSTM filters for each company's sentiment sequence. This configuration not only facilitates the retrieval and sampling of relevant news sentiment from a broader array of related companies but also allows each company in the SCRG to have its unique set of LSTM weights, effectively managing temporal IIDness. To address the complete absence pattern of sentiment scores, NIST-Cheb is equipped with a Softmax output head to ensure that the three sub-scores are generated simultaneously and the sum of them always equals one.

4.3.2.1 Spatial Sentiment Aggregation

Distinct from the original NIST-GNN setup, which utilizes a Graph Convolutional Network (GCN) filter (Kipf and Welling, 2017) designed for stock movement prediction, we have adapted NIST-Cheb to include a Chebyshev filter (Defferrard et al., 2016). This adjustment is intended to expand the reach of sentiment aggregation by capturing information from companies multiple hops away, thus enhancing the breadth of data integration.

The rationale for this modification stems from the fact that the GCN filter is essentially a first-order approximation of the Chebyshev filter (Kipf and Welling, 2017; Wu et al., 2020a). This approximation restricts its ability to stack multiple GCNs for multi-hop feature aggregation, often resulting in over-smoothing (Chen et al., 2020), which typically limits optimal GCN performance to two layers.

Given that the average length of the shortest paths between companies in our Semantic Company Relationship Graph (SCRG) is 2.86, effectively aggregating sentiment information up to three hops is crucial. Thus, we opt for the Chebyshev filter, avoiding the simplifications associated with GCN.

While the distinction between the typically optimal two-layer setting and our three-layer requirement may seem minor to readers unfamiliar with the field, it is significant in practice. In fact, most graph neural networks are inherently shallow, typically consisting of only two or three layers (Li et al., 2019b), as the number of nodes increases exponentially with the number of hops (layers). Adding an extra layer, therefore, can significantly enhance GNN performance.

Specifically, the Chebyshev filter employs the following rule for spatial sentiment aggregation:

$$X_g = g(\tilde{L}, X_t) = \sum_{k=0}^{K-1} T_k(\tilde{L}) X_t W_k, \quad (4.2)$$

where $X_t \in \mathbb{R}^{N \times d_x}$ is the sentiment feature matrix for all the companies within SCRG, and $X_g \in \mathbb{R}^{N \times d_s}$ represents the aggregated feature matrix, with $d_s = 16$ to allow for broader feature processing. The parameter K controls the breadth of spatial aggregation, extending to K -hop neighbors. In our settings, K is set to 3, reflecting the average shortest path length of 2.86 between companies in the SCRG. The weight matrix $W_k \in \mathbb{R}^{d_x \times d_s}$ of each layer facilitates the sentiment feature transformation for the k -th hop neighbors.

The term $T_k(\tilde{L}) \in \mathbb{R}^{N \times N}$ represents the Chebyshev polynomial of order k , a crucial

element in controlling the spatial aggregation process for the k -th hop. The definition of $T_k(\tilde{L})$ is grounded in the use of the weight-normalized Laplacian, $\tilde{L}$, of the SCRG, as outlined in the works of Defferrard et al. (2016) and Seo et al. (2018):

$$T_k(\tilde{L}) = 2\tilde{L}T_{k-1}(\tilde{L}) - T_{k-2}(\tilde{L}) \quad (4.3)$$

$$\tilde{L} = I - D^{-\frac{1}{2}}AD^{-\frac{1}{2}} \quad (4.4)$$

with the initial conditions set as $T_0(\tilde{L}) = I$ and $T_1(\tilde{L}) = \tilde{L}$, where I denotes the identity matrix. D and A represent the degree matrix and the adjacency matrix of SCRG. This approach, employing the weight-normalized Laplacian, $\tilde{L}$, ensures that the aggregation process accounts for the varying strengths of connections between companies. Specifically, it allows each company to differentially weigh the sentiment information from its neighbors, based on the strength of their respective connections within the SCRG. This nuanced attention allocation facilitates a more discerning spatial aggregation of sentiment features, reflective of the actual relational dynamics between companies.

4.3.2.2 Temporal Sentiment Aggregation

NIST-Cheb incorporates an FC-LSTM (Graves, 2013) as the temporal filter, setting it apart from traditional ST-GNNs (Seo et al., 2018) by assigning unique weights to each company node within the SCRG, instead of utilizing shared weights. This configuration ensures that the distinct sentiment dynamics of each company are accurately modeled, effectively accommodating the non-IID nature of the companies, which arises from their substantial diversity in factors such as industry sectors, product lines, and corporate governance.

The FC-LSTM temporal filter operates via a time-step-wise mechanism characterized by two primary states, the cell state $c_t \in \mathbb{R}^{d_h}$ and hidden state $h_t \in \mathbb{R}^{d_h}$, along with three gates, the input gate $i_t \in \mathbb{R}^{d_h}$, the forget gate $f_t \in \mathbb{R}^{d_h}$ and the output gate $o_t \in \mathbb{R}^{d_h}$.

At each timestamp t , the LSTM filter is presented with the spatially-aggregated sentiment vector $x_g \in \mathbb{R}^d$ (a single row of X_g) together with the preceding hidden state h_{t-1} and cell state c_{t-1} . The input gate determines which components of the new input data x_g and the previous hidden state h_{t-1} are to be updated, whereas the forget gate decides which parts of the existing cell state c_{t-1} should be discarded.

This architecture ensures a dynamic flow of sentiment information through the temporal filter, selectively processing and transmitting only the relevant sentiment data

to subsequent time steps. The detailed mechanism works as follows:

$$i_t = \text{sigmoid}(x_g W_{xi} + h_{t-1} W_{hi} + w_{ci} \odot c_{t-1} + b_i) \quad (4.5)$$

$$f_t = \text{sigmoid}(x_g W_{xf} + h_{t-1} W_{hf} + w_{cf} \odot c_{t-1} + b_f) \quad (4.6)$$

$$c_t = i_t \odot \tanh(x_g W_{xc} + h_{t-1} W_{hc} + b_c) + f_t \odot c_{t-1} \quad (4.7)$$

$$o_t = \text{sigmoid}(x_g W_{xo} + h_{t-1} W_{ho} + w_{co} \odot c_{t-1} + b_o) \quad (4.8)$$

$$h_t = o_t \odot \tanh(c_t) \quad (4.9)$$

The functionality of the gates is achieved via the dot products $\odot$, where the weight matrices $W_x, W_h \in \mathbb{R}^{d_s \times d_h}$ transforms the input sentiment vector x_g and the previous hidden state h_{t-1} , and weight vector $w_c \in \mathbb{R}^{d_h}$ transforms the previous cell state c_{t-1} . After the update of cell state c_t , the output gate o_t specifies which part of information from the current cell state should be kept as output hidden state h_t , eq. (4.9), for the forward-pass of next timestamp.

In generating the sentiment score, $s_t \in \mathbb{R}^{d_x}$, employs a Softmax output head. This head processes the hidden state h_t by first applying a linear transformation and then a Softmax activation:

$$s_t = \text{softmax}(h_t W_s + b_s) \quad (4.10)$$

Here, the weight matrix $W_s \in \mathbb{R}^{d_h \times d_x}$ ensures that the dimensions of the transformed hidden state match those of the sentiment scores. The Softmax function guarantees that the sum of the inferred sentiment scores equals one.

For the hyper-parameter settings, we configure the temporal filter to look back 10 timestamps. This allows NIST-Cheb to incorporate up to half a month of historical sentiment data, which reflects the prolonged influence of past sentiments on current trends identified in previous research (Tetlock, 2007; Garcia, 2013; Uhl et al., 2015). The spatial feature aggregation is set to a dimensionality of $d_s = 16$, enabling a comprehensive sampling of neighboring sentiment information. To accommodate the additional temporal dimension as a new source of information, the temporal dimensionality of the hidden states is doubled to $d_h = 32$, providing the model with sufficient capacity to perform extensive temporal feature aggregation on top of the spatial aggregation.

4.3.3 The Masked Semi-Supervised Optimization Scheme

To transplant NIST-Cheb to use the incomplete sentiment data for model optimization, we propose a masked semi-supervised training approach inspired by Kipf and Welling

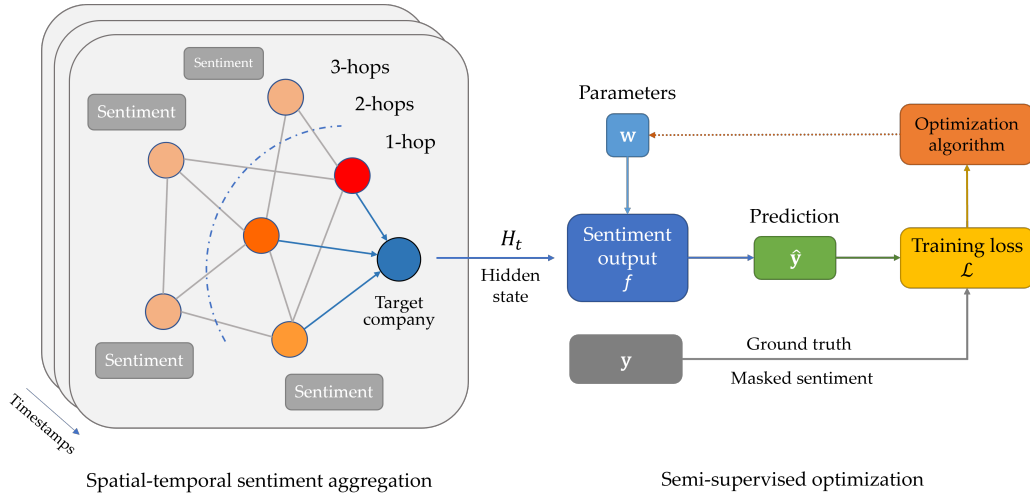


Figure 4.3: Workflow of the NIST-Cheb with semi-supervised optimization for non-existing news sentiment inference.

(2017). This strategy is particularly adept at utilizing the partially available sentiment scores, by strategically masking the known sentiment scores to simulate the absence sentiment data.

Specifically, for each company v on a given trading day t that has an associated sentiment score, we simulate the absence of data by masking these scores and task the model with inferring the sentiment for these artificially obscured instances. The discrepancy between the actual sentiment scores and the predictions made by the model is used to perform gradient descent and backpropagation, thereby fine-tuning the model's weights. In scenarios where a neighboring node lacks sentiment scores, we assign a zero vector as a placeholder, enabling the model to adapt to and learn from the presence of unlabeled neighbors.

This detailed process can be represented by the following loss function:

$$\mathcal{L} = \sum_{v \in \mathbf{y}_m} \frac{|\mathbf{s}_v - \tilde{\mathbf{s}}_v|}{N_m} \quad (4.11)$$

where $\mathbf{s}_v$ represents the actual sentiment score for company v ; $\mathbf{y}_m$ denotes the set of masked node indices with sentiment labels; and N_m counts the number of these masked samples. We employ the ℓ_1 norm in our loss function because it effectively captures the magnitude of discrepancies in sentiment scores, which are typically less than 1. Utilizing the ℓ_2 norm could minimize these discrepancies, potentially affecting gradient stability and the convergence of the Adam optimizer (Kingma and Ba, 2014). The NIST-Cheb is trained over 200 epochs with a learning rate of 0.01.

The workflow of the NIST-Cheb framework with the masked semi-supervised optimization is illustrated in Figure 4.3.

4.4 Evaluation Settings

To assess our sentiment inference method, we first report the error-based metrics following MVI conventions. We then implement a trading simulation based evaluation approach to directly link the quality of sentiment inference to measurable trading performance.

Our research period spans from January 2005, to December 2018. This period is divided into an in-sample phase from January 2008 to December 2015, and an out-of-sample phase from January 2016, to December 2018. The initial three years (2005 to 2007) serve as the bootstrap phase for the initial training of the sentiment inference model. Our evaluation of trading performance is conducted on a daily basis, with the focus on predicting the binary direction of the next day's closing price.

Our trading simulations are carried out using S&P 500 market data. The 24 features employed in our analysis include price data, technical indicators, and fundamental features. Building on the 13 features introduced in Chapter 3, we add 11 fundamental features, in line with previous studies that incorporate sentiment and fundamental information for stock market forecasting and analysis (Patatoukas, 2014; Li et al., 2015; Nti et al., 2020; Karavias et al., 2021). To ensure consistency with these studies, we focus on fundamental features that are positively correlated with future cash flows, thus providing valuable insights into share prices (Patatoukas, 2014; Karavias et al., 2021). These features include the price-to-earnings (P/E) ratio (peRatio) (Li et al., 2015), price-to book (P/B) ratio (pbRatio) (Li et al., 2015), Debt/Equity ratio (debtToEquity) (Renu and Christie, 2018), rate of return on equity (roe) (Patatoukas, 2014), etc (see Table 4.4 for the full list). Studies have shown that incorporating fundamental indicators enhances understanding of the factors driving stock price movements, offering deeper insights into market dynamics (Renu and Christie, 2018; Nti et al., 2020).

Given that fundamental features are quarterly updated from company reports, we use the preceding quarter's data for daily pattern padding. We exclude stocks missing full-length trading data and those not included in the SCRG, resulting in a total of 353 S&P 500 companies. The descriptive statistics of our employed features are detailed in Table 4.4.

	mean	std	min	25%	50%	75%	max
open	59.43	56.40	0.66	26.26	45.04	73.62	1174.00
high	60.06	56.94	0.68	26.61	45.56	74.34	1188.40
low	58.78	55.82	0.61	25.90	44.53	72.91	1150.60
close	59.52	56.50	0.66	26.28	45.11	73.68	1186.40
volume	7934190.77	33517308.14	7400.00	1353600.00	2755500.00	6085300.00	3372969600.00
vwap	57.23	55.50	0.65	24.85	42.93	70.57	1041.47
adx	12.86	5.15	3.30	9.08	11.81	15.55	46.54
ema	59.47	56.06	0.89	26.74	45.16	73.69	1182.14
rsi	51.89	6.34	19.77	47.59	51.96	56.24	82.35
sma	59.48	56.14	0.77	26.71	45.13	73.71	1167.09
standardDeviation	2.44	3.13	0.05	0.97	1.66	2.88	188.68
williams	-42.05	31.91	-100.00	-68.43	-37.87	-15.10	607.80
changePercent	0.00	0.02	-0.70	-0.01	0.00	0.01	1.02
revenuePerShare	10.65	16.18	-175.66	3.09	6.38	12.23	268.06
netIncomePerShare	0.69	4.23	-455.89	0.28	0.60	1.10	32.17
operatingCashFlowPerShare	1.43	2.94	-106.60	0.46	1.04	1.91	97.87
bookValuePerShare	22.72	26.71	-65.46	8.91	16.17	29.57	757.92
peRatio	21.36	522.21	-27596.18	10.55	16.20	23.65	47553.90
priceToSalesRatio	10.97	28.24	-528.76	3.89	7.04	12.81	2436.18
pbRatio	3.87	117.77	-6789.65	1.50	2.56	4.23	12266.18
debtToEquity	1.03	39.15	-880.47	0.26	0.56	1.07	4615.70
netDebtToEBITDA	7.66	110.46	-2653.88	0.46	5.14	12.69	8594.00
payoutRatio	0.59	26.00	-435.00	0.00	0.26	0.51	3041.62
roe	0.06	1.40	-17.68	0.02	0.04	0.06	140.34

Table 4.4: Descriptive statistics of the S&P 500 market dataset

4.4.1 Baseline sentiment inference methods

As we discussed in section 4.1 and 4.2, most of the machine learning MVI methods like k -NN are not applicable, as they depend on other non-missing attributes for imputation (Brás and Menezes, 2007; Li et al., 2014b). Only the temporal MVI methods can be used as benchmark models (Yi et al., 2016).

For a comprehensive comparison, we adopt four common temporal MVI methods as sentiment inference baselines, which are forward-filling (Ffill), mean, autoregressive integrated moving average (ARIMA), and LSTM (Yi et al., 2016; Kim and Chi, 2018; Che et al., 2018; Wang et al., 2023a).

In line with NIST-Cheb, the historical window for imputation is set to 10 trading days. The forward-filling approach fills absent sentiment scores with the most recent available data. The mean method uses the moving-average of the last 10 trading days' sentiment scores. ARIMA is a common time-series baseline in MVI tasks. We set the parameter of ARIMA to (10, 1, 0). Regarding LSTM, the hidden size is set to 16 in line with of NIST-Cheb's hidden dimension d_x , as LSTM only has a temporal dimension as the source of information.

4.4.2 Error-based evaluations

In our error-based evaluations, we employ Mean Absolute Error (MAE) and Mean Relative Error (MRE) as our performance metrics, following the methodologies described in Cao et al. (2018). These metrics are selected for their ability to quantitatively express the accuracy of sentiment inference in clear and interpretable terms.

MAE measures the average magnitude of the errors without considering their direction (i.e., it aggregates the absolute values of the individual differences between predicted and actual values). It is defined mathematically for our dataset as:

$$\text{MAE} = \frac{\sum_i \sum_p |s_i^p - \tilde{s}_i^p|}{3N_{\text{news}}}, \quad (4.12)$$

where i represents the i -th news piece and p denotes the sentiment polarity of either Positive, Neutral or Negative. s_i^p represents the ground-truth sentiment score and $\tilde{s}_i^p$ denotes the inferred sentiment score. N_{news} is the total number of news items analyzed.

MRE, on the other hand, provides a measure of error normalized to the magnitude of the actual values, making it a relative measure of error. This metric is particularly useful in contexts where we wish to understand the size of the error in relation to the typical values of the sentiment data. MRE is calculated as:

$$\text{MRE} = \frac{\sum_i \sum_p |s_i^p - \tilde{s}_i^p|}{\sum_i \sum_p |s_i^p|}. \quad (4.13)$$

Due to the lack of a designated test set within our sentiment dataset, we adhere to practices common in missing value imputation (MVI) research (Cao et al., 2018). Specifically, we simulate a testing scenario by selectively masking observed sentiment scores as if they were absent, and then substituting them with the values inferred by our models. This method allows us to internally validate the effectiveness of different sentiment reconstruction techniques under conditions that mimic real-world data absence scenarios.

4.4.3 Benchmark models for trading evaluations

We adopt five benchmark machine learning classifiers to evaluate the quality of reconstructed sentiment scores by different methods, with the consideration of covering major types of classifiers for robustness. These classifiers are Transformer (Vaswani et al., 2017), LSTM (Fischer and Krauss, 2018), multi-layer perceptron (MLP), random forest (RF) and support vector machine (SVM). The historical window for trading simulations is set to 30 trading days.

The Transformer, developed by Vaswani et al. (Vaswani et al., 2017), employs a self-attention mechanism that enables it to capture complex, long-range dependencies within data. This capability makes the Transformer particularly effective for analyzing financial market data, which involves non-linear and extended temporal relationships. For our experiments, we configure the Transformer with a hidden size of 12, utilizing 4 attention heads across 3 encoder layers. It is trained incrementally: starting with an initial training phase of 50 epochs at the outset of the out-of-sample period, followed by a prediction phase for the next 50 trading days. After each prediction phase, the model is updated with an additional 50-day training sample over 10 epochs.

The LSTM is a sequential machine learning model designed for sequence prediction tasks (Fischer and Krauss, 2018). It features a gating mechanism that regulates the flow of information, enabling the retention or dismissal of data based on its current relevance. This functionality allows the LSTM to excel in applications requiring the recall of long-term dependencies, such as predicting stock movements. We set the LSTM's hidden size to 12 and train it for 200 epochs, updating it at each prediction interval.

The Multi-layer Perceptron (MLP) serves as our benchmark neural network classifier. It is structured with two hidden layers sized at (27, 8) and (8, 4), and undergoes training for 200 epochs. The Random Forest (RF) classifier is a tree-based classifier consisting of numerous uncorrelated decision trees, each constructed from different subsets of the training data features. The Support Vector Machine (SVM) is another robust machine learning model frequently used in financial forecasting (Lee, 2009). We maintain the default settings for both RF and SVM as specified in scikit-learn (Pedregosa et al., 2011).

Please note that our objective is not to directly compare the performance of different trading classifiers against each other. Instead, we conduct performance comparisons within each classifier category to evaluate the quality of the inferred sentiment scores generated by various methods. To ensure fairness and eliminate potential biases favoring specific sentiment inference methods, hyperparameter tuning for the classifiers is performed based on validation performance without incorporating the inferred sentiment features. This neutral tuning process ensures that the evaluation reflects the inherent quality of the inferred sentiment scores, independent of the hyperparameter choices for the trading models themselves.

4.4.4 Trading evaluation measures

Regarding trading evaluations, we adopt a static portfolio based trading simulation scheme where the component stocks are the S&P 500 companies. We assign equal weights to all companies for simplicity as we are not aiming at finding the optimal portfolios. We adopt a long-short trading strategy. For a given day t , based on the current prediction result y_{t+1} , we buy or sell the target stock using 1 US dollar. At the day $t + 1$, we can calculate the profit and loss (PnL) we gain for that stock (Kim et al., 2020). We sum up the total profits and losses of the component stocks to get the profit and loss of the static portfolio. PnL is calculated by the following equation,

$$\text{PnL}_t = \frac{1}{n_t} \sum_{i=1}^{n_t} \text{sign}_t^i * r_t^i \quad (4.14)$$

where t is one timestamp in the out-of-sample period; i is one component of the daily dynamic portfolio; n_t is the total number of the dynamic components of the portfolio at day t ; sign_t^i is the model's prediction of stock i and r_t^i is the daily return of stock i at day t .

We also use Sharpe Ratio (SR) to measure of performance by adjusting risks (Fischer and Krauss, 2018). We use annualized Sharpe Ratio which is calculated by the following equation

$$\text{SR} = \frac{\overline{\text{PnL}}_p - r_f}{\sigma_{pf}} * \sqrt{252} \quad (4.15)$$

where $\overline{\text{PnL}}_p$ is the mean daily PnL of the portfolio; r_f is the risk-free rate which we set to 1-year US treasury rate (T-bill) to match our yearly based evaluation scheme; σ_{pf} is the standard deviation of the PnL series subtracted by r_f . Because the ultimate goal of the trading simulation is to test the effectiveness of the inferred sentiment instead of portfolio optimization nor trading strategy building, we set the transaction fee to zero for simplicity.

4.5 Results

In this section, we present the results of our experiments based on the evaluation criteria previously described. First, we assess the error-based performance of the 5 sentiment methods. We then evaluate the profitability of these methods with the 5 aforementioned stock movement prediction classifiers combinations. The consistency and robustness of the trading performance is also evaluated across three distinct market

	Mean	Ffill	ARIMA	LSTM	NIST-Cheb	Dummy
MAE	0.176	0.180	0.219	0.161	0.147	0.163
MRE	0.528	0.541	0.657	0.485	0.441	0.490

Table 4.5: Performance of error-based metrics of the sentiment inference models.

structures observed during 2016 to 2018. Lastly, we discuss the theoretical contribution to the current literature of spillover effect of sentiment information in the financial market.

4.5.1 Error-based Evaluation Performance

Table 4.5 presents the performance of various sentiment inference models based on the error metrics Mean Absolute Error (MAE) and Mean Relative Error (MRE). Among the models evaluated, our proposed NIST-Cheb achieves the lowest MAE and MRE scores, 0.147 and 0.441, respectively, highlighting its superior ability to reconstruct accurate sentiment scores for days without news coverage. This performance demonstrates the effectiveness of leveraging spatial and temporal relationships within NIST-Cheb compared to both traditional and modern imputation methods.

The Dummy model, which uses the average sentiment from the bootstrap phase as a baseline value, provides a relatively robust benchmark with an MAE of 0.163 and an MRE of 0.490. Notably, while simpler methods such as Mean Imputation and Forward Filling (Ffill) perform worse, only our model and the LSTM surpass the Dummy in both MAE and MRE, further validating their effectiveness. The LSTM model, which is trained individually for each company, achieves an MAE of 0.161 and an MRE of 0.485, demonstrating its ability to capture temporal dependencies but falling short of NIST-Cheb's performance.

Table 4.6 provides a deeper look into the distribution of absolute errors across the models, offering additional insight into how well the inferred sentiment aligns with actual sentiment when mocked as absent. NIST-Cheb achieves not only the lowest mean absolute error (MAE) with a standard deviation of 0.092 but also the lowest 25th percentile (0.077), median (50th percentile, 0.130), and 75th percentile (0.199) values. These results indicate that the majority of NIST-Cheb's error distribution is concentrated within a narrower range, showcasing its robustness and consistent accuracy compared to other models. Furthermore, Table 4.7 presents the statistical distribution of the inferred sentiment scores generated by NIST-Cheb. The comparison with

	Mean	Std	Min	25%	50%	75%	Max
ARIMA	0.219	0.074	0.000	0.167	0.223	0.274	0.514
Mean	0.176	0.079	0.000	0.114	0.170	0.233	0.486
Ffill	0.180	0.126	0.000	0.082	0.161	0.260	0.555
LSTM	0.158	0.102	0.000	0.078	0.137	0.218	0.563
NIST-Cheb	0.147	0.092	0.000	0.077	0.130	0.199	0.511
Dummy	0.163	0.088	0.000	0.102	0.138	0.222	0.394

Table 4.6: Statistical description of the mean absolute errors of the sentiment inference models.

	Mean	Std	Min	25%	50%	75%	Max
Negative	0.234	0.180	0.006	0.117	0.177	0.289	0.998
Neutral	0.391	0.194	0.002	0.232	0.365	0.547	0.987
Positive	0.375	0.220	0.0000	0.187	0.340	0.531	0.857

Table 4.7: Statistical description of NIST-Cheb inferred news sentiment scores.

the original sentiment score distribution in Table 4.1 reveals that the inferred scores closely resemble the original distribution, further validating NIST-Cheb’s capability for sentiment inference.

It is important to note that the error-based evaluation provides an indirect measure of the quality of inferred sentiment scores, as it relies on known sentiment scores being mocked as absent. To directly assess the utility of the inferred sentiment scores, we use them to fill media-silent days and apply them in trading simulations.

4.5.2 Trading Evaluation Performance

Table 4.8 displays the trading performance of 25 benchmark combinations in terms of profitability, highlighting a consistent trend where models integrated with NIST-Cheb inferred sentiment scores outperform their counterparts in Sharpe Ratio (SR) and Profit and Loss (PnL).

Overall, the NIST-Cheb method combined with the Transformer classifier demonstrates superior performance with the highest Sharpe Ratio of 1.61, and a Profit and Loss of 0.17, yielding 2.28 basis points (bps). The LSTM classifier also shows notable improvements when paired with NIST-Cheb, achieving an SR of 0.80 and a PnL of 0.30, which is the highest among the LSTM combinations. Additionally, the LSTM

Classifier	Method	Overall			2016			2017			2018		
		SR	PnL	PnL_bps	SR	PnL	PnL_bps	SR	PnL	PnL_bps	SR	PnL	PnL_bps
Transformer	ARIMA	0.90	0.12	1.64	0.98	0.05	1.95	2.74	0.05	2.09	-0.02	0.02	0.88
	Ffill	1.07	0.13	1.73	1.42	0.06	2.44	2.96	0.05	2.14	-0.27	0.02	0.61
	Mean	0.94	0.13	1.71	1.47	0.07	2.84	2.73	0.05	1.95	-0.50	0.01	0.33
	Raw	0.84	0.12	1.54	1.33	0.06	2.51	2.28	0.04	1.70	-0.45	0.01	0.42
	LSTM	1.23	0.14	1.91	1.60	0.07	2.75	2.08	0.04	1.70	0.39	0.03	1.28
	NIST-Cheb	1.61	0.17	2.28	1.91	0.08	3.13	3.27	0.06	2.30	0.53	0.04	1.40
LSTM	ARIMA	0.53	0.17	2.30	1.37	0.14	5.46	2.00	0.10	4.07	-0.87	-0.07	-2.61
	Ffill	0.53	0.22	2.92	1.20	0.15	5.98	2.06	0.14	5.44	-0.62	-0.07	-2.65
	Mean	0.44	0.18	2.39	1.48	0.18	7.21	1.84	0.12	4.68	-1.12	-0.12	-4.73
	Raw	0.34	0.15	1.97	0.95	0.10	4.15	2.27	0.14	5.63	-0.86	-0.10	-3.88
	LSTM	0.76	0.20	2.63	1.49	0.12	4.81	0.79	0.05	1.82	0.11	0.03	1.27
	NIST-Cheb	0.80	0.30	3.94	1.55	0.18	7.16	2.37	0.15	6.02	-0.43	-0.03	-1.36
MLP	ARIMA	0.54	0.08	1.08	2.09	0.05	2.03	1.15	0.03	1.15	-0.59	0.00	0.07
	Ffill	0.55	0.09	1.21	1.71	0.05	1.98	1.51	0.04	1.52	-0.44	0.00	0.14
	Mean	0.59	0.09	1.21	0.88	0.03	1.25	2.68	0.05	2.15	-0.46	0.01	0.22
	Raw	0.53	0.09	1.18	1.55	0.05	1.90	1.99	0.05	1.82	-0.65	0.00	-0.18
	LSTM	0.68	0.10	1.32	1.37	0.04	1.61	2.27	0.05	1.97	-0.34	0.01	0.37
	NIST-Cheb	0.95	0.13	1.68	1.86	0.06	2.19	2.75	0.06	2.22	-0.17	0.02	0.62
RF	ARIMA	0.42	0.07	0.99	1.63	0.05	1.89	0.69	0.02	0.85	-0.46	0.01	0.23
	Ffill	0.24	0.07	0.91	0.63	0.03	1.31	0.29	0.02	0.71	-0.11	0.02	0.70
	Mean	0.32	0.08	1.02	0.97	0.05	1.86	0.11	0.01	0.56	-0.14	0.02	0.65
	Raw	0.37	0.08	1.10	0.80	0.04	1.68	0.48	0.02	0.87	-0.08	0.02	0.77
	LSTM	0.56	0.08	1.11	2.10	0.05	2.15	1.31	0.03	1.24	-0.67	0.00	-0.05
	NIST-Cheb	0.63	0.10	1.27	1.24	0.04	1.52	2.19	0.05	1.87	-0.31	0.01	0.41
SVM	ARIMA	0.59	0.11	1.49	1.79	0.06	2.53	1.93	0.05	2.08	-0.45	0.00	-0.13
	Ffill	0.82	0.14	1.90	2.00	0.07	2.94	2.08	0.06	2.36	-0.21	0.01	0.41
	Mean	0.55	0.11	1.44	1.17	0.05	1.86	2.17	0.06	2.35	-0.35	0.00	0.10
	Raw	0.79	0.14	1.83	1.50	0.06	2.29	2.42	0.07	2.73	-0.19	0.01	0.47
	LSTM	0.82	0.14	1.90	2.02	0.08	2.99	2.08	0.06	2.29	-0.20	0.01	0.43
	NIST-Cheb	0.98	0.16	2.17	2.12	0.08	3.06	2.56	0.07	2.77	-0.10	0.02	0.67
Market	-	0.41	0.22	2.91	0.62	0.10	4.00	2.37	0.18	7.12	-0.47	-0.06	-2.37

Table 4.8: Trading performance of all benchmark combinations across different time-periods in terms of profitability and predictability.

classifier incorporating our inferred sentiment data uniquely outperforms market profitability, with a daily PnL increase of 1.02 basis points, indicating an excess return over the market. In comparing to the raw sentiment scores, the PnL achieved is also doubled. Regarding the classical machine learning classifiers of MLP, RF and SVM, this pattern persists. Models paired with NIST-Cheb achieves the best Sharpe Ratio and PnL throughout the out-of-sample period.

This overall pattern underscores our inference method's capability to accurately reconstruct the absent news sentiment, which in turn, enhances the performance of

assorted classifiers, yielding superior risk-adjusted returns when compared to those using the baseline sentiment.

To ensure the robustness of our inference approach, we conduct a more granular analysis of the evaluation period’s sub-years, with distinct market dynamics in 2016, 2017, and 2018. The market in 2016 and 2017 concluded with bullish trends, showcasing growths of 11% and 19%, respectively, in contrast to 2018’s bearish trend with a 7% decline.

In the year-specific analysis, the robustness of NIST-Cheb is evident. We observe that classifiers using our inferred sentiment generally maintained superior performance over those with baseline sentiment inferences. During the bullish years of 2016, three out of five classifiers realize their highest risk-adjusted returns utilizing our inferred sentiment. In the bearish climate of 2018, this observation also stands for three out of five classifiers. In 2017, all classifiers achieve their best profitability using NIST-Cheb sentiment scores.

Conversely, traditional sentiment inference methods like ARIMA, Ffill, and Mean generally posted lower performance metrics across the board. For instance, using ARIMA with the Transformer in 2018 resulted in a negative SR of -0.02 and a PnL of 0.02, only garnering 0.88 bps. Same conclusion hold for Ffill and Mean. This pattern suggests that while traditional methods have some ability to infer the absent sentiment, they are significantly outperformed by the advanced processing capabilities of NIST-Cheb.

4.5.3 Ablation study

To understand the driving factors behind the performance of our NIST-Cheb model, similar to Section 3.5.5, we conduct a model component ablation study to analyze its sources of effectiveness. Specifically, we benchmark three model components. The first is a spatial-only model that uses the Chebyshev graph neural network (Cheb) as defined in equation 4.2. The second is an LSTM model that processes only the temporal sentiment series of each individual company. The third is the GCRN model (GCRN-Cheb) proposed by Seo et al. (2018), which is a ST-GNN that integrates a Chebyshev spatial filter with an LSTM for temporal processing. However, unlike NIST-Cheb, GCRN-Cheb employs shared LSTM weights across all companies, limiting its ability to address temporal non-IIDness among companies.

Table 4.9 presents the error-based results of the ablation models, while Table 4.10

	LSTM	Cheb	GCRN-Cheb	NIST-Cheb
MAE	0.161	0.154	0.152	0.147
MRE	0.485	0.462	0.455	0.441

Table 4.9: Performance of error-based metrics of the ablation models.

shows the trading performance of five classifiers using sentiment scores inferred by these models. Together, these results reveal the incremental improvements brought by each component of our NIST-Cheb model, highlighting the driving forces behind its superior performance.

Starting with the LSTM model, we see a slight improvement over the Dummy baseline, with an MAE of 0.161 and an MRE of 0.485. This indicates that temporal modeling of individual company sentiment series provides some benefit in capturing trends, although the improvement is modest. This trend extends to trading performance, where LSTM-driven classifiers achieve moderate Sharpe Ratios (SR) and Profit and Loss (PnL), with the Transformer classifier yielding the best performance among LSTM-driven setups (SR = 1.23, PnL = 0.14).

The Chebyshev (Cheb) model, which applies a spatial-only approach, yields a more substantial improvement over both the Dummy and LSTM models, with an MAE of 0.154 and an MRE of 0.462. This stronger performance underscores the importance of spatial relationships, indicating that sentiment from neighboring companies plays a critical role in accurately inferring sentiment during media-silent periods. The trading results further support this observation, as Cheb-driven classifiers outperform their LSTM-driven counterparts.

The GCRN-Cheb model, which combines the Chebyshev spatial filter with a shared LSTM for temporal processing, further enhances performance with an MAE of 0.152 and an MRE of 0.455. This improvement suggests that incorporating temporal information via the shared LSTM weights is beneficial, allowing for more nuanced sentiment inference across time. However, because the LSTM weights are shared across companies, GCRN-Cheb is limited in its ability to address temporal non-IIDness among companies, which restricts its overall effectiveness. In trading simulations, GCRN-Cheb-driven classifiers show notable improvements, with the Transformer classifier achieving the highest performance among these setups (SR = 1.44, PnL = 0.18).

Finally, our NIST-Cheb model, which integrates the Chebyshev filter with individually assigned LSTM weights for each company, achieves the lowest MAE of 0.147 and

Classifier	Method	SR	PnL	PnL_bps
Transformer	LSTM	1.23	0.14	1.91
	Cheb	1.35	0.15	1.97
	GCRN-Cheb	1.44	0.18	2.39
	NIST-Cheb	1.61	0.17	2.28
LSTM	LSTM	0.76	0.20	2.63
	Cheb	0.77	0.11	1.46
	GCRN-Cheb	0.79	0.14	1.83
	NIST-Cheb	0.80	0.30	3.94
MLP	LSTM	0.68	0.10	1.32
	Cheb	0.72	0.11	1.40
	GCRN-Cheb	0.82	0.11	1.51
	NIST-Cheb	0.95	0.13	1.68
RF	LSTM	0.56	0.08	1.11
	Cheb	0.54	0.09	1.18
	GCRN-Cheb	0.59	0.09	1.21
	NIST-Cheb	0.63	0.10	1.27
SVM	LSTM	0.82	0.14	1.90
	Cheb	0.86	0.12	1.53
	GCRN-Cheb	0.94	0.13	1.68
	NIST-Cheb	0.98	0.16	2.17

Table 4.10: Trading performance of ablation study.

MRE of 0.441 and highest SR of 1.61 among all the model combinations. This superior performance demonstrates the combined value of capturing both spatial and temporal relationships while addressing temporal non-IIDness among companies. By assigning unique LSTM weights, NIST-Cheb effectively accommodates company-specific temporal patterns, enabling more accurate sentiment inference.

4.5.4 Discussions and implications

Our empirical evidence demonstrates that the proposed NIST-Cheb model significantly enhances sentiment inference accuracy, achieving lower MAE and MRE scores. This improvement is attributed to NIST-Cheb’s ability to incorporate sentiment information from neighboring companies through their interconnections, surpassing traditional

temporal-based MVI methods that rely solely on a company's historical data. The component ablation study further validates the importance of spatial sentiment signals, as the spatial-only model (Cheb) outperforms the LSTM model by effectively utilizing sentiment data from related companies.

Our findings contribute to the study of *information spillover effect* in the context of how sentiment information about one company can inform or affect the others. Prior studies in this area often explore 1) how information from one entity can affect the valuation and operation behavior related to other entities, and 2) the types of involved information for spillover. The reviewed financial entities include stock markets (Lang and Stulz, 1992; Kim, 2003; Tsai, 2014; Singh et al., 2010; Long et al., 2021; Laughlin et al., 2014), industry clusters (Engelberg et al., 2018; Guiso and Schivardi, 2007; Audretsch et al., 2019), and related companies (Lang and Stulz, 1992; Wang et al., 2021a; Chan and Hameed, 2006). Types of information for spillover include public announcements (Kim, 2003; Lang and Stulz, 1992), fear and risks (Tsai, 2014; Long et al., 2021; Wang et al., 2021a), as well as price and volatility (Singh et al., 2010; Engelberg et al., 2018; Wang et al., 2021a).

Our study enriches previous literature by confirming the existence of the news sentiment as a novel type of information in spillover effect. Our study demonstrated that news sentiment data from companies with a high degree of inter-connections can be instrumental in inferring the latent sentiment towards firms that are less exposed in news media. Our empirical results reveal that this inferred news sentiment is predictive of asset price movements, thereby underscoring the practical value of information spillover. This finding has profound implications for asset management and trading strategies, suggesting that investor sentiment, even from external sources, should be factored into predictive models to enhance their accuracy and reliability.

Our empirical trading simulations showed that stock movement classifiers equipped with our inferred sentiment can boost their performance in terms of profitability. This improvement underscores the practical value of our NIST-Cheb model in generating high-quality sentiment predictions that effectively complement traditional stock features. By offering a reliable mechanism to fill sentiment gaps on media-silent days, our model addresses a common limitation in real-world trading scenarios, where incomplete information often hinders decision-making. Consequently, the integration of NIST-Cheb into stock movement prediction frameworks not only enhances the predictive capabilities of trading systems but also demonstrates its potential to generate tangible financial benefits for market practitioners.

4.6 Conclusion

In this chapter, we proposed a spatial-temporal graph neural network based absent news sentiment inference framework to assess investor sentiment on media silent days. Our method comprises the Semantic Company Relational Graph measuring the company relationships via a relational vector embedding scheme, and the Non-IID Chebyshev Network propagating both the spatial sentiment information from the neighboring companies and the domestic historical sentiment features. Experiments showed that our proposed method outperforms the benchmarks in terms of quality of sentiment inference as it can effectively contribute to lower error measures and asset pricing with better risk adjusted returns.

Chapter 5

Hierarchical Graph Pooling for Intraday Market Index Prediction

The field of market index prediction has traditionally focused on methodologies ranging from classic machine learning to deep learning techniques. These methods typically emphasize the temporal characteristics of market index, thereby overlooking the rich insights available from market microstructures of constituent industries and sectors. Recently, the emergence of hierarchical graph pooling techniques offers a new perspective to address this research gap.

In this vein, we introduce FinPool, an innovative graph pooling operator designed for the hierarchical feature representation of stock markets. To harness the full potential of FinPool in index prediction, we propose two innovative prediction frameworks: Stacked FinPool and Multi-tier Attention FinPool. These frameworks leverage the Semantic Company Relationship Graph (SCRG) as the foundational knowledge of companies and their business connections.

Our empirical analysis reveals significant enhancements in trading profits and risk-adjusted returns, with gains reaching up to 20.04 basis points over the market. Such achievement signifies a substantial advancement compared to existing benchmark models. By focusing exclusively on the data from constituent stocks without domestic index features, our experiment results underscore the value of leveraging lagged intraday constituent information for index prediction. This finding not only challenges the Efficient Market Hypothesis but also demonstrates the untapped predictive power inherent in the microstructural details of market constituents.

* This work has been accepted and will be presented on the Economics of Financial Technology Conference 2024.

5.1 Introduction

The research field of financial market index prediction, with a rich history spanning several decades, has predominantly focused on methodologies ranging from traditional machine learning methods to recent deep learning techniques, due to their ability to model complex temporal relationships between financial variables (Hájek, 2011). Among all the traditional machine learning methods, the multi-layer perceptron (MLP) and support vector machine (SVM) are the ones that were mostly utilized (Moghaddam et al., 2016; Zhong and Enke, 2017; Hoseinzade and Haratizadeh, 2019; Kumbure et al., 2022). For instance, Moghaddam et al. (2016) adopted a MLP based regression model to predict the daily movements of NASDAQ index, achieving good mean square error (MSE) performance. Ni et al. (2011) presented a hybrid approach combining fractal feature selection with a support vector machine for predicting daily stock price index trends, demonstrating its superiority in both feature minimization and average prediction accuracy over five other common feature selection methods.

Regarding the deep learning methods, Bhandari et al. (2022) proposed to explore Long Short-Term Memory (LSTM) on the task of next-day closing price prediction of the S&P 500 index with the combination of fundamental market data, macroeconomic data, and technical indicators to capture the behavior of the stock market. Their experiment results showed that single layer LSTM model provides a superior fit and high prediction accuracy compared to multi-layer LSTM models. Moreover, Hoseinzade and Haratizadeh (2019) introduced a Convolutional Neural Network (CNN) based market index prediction framework, CNNpred, for feature extraction from diverse financial data sources, including multiple markets, to predict market trends, with a specific focus on the indices of S&P 500, NASDAQ, DJI, NYSE, and RUSSELL. The results demonstrate a notable enhancement in prediction accuracy over current leading baseline algorithms. Wang et al. (2022) adopted a Transformer model (Vaswani et al., 2017) to implement daily index prediction task on the worldwide stock market indices, including CSI 300, S&P 500, Hang Seng Index, and Nikkei 225. Their experiments showed that Transformer outperforms classic methods like LSTM and CNN, and can gain excess returns for investors. Similarly, Tao et al. (2024) modified the Transformer architecture with period-correlation mechanism and series decomposition layers for better daily predictions on Shanghai Composite Index, Shenzhen Component Index and Hang Seng Index, from 1991 to 2022. Their experiment results confirmed an improved mean errors compared to classical LSTM and CNN methods.

Though these previous studies have showed some extent of performance significance in market index perdition, still, there are several research challenges remain unsolved.

First, prior studies' focus on temporal characteristics of market indices restricts their exploration of rich insights available from the constituent industrial clusters. The financial market is characterized by inherent hierarchical structures (Mantegna, 1999), with companies organized into various levels of industry clusters based on their primary business operations. For example, the Global Industry Classification Standard (GICS)¹ offers a detailed four-tiered hierarchical industry classification, encompassing 11 sectors, 24 industry groups, 69 industries, and 158 sub-industries as of 2019 (Bhojraj et al., 2003; Schumaker and Chen, 2009).

Given market indices, like the S&P 500, are calculated as market cap weighted averages of constituent stock prices, the sole use of index temporal features would therefore overlook the nuanced market microstructures of sector-specific dynamics. While market microstructure is essential for understanding price discovery and liquidity (Bhojraj et al., 2003; Schumaker and Chen, 2009), this loss of critical information limits the predictive accuracy of these traditional methods and omits the valuable context provided by the hierarchical organization of market sectors and industries.

Second, a significant number of previous studies focus on the daily interval of index predictions (Cervelló-Royo et al., 2015), thus missing the advantages offered by more information-intensive intraday frequencies (Andersen and Bollerslev, 1997; Bollen and Inder, 2002; Dungey et al., 2022). In fact, prior studies have demonstrated that the extra information from intraday high-frequencies can contribute to better predictability and profitability in tradings (Shang, 2017; Batten et al., 2018; Karpman et al., 2023). Schulmeister (2009) examined technical trading systems in S&P 500 using daily and half-hourly data from 1960 to 2007. They found that when based on daily data, profitability declined since 1960 and was absent since the early 1990s. However, when based on 30-min (half-hourly) data, the rules were profitable with no decline over time.

Specially, previous research has identified three types of additional information contained in high-frequency intraday data. Dungey et al. (2022) highlighted that high-frequency data provides more insights into the data-generating processes during crisis periods. Effectively capturing this information enables models to better understand market microstructure changes during special periods.

¹<https://www.msci.com/our-solutions/indexes/gics>

Additionally, Sun and Ibikunle (2017) pointed out that high-frequency data is essential for capturing the immediate effects of dynamics of block trades, which might be missed with lower frequency data. They found that the price impact is stronger during the first hour of trading, suggesting that information accumulated overnight is quickly incorporated into prices once trading begins.

Furthermore, high-frequency intraday data is crucial for detecting sudden price changes, which are not discernible with lower frequency data (Kirilenko et al., 2017; Kong et al., 2021). Using intraday data allows us to capture these instant movements before significant stock price jumps, making it essential for tracking and predicting dynamic changes that occur within short time intervals.

Contribution. By exploiting the information intensive high-frequency market data, this study innovatively frames intraday market index prediction as a graph classification task to address the identified research challenges. Our contributions are as follows:

- We introduce FinPool, a graph pooling operator tailored for hierarchical feature integration of financial market sectors. We leverage hierarchical graph pooling techniques to extract insights from constituent industrial clusters, aiming to achieve a deeper understanding of market dynamics in the intraday frequency. FinPool is adapted from the DiffPool method (Ying et al., 2018b), integrating domain-specific architectural modifications. By replacing the original GraphSAGE filter (Hamilton et al., 2017) with a two-layer GCN filter (Kipf and Welling, 2017), we address the problem of redundant node sampling to more effectively manage the diverse counts of neighboring companies, which are critical due to the distinct cross-relationship patterns among companies. Furthermore, to mitigate information loss typically encountered during feature concatenation (Li et al., 2019b), we implement skip-connections across layers (He et al., 2016). These targeted adaptations empower FinPool with enhanced ability to more effectively process and integrate hierarchical features of the financial market.
- To optimally apply FinPool for index movement forecasting, we develop two prediction frameworks working independently: Stacked FinPool and Multi-tier Attention FinPool. Stacked FinPool straightforwardly stacks FinPool operators in a sequential manner across the market tiers, while Multi-tier Attention FinPool introduces a multi-tier attention mechanism to assign different weights to the levels of market hierarchies. These configurations enable FinPool to progressively

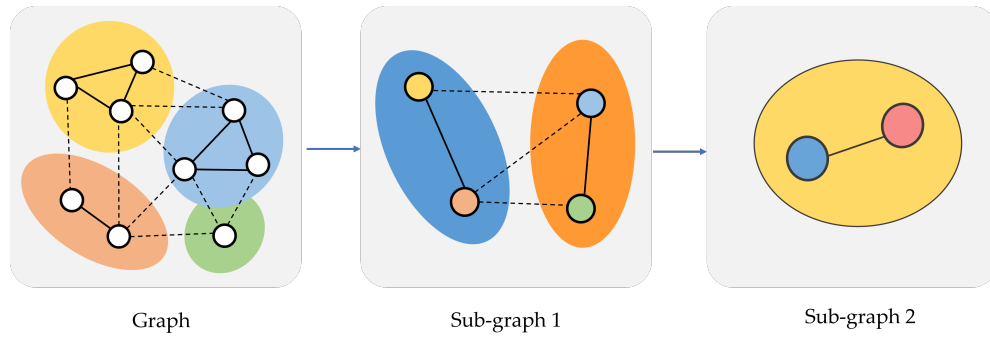


Figure 5.1: Illustration of graph pooling.

capture market microstructures and sector-specific dynamics, revealing granular insights into the rich information in the intraday frequency that significantly diverges from those observed at daily intervals (Andersen and Bollerslev, 1997; Schulmeister, 2009; Batten et al., 2018; Dungey et al., 2022).

5.2 Methodology

We detail our methodological designs in this section. First, we describe the architecture of DiffPool, which forms the foundation of our FinPool operator. We give an in-depth analysis of why this architectural foundation fits the research context of market index prediction. Additionally, we detail our FinPool operator by introducing how and why we implement the domain-specific modifications on the DiffPool by taking the financial contextual knowledge into consideration. Furthermore, we introduce two novel hierarchical feature aggregation frameworks that effectively deploy FinPool for index predictions: Stacked FinPool and Multi-tier Attention FinPool.

5.2.1 The DiffPool foundation

Graph pooling is a technique to reduce the size and complexity of a graph while preserving its essential structural and feature information, enabling hierarchical representation learning in graph neural networks. As depicted in Figure 5.1, the graph is iteratively coarsened into smaller sub-graphs, with hierarchical node representations generated at each stage of the pooling process.

There are two types of hierarchical graph pooling methods, namely the node clustering pooling and node dropping pooling. Node clustering pooling, as proposed by Ying et al. (2018b), clusters nodes within the same community into new, singular nodes

to form coarsened sub-graphs. Conversely, node dropping pooling, described by Lee et al. (2019), involves selectively retaining a subset of nodes from the original, larger graph while discarding the rest, thereby creating scaled-down sub-graphs.

In this research, the node clustering pooling method, specifically the DiffPool technique as proposed by Ying et al. (2018b), has been chosen as the foundation of our FinPool. This selection is grounded in the inherent nature of companies that can be clustered into different industrial communities based on their operating and marketing activities. For instance, the widely adopted Global Industry Classification Standard (GICS) (Cavaglia et al., 2001; Conover et al., 2008; Schumaker and Chen, 2009; Seong and Nam, 2021) organizes companies into a four-tier hierarchy: 11 sectors, 24 industry groups, 69 industries, and 158 sub-industries, as in 2019.

By leveraging the hierarchical structures within companies, DiffPool can gradually gain a comprehensive understanding of the fine-grained market microstructures of different industries and sectors by integrating insights from component stocks.

The core of DiffPool lies in its four matrices and two graph neural networks. The four matrices are the adjacency matrix $A^{(l)} \in \mathbb{R}^{n \times n}$, the node feature matrix $X^{(l)} \in \mathbb{R}^{n \times d}$, the node embedding matrix $Z^{(l)} \in \mathbb{R}^{n \times 2d}$, and the cluster assignment matrix, $S^{(l)} \in \mathbb{R}^{n \times m}$. The two graph neural networks are *embedding GNN* - $\text{GNN}_{emb}^{(l)}$, and *pooling GNN* - $\text{GNN}_{pool}^{(l)}$, responsible to generate the four matrices.

At each pooling level l (a tier of the industry hierarchy), the adjacency matrix $A^{(l)}$ denotes the connections between the n nodes, which can represent either individual companies or industry clusters at level l . The initial adjacency matrix $A^{(0)}$ starts with the adjacency matrix of SCRG and evolves at each pooling level l to reflect the changing graph structure derived from graph pooling. The node feature matrix $X^{(l)}$ represents the the d -dimensional company (cluster) features at level l , and begins with the input features of the S&P 500 companies.

The node embedding matrix $Z^{(l)}$ denotes the updated node features after spatial message propagation (Gilmer et al., 2017), via the *embedding GNN*, denoted as $\text{GNN}_{emb}^{(l)}$:

$$Z^{(l)} = \text{GNN}_{emb}^{(l)}(A^{(l)}, X^{(l)}). \quad (5.1)$$

The cluster assignment matrix $S^{(l)}$ at each level is the critical step for grouping the n nodes into m clusters for the subsequent level $l + 1$. $S^{(l)}$ is obtained through the *pooling GNN*, $\text{GNN}_{pool}^{(l)}$, as follows:

$$S^{(l)} = \text{Softmax} \left(\text{GNN}_{pool}^{(l)}(A^{(l)}, X^{(l)}) \right). \quad (5.2)$$

Both $\text{GNN}_{emb}^{(l)}$ and $\text{GNN}_{pool}^{(l)}$ take the same input of adjacency matrix $A^{(l)}$ and feature matrix $X^{(l)}$. Technical details of both GNNs will be introduced in section 5.2.2.

The Softmax in equation (5.2) enables DiffPool to perform soft clustering that proportionally assigns each company (or cluster) at level l into all clusters at level $l + 1$. Such mechanism is particularly vital in the financial context, as S&P 500 companies are the largest ones with highest market caps. These companies typically own diverse business scopes, often referred to as conglomerates (Edwards, 1955; Maksimovic and Phillips, 2002). For example, Tesla Inc. is a company that operates in multiple industries - not only as an automotive manufacturer specializing in electric vehicles but also as a clean energy company producing solar panels and batteries. Therefore, it is overly simplistic to assign a conglomerate like Tesla to a single, fixed industry category. Instead, the more nuanced soft assignments of companies to various industries is more suitable to meet the requirements. This is where the $S^{(l)}$ plays a crucial role, providing a percentage-based assignment to different industries. The softmax function is applied row-wise, ensuring that the sum of assignment weights for each company to the next-level industry clusters sums up to 1.

With the cluster assignment matrix $S^{(l)}$, the computation of the next-level adjacency matrix $A^{(l+1)}$ and the node feature matrix $X^{(l+1)}$ is performed using the following equations:

$$X^{(l+1)} = S^{(l)T} Z^{(l)} \in \mathbb{R}^{m \times d}, \quad (5.3)$$

$$A^{(l+1)} = S^{(l)T} A^{(l)} S^{(l)} \in \mathbb{R}^{m \times m}. \quad (5.4)$$

In equation (5.3), the node embeddings $Z^{(l)}$ from the previous layer are aggregated in accordance with the cluster assignments from $S^{(l)}$, resulting in the new feature matrix $X^{(l+1)}$ for the m nodes at the next level. Meanwhile, equation (5.4) transforms the adjacency matrix $A^{(l)}$ into a coarsened adjacency matrix $A^{(l+1)}$, which represents the connections between each pair of the new m nodes. Given the soft-clustering approach utilized, $A^{(l+1)}$ emerges as a fully connected graph where the sum of the edge weights for each node equates to 1. This approach ensures that the relationships between nodes in the coarsened graph are accurately represented, taking into account the probabilistic nature of the soft cluster assignments.

5.2.2 The FinPool operator

In the foundational design of DiffPool, as outlined by Ying et al. (2018b), both the *embedding GNN* and the *pooling GNN* employ a same GraphSAGE GNN architecture

(Hamilton et al., 2017), with different parameterizations as they play separate roles. The *embedding GNN* is responsible for generating new embeddings for the company (or cluster) nodes at the same layer. The *pooling GNN* is tasked with generating a probabilistic assignment of the input nodes to clusters at the subsequent layer.

Specially, GraphSAGE employs a fixed-number sampling scheme which works as follows:

$$\mathbf{h}_v^{(k)} = \sigma \left(\mathbf{W}^{(k)} \cdot f_k \left(\mathbf{h}_v^{(k-1)}, \left\{ \mathbf{h}_u^{(k-1)}, \forall u \in S_{\mathcal{N}(v)} \right\} \right) \right) \quad (5.5)$$

where the hidden state $\mathbf{h}_v^{(k)}$ of the given node v is based on the feature aggregation of a fixed-size sampled subset $S_{\mathcal{N}(v)}$ of the v 's neighbor set $\mathcal{N}(v)$. f_k is an aggregation function with multiple choices, such as mean, pooling, and LSTM aggregators (Hamilton et al., 2017). The exact number of the fixed-size sampling is a hyper-parameter of GraphSAGE.

Although the fixed-sampling of neighbors has been successfully applied in generic domains, such as those used in protein structures and citation networks (Hamilton et al., 2017), this strategy presents certain concerns in the market index prediction context due to the following reasons:

Pre-existing node sampling in SCRG. The Semantic Company Relationship Graph (SCRG) inherently incorporates a threshold-based pre-sampling of connections between company nodes, as elaborated in chapter 2. Introducing an additional layer of node sampling, as necessitated by GraphSAGE, risks the issue of oversampling. This could lead to a significant loss of vital information, compromising the structural integrity of the graph in our analytical framework.

Wide variability in company neighbor counts. The diversity in market scopes and business types among companies results in a broad spectrum of neighbor counts. Consequently, determining a universally suitable fixed number of neighbors for GraphSAGE's sampling process poses a considerable challenge. This fixed sampling approach could inadvertently introduce sampling bias, as it cannot equally represent the diversity of relationships and structures across different companies.

Limitations in unweighted feature aggregation. GraphSAGE's methodology predominantly caters to unweighted graphs, assuming equal importance across all sampled neighboring nodes during feature aggregation, as shown in equation (5.5). However, this assumption is problematic in the financial domain, where the intensity of company relationships varies: some connections are marked by dense interactions, while others are more tenuous. An indiscriminate weighting of features overlooks

the inherent disparity in company's relational significance, failing to skew the model's sensitivity to the diverse influence levels among neighbors.

Given these considerations, the Graph Convolutional Network (GCN) as proposed by Kipf and Welling (2017), emerges as a fitting choice for our study. It facilitates consistent and unbiased feature aggregation across diverse company nodes:

$$H^{(l)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A}^{(l)} \tilde{D}^{-\frac{1}{2}} X^{(l)} W \right), \quad (5.6)$$

where $\tilde{A}^{(l)}$ represents the adjacency matrix at layer l with self-loops to incorporate each node's own features. $\tilde{D} \in \mathbb{R}^{n \times n}$ is the diagonal degree matrix corresponding to $\tilde{A}^{(l)}$, with diagonal entries $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}^{(l)}$. $X^{(l)}$ denotes feature matrix of nodes at pooling level l . Unlike the equal weighting in GraphSAGE, the symmetric normalization operator, $\tilde{D}^{-\frac{1}{2}} \tilde{A}^{(l)} \tilde{D}^{-\frac{1}{2}}$, differentially weights the contribution of neighboring companies' features based on their connection strengths, thus placing more emphasis on features from closely connected companies during aggregation. In this way, the three limitations of GraphSAGE have been properly tackled.

FinPool's other enhancements over DiffPool comes with the architectural designs. In FinPool, we utilize a dual-layered GCN filter and incorporate the Gaussian Error Linear Unit (GELU) activation function (Hendrycks and Gimpel, 2016), replacing the traditional ReLU activation used in the original GCN (Kipf and Welling, 2017). The dual-layer structure enables the effective sampling and aggregation of features from two-hop neighboring companies (Kipf and Welling, 2017), which is shown to be the best-performing configuration on many real-world graphs (Li et al., 2019b; Chen et al., 2020). The GELU activation function is particularly advantageous compared to ReLU. While ReLU completely blocks negative gradients, GELU allows the passage of some negative information, thereby fostering the development of richer feature representations. This attribute of GELU has been empirically proven to enhance performance across a variety of tasks (Kenton and Toutanova, 2019; Lewis et al., 2020; Tolstikhin et al., 2021; Radford et al., 2023). Moreover, we integrate batch normalization following each GCN layer after GELU activation, as recommended by Ioffe and Szegedy (2015). This step helps to maintain a stable distribution of activations throughout the GCN filters, contributing to a faster convergence and prevents overfitting during the training process (Ioffe and Szegedy, 2015).

The overall architecture of FinPool is defined by the following sequence of opera-

tions:

$$H_1^{(l)} = \text{GCN}_1(A^{(l)}, X^{(l)}), \quad (5.7)$$

$$Z_1^{(l)} = \text{BatchNorm}\left(\text{GELU}(H_1^{(l)})\right), \quad (5.8)$$

$$H_2^{(l)} = \text{GCN}_2(A^{(l)}, Z_1^{(l)}), \quad (5.9)$$

$$Z_2^{(l)} = \text{BatchNorm}\left(\text{GELU}(H_2^{(l)})\right), \quad (5.10)$$

$$Z^{(l)} = \text{Concat}(Z_1^{(l)}, Z_2^{(l)}), \quad (5.11)$$

$$S^{(l)} = \text{Softmax}(Z^{(l)} W_s). \quad (5.12)$$

where equation (5.11) and (5.12) are the output head of FinPool's *embedding GNN* and *pooling GNN*, respectively. For equation (5.11), in correspondence to DiffPool's equation (5.1), FinPool integrates a critical enhancement: a skip-connection that combines the feature matrices from two hops, $Z_1^{(l)}$ and $Z_2^{(l)}$. This skip-connection, inspired from He et al. (2016) and Hamilton et al. (2017), is designed to mitigate potential information loss, an issue that might arise when exclusively relying on $Z_2^{(l)}$. Furthermore, the pooling output head (equation (5.12)), executes a soft-clustering approach on the concatenated features, $Z^{(l)}$, allowing for a nuanced inclusion of clustering information from two-hop neighbors. This contrasts with DiffPool's method (equation (5.2)), offering a refined mechanism that incorporates broader contextual data for more accurate node clustering.

5.2.3 FinPool index prediction frameworks

Please note that the FinPool operator is an intermediate module that groups the n companies (clusters) at level l into m clusters with feature aggregation. To make index movement predictions, a framework that deploys FinPool in a prediction pipeline is needed. Inspired by Mikolov et al. (2013b) and Seo et al. (2018), this study proposes two hierarchical aggregation frameworks to optimally apply the FinPool operators in pipelines for effective market index predictions. These frameworks, namely Stacked FinPool and Multi-tier Attention FinPool, are specifically designed to function in conjunction with the SCRG.

Targeting the S&P 500 constituents, our study focuses on the top three tiers of the Global Industry Classification Standard (GICS) hierarchy: the 11 sectors, 24 industry groups, and 69 industries as of 2019. The fourth tier, which consists of 158 sub-industries, is omitted from our analysis, as the sub-industry represents a level

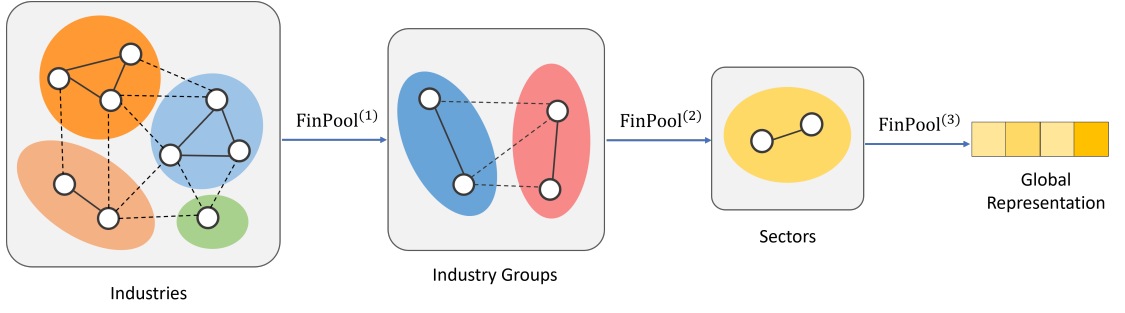


Figure 5.2: Pipeline visualization of the Stacked FinPool framework.

of granularity that is excessively fine for effectively analyzing those 500 component companies. Consequently, both the Stacked FinPool and Multi-tier Attention FinPool frameworks utilize three FinPool operators. It's important to note that the frameworks utilize only the number of clusters at each GICS hierarchy level (11, 24, and 69) as a design hyper-parameter, without applying the detailed GICS classifications directly. Instead, the frameworks allow FinPool to autonomously learn the mapping, $S^{(l)}$, during the training phase, thereby discovering the optimal cluster assignments.

Stacked FinPool. The Stacked FinPool framework represents a straightforward yet effective approach. It involves vertically stacking three FinPool operators to gradually aggregate the tier-wise feature information of the GICS hierarchy to form the global feature representation of the S&P 500, which is denoted as $z_G \in \mathbb{R}^d$. The operational process of this framework is depicted through the following sequence of equations:

$$\left[A^{(1)}, X^{(1)} \right] = \text{FinPool}^{(1)}(A^{(0)}, X^{(0)}), \quad (5.13)$$

$$\left[A^{(2)}, X^{(2)} \right] = \text{FinPool}^{(2)}(A^{(1)}, X^{(1)}), \quad (5.14)$$

$$\left[A^{(3)}, X^{(3)} \right] = \text{FinPool}^{(3)}(A^{(2)}, X^{(2)}), \quad (5.15)$$

where $A^{(0)}$ and $X^{(0)}$ are the initial adjacency matrix and raw node feature matrix of the SCRG, respectively. These three FinPool operators correspond to the top three tiers of the GICS hierarchy: the 69 industries, 24 industry groups, and 11 sectors. The final global feature representation of the S&P 500, z_G , is obtained by averaging the aggregated features of the 11 sectors through an *embedding GNN*,

$$z_G = \text{mean} \left(\text{GNN}_{\text{emb}}(A^{(3)}, X^{(3)}) \right). \quad (5.16)$$

Subsequently, a multi-layer perceptron head with two linear layers (64, 32) is attached

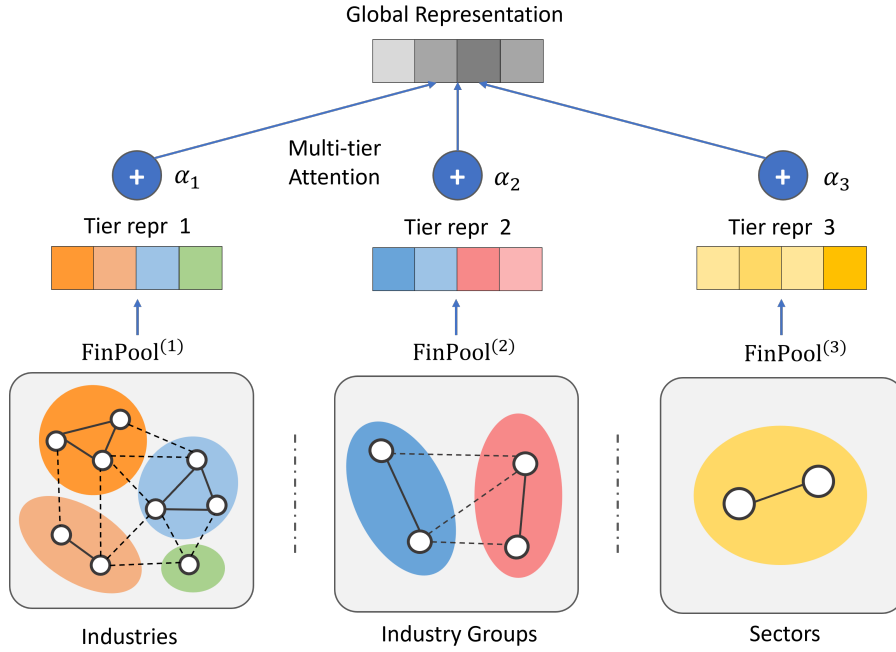


Figure 5.3: Pipeline visualization of the Multi-tier Attention FinPool framework.

to this global representation for binary predictions of index movements,

$$\hat{y}_{t+1} = \begin{cases} 1 \\ 0 \end{cases}, \quad (5.17)$$

where 1 indicates a predicted increase, and 0 a decrease in the index's closing price from time t to $t + 1$. A detailed visualization of the entire stacked pooling pipeline is provided in Figure 5.2.

To train the model for binary predictions of index movements, we employ the Binary Cross-Entropy (BCE) loss function, a widely used objective function for binary classification tasks. The BCE loss measures the difference between the predicted probabilities and the true labels, penalizing incorrect predictions more heavily as the confidence in the wrong prediction increases. For a single prediction, the BCE loss is given by:

$$\mathcal{L}_{\text{BCE}} = -(y \cdot \log(\hat{p}) + (1 - y) \cdot \log(1 - \hat{p})), \quad (5.18)$$

where $y \in \{0, 1\}$ is the ground truth label, and $\hat{p}$ is the predicted probability of the label being 1, and $(1 - \hat{p})$ is the predicted probability of the label being 0.

Multi-tier Attention FinPool. Inspired by the attention mechanism proposed by Bahdanau et al. (2014) and the multi-head attention mechanism in Transformers (Vaswani et al., 2017), we propose a Multi-tier Attention FinPool framework that dis-

tinctly aggregates feature information from the three GICS tiers based on their degrees of importance. The architecture of this framework is outlined as follows:

$$\left[A^{(1)}, X^{(1)} \right] = \text{FinPool}^{(1)}(A^{(0)}, X^{(0)}), \quad (5.19)$$

$$\left[A^{(2)}, X^{(2)} \right] = \text{FinPool}^{(2)}(A^{(0)}, X^{(0)}), \quad (5.20)$$

$$\left[A^{(3)}, X^{(3)} \right] = \text{FinPool}^{(3)}(A^{(0)}, X^{(0)}), \quad (5.21)$$

Diverging from Stacked FinPool, the three FinPool operators, $\text{FinPool}^{(1)}$, $\text{FinPool}^{(2)}$ and $\text{FinPool}^{(3)}$, in equation (5.19) to (5.21), take the same inputs of $A^{(0)}$ and $X^{(0)}$. They operate in parallel to perform distinct feature pooling to generate $\{A^{(i)}, i = 1, 2, 3\}$ and $\{X^{(i)}, i = 1, 2, 3\}$. These matrices are then fed into *embedding GNNs* to yield feature representations, $\{z_{\mathcal{G}}^{(i)}, i = 1, 2, 3\}$, corresponding to the three tiers of the GICS hierarchy:

$$z_{\mathcal{G}}^{(1)} = \text{mean} \left(\text{GNN}_{emb}^{(1)}(A^{(1)}, X^{(1)}) \right), \quad (5.22)$$

$$z_{\mathcal{G}}^{(2)} = \text{mean} \left(\text{GNN}_{emb}^{(2)}(A^{(2)}, X^{(2)}) \right), \quad (5.23)$$

$$z_{\mathcal{G}}^{(3)} = \text{mean} \left(\text{GNN}_{emb}^{(3)}(A^{(3)}, X^{(3)}) \right), \quad (5.24)$$

The final d -dimensional global representation of S&P 500, $z_{\mathcal{G}}$, is derived from the weighted sum of these tier representations via an attention mechanism (Bahdanau et al., 2014), which evaluates the importance of the feature information of that industry tier,

$$z_{\mathcal{G}} = \begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_3 \end{bmatrix} \begin{bmatrix} z_{\mathcal{G}}^{(1)} \\ z_{\mathcal{G}}^{(2)} \\ z_{\mathcal{G}}^{(3)} \end{bmatrix}. \quad (5.25)$$

The attention weight α_i within the weight vector α , is computed through an attention transformation of the corresponding $z_{\mathcal{G}}^{(i)}$,

$$v_i = \text{GELU}(z_{\mathcal{G}}^{(i)} \cdot w_a) \cdot w_b, \quad i = 1, 2, 3 \quad (5.26)$$

$$\alpha = \text{Softmax}(v_1, v_2, v_3) \quad (5.27)$$

where w_a and w_b are learnable weights that transform each tier representation, $z_{\mathcal{G}}^{(i)}$, into attention scores, v_i . The softmax function then normalizes these scores, ensuring their sum equals 1.

In comparison to the straightforward Stacked FinPool, this Multi-tier Attention framework allows the prediction model to jointly attend to information from different

representation sub-spaces across various GICS tiers, avoiding potential information loss that might occur in the Stacked FinPool’s layer-by-layer information propagation. The visualization of the whole pipeline can be found in Figure 5.3.

In summary, the methodology section outlines the technical details of FinPool, revealing its function as an intermediate module for integrating feature information across each industry hierarchy, in conjunction with the Semantic Company Relationship Graph (SCRG). This section then elaborates on two proposed frameworks designed to deploy FinPool operators. These frameworks effectively aggregate the integrated feature information processed by FinPool to construct a comprehensive global representation of the S&P 500 for market index prediction. The implementation of these frameworks closely follows the structure of the industry tiers as defined by the Global Industry Classification Standard (GICS), ensuring alignment with established standard industry hierarchies.

5.3 Evaluation

The two FinPool index prediction frameworks are systematically evaluated through a trading simulation-based scheme, divided into three key parts. Initially, we describe the intraday stock data used in our analysis, setting the stage for understanding the data context. Next, the experiment settings and baseline benchmarks are outlined to provide clarity on our comparative approach. Finally, we detail the training process and the trading strategies employed, offering insights into practical applications and performance evaluation in market scenarios.

5.3.1 Data description

In this study, the S&P 500 index is utilized as the target for movement predictions. Given that the components of the S&P 500 index changes periodically over the trading years, and considering that FinPool’s soft assignment scheme, $S^{(l)}$, necessitates a fixed set of input nodes in the SCRG, a consistent list of S&P 500 components is established. This list comprises the intersecting components across the beginning and the end of our in-sample period. To ensure a sufficiently large pool of intersected stocks, the in-sample period is set to six years, from 2014-01-01 to 2018-12-31, resulting in 367 intersected stocks. These stocks represent 88% of the total market capitalization of the S&P 500 as of December 31, 2018. Such a substantial portion of the index’s market

capitalization affirms that the selected components effectively represent the overall index. The in-sample year 2018 is used as validation set for hyperparameter tuning.

Inspired by Schulmeister (2009), this study adopts a 30-min (half-hourly) stock data interval, as intraday frequency is more challenging and tends to contain valuable information that can be effectively diffused for better profitability in tradings (Andersen and Bollerslev, 1997; Batten et al., 2018). Such 30-min interval is also widely used in relevant financial time series analysis (Sun et al., 2016; Renault, 2017; Gao et al., 2018).

For each trading day, from 09:30 AM to 16:00 PM, there are 14 half-hourly timestamps, accumulating to 21,072 data points over the in-sample period. For each timestamp, 12 technical and numerical features are collected, as recommended by previous studies (Campbell and Thompson, 2008; Chen et al., 2015; Bao et al., 2017), including open, high, low, close, volume, sma, ema, wma, williams, rsi, adx and standard deviation. The feature selection is consistent with Chapter 3, adhering to common practices in financial time series predictions (see Section 3.4). The close change percentage feature is omitted due to redundancy. Unlike Chapter 4, fundamental features are not included in this analysis because the temporal benchmarking models rely on index features, and indices do not have fundamental attributes (see Section 5.3.2).

To thoroughly assess the effectiveness and robustness of the proposed models, the models are evaluated over three distinct years: 2019, 2020, and 2021. This allows for a comprehensive understanding of the models' performance across different market conditions. The descriptive statistics of the selected features are presented in Table 5.1.

5.3.2 Experiment settings and benchmark models

In line with recommendations by Fischer and Krauss (2018), for financial time-series predictions, a longer horizon is preferred in practice. Considering the memory limitations of our computational resources, we set the sliding window length to 30 timestamps. This results in 360 input feature dimensions for X_G , with 12 features per timestamp. Counting the number of S&P 500 companies, the overall memory consumption leads to 16 GB. To facilitate effective feature transformation, the feature dimension d is configured to be 64, determined through grid search based on model performance on the validation set (Feng et al., 2019), with candidate values drawn from {24, 32, 64, 96}. For convenience, we use the abbreviation FinPool-Stacked and FinPool-Multitier

	mean	std	min	25%	50%	75%	max
open	88.48	83.99	3.07	41.81	67.45	107.51	1434.55
high	88.70	84.21	3.08	41.92	67.60	107.76	1435.52
low	88.27	83.77	3.02	41.71	67.29	107.26	1426.30
close	88.48	83.99	3.07	41.82	67.45	107.51	1426.30
volume	381883.56	1157471.58	75.00	62678.00	140107.00	329978.00	263600904.00
sma	88.47	83.97	3.13	41.81	67.44	107.50	1421.84
ema	88.47	83.97	3.15	41.82	67.44	107.51	1421.66
wma	88.48	83.97	3.12	41.81	67.44	107.51	1422.72
williams	-48.53	30.27	-100.00	-75.56	-47.75	-21.13	0.00
rsi	51.65	202.91	-34732.80	39.50	51.48	63.07	41180.60
adx	39.36	23.39	0.00	24.07	33.08	46.67	169.43
std	0.38	0.65	0.00	0.11	0.21	0.42	53.11

Table 5.1: Descriptive statistics of the 12 technical and numerical stock features.

to represent the two market index prediction frameworks.

Essentially, FinPool functions by assigning individual stocks to higher-level industry clusters. Similarly, Principal Component Analysis (PCA) can be used to achieve this goal by iteratively reducing the number of the S&P 500 companies and identifying the major components (Dunteman, 1989; Adebiyi et al., 2014), aligning with the three tiers of the GICS. An empirical study conducted by Zhong and Enke (2017) proved that, among the three mature dimensionality reduction techniques of Principal Component Analysis (PCA), Fuzzy Robust Principal Component Analysis (FRPCA), and Kernel-based Principal Component Analysis (KPCA), PCA combined with a MLP yielded the best results for stock movement prediction. Thus, PCA is included as a benchmark model in our study to provide a comparative evaluation of FinPool’s performance. Specifically, for each dimension of the 360 features, a sequence of PCA models, following the GICS hierarchies, are fitted to project the 367 companies of that feature dimension into one dimension of the global feature representation. The 360 individual feature dimensions are then concatenated to form the final representation of SCRG. A same MLP classification head of our hierarchical graph pooling frameworks is connected for binary index movement prediction.

To ensure a comprehensive evaluation, Long Short-Term Memory (LSTM) networks are incorporated as a strong deep learning baseline. Recognized for their effectiveness in financial market predictions (Fischer and Krauss, 2018; Bhandari et al., 2022), LSTM networks benefit from a temporal position-aware architecture, offering

distinct advantages in time-series data processing compared to our proposed models. To align LSTM with our model settings, we maintain the same hidden dimension and sliding window length. Additionally, three classical machine learning models are also included as baselines: Random Forest (RF), Multi-Layer Perceptron (MLP), and Support Vector Machine (SVM). These models are selected for their demonstrated robustness and relatively high performance in stock market prediction contexts (Moghaddam et al., 2016; Zhong and Enke, 2017; Fischer and Krauss, 2018). For hyperparameter settings, the MLP uses hidden sizes of (64, 32), consistent with FinPool’s MLP prediction head, while SVM and RF are configured with their default settings in scikit-learn (Lin et al., 2022).

Additionally, in light of recent studies that have successfully applied the Transformer model to market index movement predictions with promising results (Wang et al., 2022; Tao et al., 2024), we also incorporate a Transformer model as a strong baseline in our evaluation. To ensure a fair comparison, we align the hidden dimension and training settings of the Transformer with those used in both the LSTM and our proposed models. Following Wang et al. (2022), we employ sine and cosine functions of varying frequencies to encode positional information in the index quote sequences. This temporal positional embedding, similar to the LSTM, offers distinct position-aware advantage for time-series data processing compared to our proposed models.

To evaluate the effectiveness of our innovations against other pooling methods, we incorporate Self-Attention Graph Pooling (SAGPool) (Lee et al., 2019), MinCutPool (Bianchi et al., 2020) and DiffPool, which serves as the foundation of our work, following Arya et al. (2023). SAGPool (Lee et al., 2019) is a node dropping based pooling method. It introduces a hierarchical graph pooling mechanism based on self-attention, leveraging graph convolution to compute attention scores that consider both node features and graph topology. The scores are then used to rank nodes with nodes with the top scores are retained, effectively reducing the graph size while preserving both key node features and structural topology. MinCutPool (Bianchi et al., 2020), in contrast, is a node clustering-based pooling method with soft assignments, similar to DiffPool. However, it differentiates itself by leveraging a relaxed formulation of the normalized minCUT problem, optimized via a GNN and a multi-layer perceptron (MLP), to compute cluster assignments that account for both graph topology and node features. The soft cluster assignment matrix groups strongly connected nodes with similar features, enabling the generation of coarsened graphs by aggregating node features and edges within and across clusters. For all three pooling models, we maintain consistency with

FinPool by using the same hidden size of 64. Additionally, all models perform hierarchical pooling with the same GICS-based number of nodes as FinPool, ensuring a uniform basis for comparison.

It is important to note the difference in feature usage between the various models. While LSTM, Transformer and the machine learning baselines utilize the S&P 500's domestic 12 features directly, the graph pooling models and the PCA baseline employ these features at the component stock level only. This distinction underscores the unique approach of our models that obtains insights from component stocks only. The inclusion of these diverse baseline models, spanning both deep learning and traditional machine learning approaches, provides a comprehensive framework to evaluate the performance of our proposed models across a range of methodologies.

5.3.3 Evaluation scheme

In this study, the performance of the models is evaluated through trading simulations employing a long-short trading strategy, following the approach outlined by Hsu et al. (2016). For each stock predicted to experience a rise (+1) at the half-hourly timestamp $t + 1$, we simulate a buy trade at time t and a sell trade at $t + 1$. Conversely, for stocks anticipated to decline (-1), a short position is simulated at t , followed by a repurchase at $t + 1$.

For effective model training, a step-by-step time-series cross-validation scheme is implemented, as recommended by Bergmeir and Benítez (2012). Each training step encompasses 30 timestamps, striking a balance between training duration and data utilization efficiency. At every step, the model is retrained using all available data up to the current timestamp. It then predicts the outcomes for the upcoming 30 timestamps. This iterative process enables continuous model refinement and assessment against evolving data.

We measure the profitability performance using profit and loss (PnL) and risk-adjusted return (Sharpe Ratio). For a better description, these metrics are reported on a daily basis. The daily profit and loss is calculated using the following formula:

$$\text{PnL} = \sum_T \sum_{t=0}^{13} \text{sign}_t * r_t, \quad (5.28)$$

where r_t represents the half-hourly index return at timestamp t , and T denotes the total number of trading days within each individual year. The PnL is also reported in basis points, PnL_{bps} , which is multiplied by 10,000 and then divided by the total number of

trading days, T .

The Sharpe Ratio (SR) is utilized to provide a risk-adjusted measure of profitability. We calculate the annualized Sharpe Ratio using the formula:

$$SR = \frac{\overline{\text{PnL}}}{\sigma_{pf}} * \sqrt{T} \quad (5.29)$$

where $\overline{\text{PnL}}$ is the mean daily PnL achieved by the models, and σ_{pf} is the standard deviation of the PnL series. We prioritize the Sharpe Ratio as the main performance metric as it effectively adjusts for the risk, offering a comprehensive view of the profitability of the models.

For statistical predictability metrics, we adopt Accuracy (Acc), Precision, Recall, and Matthews Correlation Coefficient (MCC) following Chapter 3 (Section 3.4.3). These metrics are derived from the confusion matrix and are calculated as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5.30)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5.31)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5.32)$$

$$\text{MCC} = \frac{(TP \cdot TN) - (FP \cdot FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (5.33)$$

Here, TP represents True Positives, TN is True Negatives, FP denotes False Positives, and FN stands for False Negatives.

5.4 Results

In this section, we present the findings from our trading simulation scheme. Our detailed analysis offers insights into the effectiveness of our model's performance in simulated trading scenarios, thereby providing a clear understanding of its potential impact and utility in financial market predictions.

5.4.1 Experiment results

Table 5.2 and 5.3 comprehensively report the profitability and predictability performance characteristics of all models. The selected out-of-sample years are crucial for verifying the robustness of our proposed models across diverse range of market conditions.

Models	2019			2020			2021		
	SR	PnL	PnL_bps	SR	PnL	PnL_bps	SR	PnL	PnL_bps
FinPool-Stacked	2.13	0.25	9.81	1.58	0.48	18.90	1.88	0.24	9.62
FinPool-Multitier	2.78	0.31	12.15	2.34	0.70	27.54	2.40	0.29	11.67
DiffPool	1.60	0.19	7.26	1.34	0.42	16.48	1.43	0.19	7.39
MinCut	1.84	0.21	8.36	1.33	0.35	13.80	1.78	0.23	9.21
SAGPool	1.53	0.18	7.11	1.07	0.33	12.75	0.92	0.13	5.05
LSTM	1.47	0.18	7.13	1.34	0.38	14.99	1.21	0.17	6.67
PCA	1.69	0.18	7.16	0.48	0.16	6.20	0.61	0.08	3.25
MLP	1.19	0.15	5.99	0.34	0.12	4.71	1.62	0.22	8.72
RF	1.31	0.15	5.90	0.67	0.20	7.75	0.75	0.09	3.72
SVM	1.80	0.22	8.46	1.09	0.32	12.68	1.32	0.17	6.82
Transformer	1.16	0.13	5.26	0.82	0.23	9.15	1.69	0.23	9.02
MKT	2.11	0.27	10.65	0.58	0.19	7.50	1.83	0.25	9.90

Table 5.2: Profitability Performance characteristics in each of the out-of-sample years. MKT denotes the metrics of S&P 500 index.

5.4.1.1 Profitability results

A clear trend observed from Table 5.2 is that both our models, FinPool-Stacked and FinPool-Multitier, consistently outperform the baseline models across each of these individual years. Most impressively, FinPool-Multitier distinguishes itself as the stand-out model, being the only one to consistently achieve Sharpe Ratios higher than 2.0.

Breaking down the analysis year by year, in 2019, both FinPool-Stacked and FinPool-Multitier not only outperform the baseline models but also achieve higher Sharpe Ratios than the S&P 500 index itself. This is a noteworthy accomplishment, highlighting the effectiveness of these models in capturing market movements. In terms of total profits, FinPool-Multitier takes the lead, as it is the only model generating a positive surplus of 1.5 basis points over the market, further solidifying its top position.

The extraordinary volatility of 2020 presented a formidable test for predictive models. The baseline models, typically robust in more stable conditions, encounter difficulties in delivering profitable predictions while balancing the risks involved, as evidenced by their generally low Sharpe Ratios, most falling below the threshold of 1. In contrast, our proposed models exhibit significant profitability gains over the market. Notably, the FinPool-Multitier model achieves an exceptional profit surplus of 20.04 basis points. This notable success underscores the model's capacity for not only securing profits but also managing risks effectively, as reflected in its impressive Sharpe

Ratio of 2.34.

Regarding the year 2021, FinPool-Multitier manages to outshine the market with a PnL surplus of 1.77 basis points. This consistent performance over multiple years underscores the model's adaptability and robustness in varying market conditions. Contrarily, the best-performing baseline model in this period, Transformer, demonstrates a performance that falls short of the market's, highlighting the challenging nature of market prediction and the distinctive success of our proposed models.

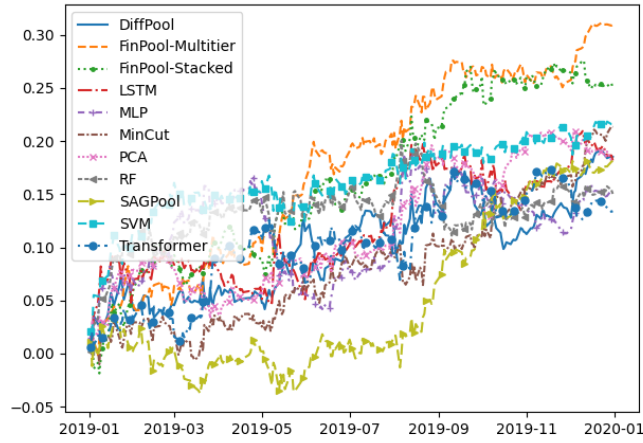
In comparison to our models, the pooling baselines — DiffPool, MinCut, and SAGPool — demonstrate varying levels of performance across the out-of-sample years. While DiffPool and MinCut outperform several traditional baselines like PCA, MLP, and RF, they still fall short when compared to FinPool-Stacked and FinPool-Multitier in both profitability and risk-adjusted returns. DiffPool achieves a maximum Sharpe Ratio of 1.60 in 2019, with modest PnL values across all years, indicating some effectiveness in pooling graph features but limited capacity to fully capture hierarchical relationships in financial data.

MinCut exhibits a slight advantage over DiffPool in terms of profitability, especially in 2019 and 2021, where its Sharpe Ratios reach 1.84 and 1.78, respectively. This suggests that the clustering mechanism of MinCut is better suited to identify meaningful node groupings. However, in volatile market conditions such as 2020, MinCut struggles to maintain consistent performance, highlighting its sensitivity to market fluctuations.

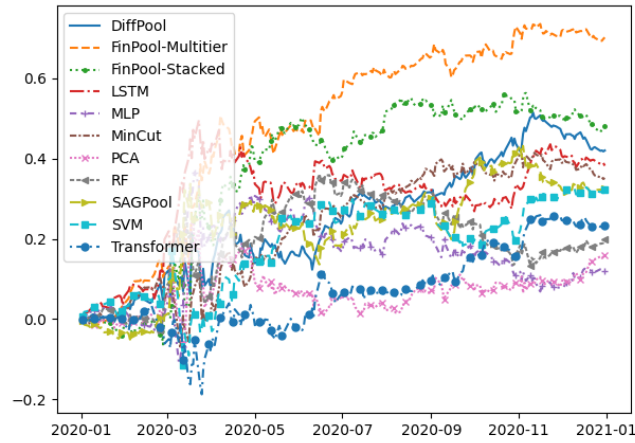
SAGPool, on the other hand, demonstrates the weakest performance among the pooling methods, with Sharpe Ratios consistently below 1.6 across all years. Its node-dropping mechanism loses critical information necessary for market profitability, especially in 2021, where it achieves a Sharpe Ratio of only 0.92. This limitation underscores the importance of retaining detailed graph structure, as exemplified by the superior performance of FinPool-Multitier.

5.4.1.2 Yearly cumulative PnL trends

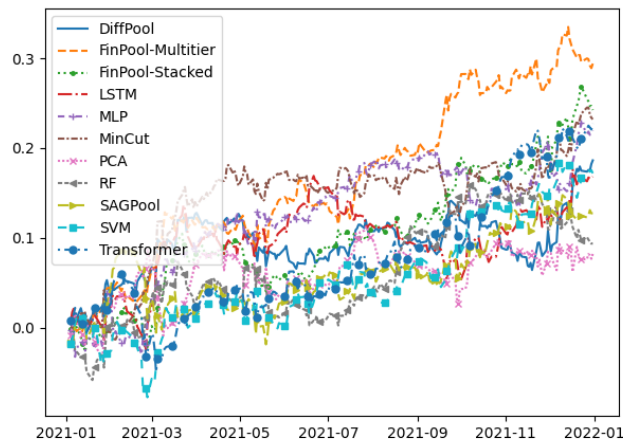
Figure 5.4 provides a detailed view of the cumulative PnL for each model across the individual years. In 2019, a notable observation is that while the SVM, RF, and MLP baselines initially demonstrate strong PnL performance, this trend diminishes over a more extended period. This is particularly evident with the MLP model, which exhibits a period of high volatility towards the end of the year. In contrast, both DiffPool models display a steady upward trend throughout the year, indicating a sustained positive



(a) Cumulative PnL in 2019



(b) Cumulative PnL in 2020



(c) Cumulative PnL in 2021

Figure 5.4: Cumulative PnL in each individual year from 2019 to 2020 of all benchmark models.

performance in the trading simulation.

The year 2020 sees FinPool-Multitier distinctively outperforming all other models, with an increasingly wide margin in PnL. This consistent superiority showcases the model's robustness and effectiveness in dynamic market environments like 2020.

However, the year 2021 presents a more mixed picture. The FinPool-Stacked model experiences a downturn at the start of the year, maintaining a negative PnL for approximately half of the year. Though it eventually recovers and slightly surpasses the baseline models, including the S&P 500, this fluctuation in performance raises questions about the robustness of the FinPool-Stacked model. While its counterpart, FinPool-Multitier, maintains a solid PnL advantage across the whole year.

5.4.1.3 Predictability results

Models	2019				2020				2021			
	Acc	Precision	Recall	MCC	Acc	Precision	Recall	MCC	Acc	Precision	Recall	MCC
FinPool-Stacked	0.516	0.543	0.553	0.026	0.507	0.540	0.549	0.006	0.512	0.546	0.533	0.020
FinPool-Multitier	0.506	0.535	0.536	0.008	0.527	0.563	0.536	0.052	0.525	0.557	0.553	0.045
DiffPool	0.503	0.533	0.517	0.005	0.509	0.545	0.520	0.016	0.502	0.536	0.539	-0.002
MinCut	0.501	0.533	0.480	0.005	0.503	0.538	0.536	0.002	0.506	0.542	0.510	0.012
SAGPool	0.508	0.535	0.568	0.008	0.511	0.546	0.533	0.017	0.520	0.539	0.733	0.008
LSTM	0.517	0.546	0.534	0.031	0.494	0.532	0.491	-0.011	0.502	0.537	0.521	0.001
PCA	0.508	0.536	0.549	0.012	0.508	0.541	0.557	0.007	0.505	0.536	0.566	0.000
MLP	0.506	0.536	0.525	0.010	0.502	0.538	0.523	0.001	0.509	0.542	0.541	0.013
RF	0.504	0.530	0.583	-0.003	0.514	0.543	0.613	0.013	0.511	0.538	0.623	0.005
SVM	0.511	0.531	0.669	0.001	0.528	0.547	0.699	0.029	0.519	0.540	0.703	0.011
Transformer	0.518	0.532	0.758	0.005	0.523	0.540	0.759	0.009	0.523	0.538	0.788	0.006

Table 5.3: Predictability performance in terms of Accuracy, Precision, Recall and MCC.

Table 5.3 summarizes the predictability performance of various models across the out-of-sample years, evaluated using Accuracy (Acc), Precision, Recall, and Matthews Correlation Coefficient (MCC). Our proposed FinPool-Multitier consistently achieves the highest Precision and MCC scores across most years while maintaining competitive Accuracy levels. These results emphasize FinPool-Multitier's ability to make precise predictions with minimal false positives and its robustness in capturing the overall relationship between predicted and actual stock movements.

Transformer exhibits consistently higher Recall scores across all years, with values peaking at 0.758 in 2019 and 0.788 in 2021. This suggests that Transformers excel in capturing true positives due to the ability to model complex temporal patterns. However, its low MCC values suggest a trade-off in its ability to balance true positives and true negatives, limiting its robustness in overall predictability.

Models	5min			15min		
	SR	PnL	PnL_bps	SR	PnL	PnL_bps
FinPool-Stacked	4.17	0.14	22.95	2.12	0.25	9.81
FinPool-Multitier	4.90	0.19	32.37	2.49	0.32	12.82
DiffPool	3.41	0.12	20.18	1.16	0.15	6.00
MinCut	2.60	0.08	13.19	1.49	0.20	7.82
SAGPool	2.18	0.08	13.43	0.92	0.12	4.74
Transformer	3.27	0.10	17.32	0.80	0.11	4.19
LSTM	3.03	0.08	13.47	0.54	0.07	2.78
PCA	2.37	0.09	15.39	0.76	0.09	3.67
MLP	2.03	0.07	11.90	1.29	0.16	6.43
RF	1.92	0.07	11.20	0.23	0.03	1.11
SVM	0.65	0.02	3.60	1.77	0.21	8.44
MKT	3.03	0.10	16.81	1.89	0.25	9.83

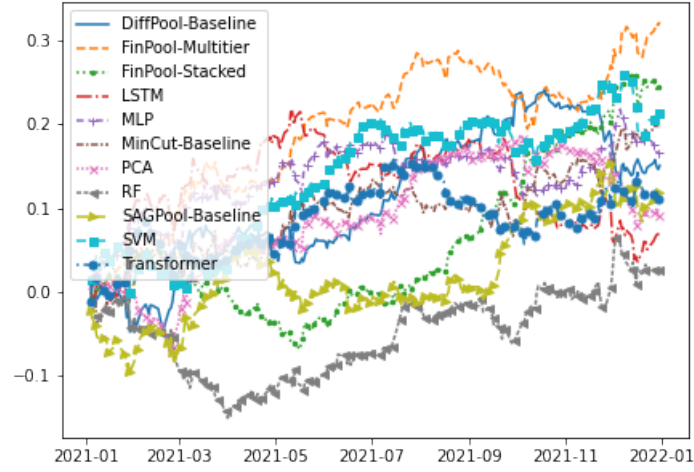
Table 5.4: Performance characteristics of profitability and predictability in each of the intraday intervals. MKT denotes the metrics of S&P 500 index.

SVM and LSTM also emerge as strong baselines. SVM particularly excels in Recall, with scores reaching 0.699 in 2020 and 0.703 in 2021. LSTM achieves the highest MCC score of 0.031 in 2019 and competitive Precision scores, such as 0.546 in the same year. However, its relatively low Recall scores, particularly in 2020, suggest that it misses some true positives compared to models like Transformers or SVM.

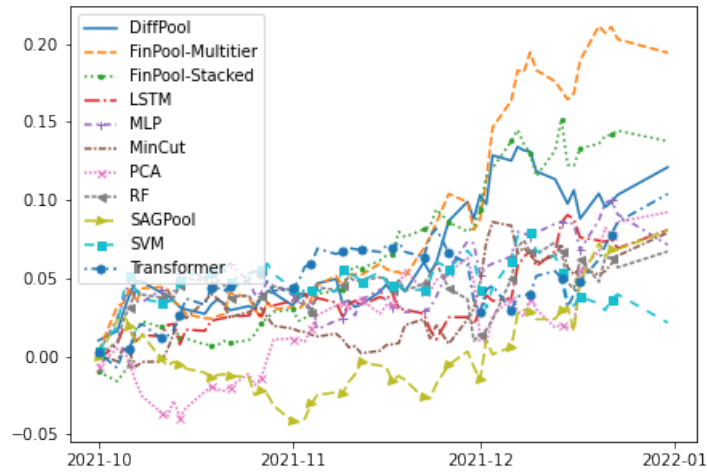
5.4.2 Robustness to intraday data intervals

In addition to evaluating robustness across different market conditions, we extend our analysis to test our proposed models and benchmarks over varying intraday data intervals. Specifically, we conduct further tests at 15-minute and 5-minute intervals. To ensure comparability, we adjust the out-of-sample periods so that the volume of data points remains a consistent level with the half-hourly interval: the entire year of 2021 for the 15-minute interval and the last quarter of 2021, from October to December 2021, for the 5-minute interval.

Table 5.4 provides a detailed overview of the models' performance across 15-minute and 5-minute intervals, emphasizing aspects of profitability and predictability. At the 15-minute interval, the FinPool-Multitier model stands out with the highest Sharpe Ratio of 2.49 and a PnL of 12.82 basis points, showcasing strong profitability



(a) Cumulative PnL of 15-min interval



(b) Cumulative PnL of 5-min interval

Figure 5.5: Cumulative PnL in 15-min and 5-min data intervals.

and effective risk management. The accuracy of this model is slightly lower than SVM, which shows an accuracy of 52.15%. The other proposed model, FinPool-Stacked, follows closely with a Sharpe Ratio of 2.12, positioning it as the second-best in terms of risk-adjusted returns, despite a slightly lower PnL compared to the market baseline. Other models, including the Transformer, LSTM, PCA, MLP, RF, and SVM, display varied levels of performance, generally below that of the market baseline, with SVM notably achieving a Sharpe Ratio of 1.77.

At the 5-minute interval, the two proposed FinPool models continue to outperform other benchmarks, with the FinPool-Multitier reaching an impressive Sharpe Ratio of 4.90, PnL of 32.37 basis points. This interval highlights the FinPool models' capability to effectively manage and exploit the high-frequency data, showcasing significant gains in market predictive power and financial returns. The Transformer and LSTM models also improve their performance at this shorter interval but remain less effective compared to the FinPool offerings.

Among the graph pooling benchmarks, DiffPool, MinCut, and SAGPool exhibit moderate performance. At the 15-minute interval, MinCut achieves a Sharpe Ratio of 1.49, slightly outperforming DiffPool (1.16) and SAGPool (0.92). At the 5-minute interval, DiffPool achieves a Sharpe Ratio of 3.41, outperforming MinCut (2.60) and SAGPool (2.18). The PnL values of these models remain inconstantly lower than the FinPool models, indicating their relatively limited effectiveness in capturing short-term market trends.

Figure 5.5 showcases the cumulative PnL for prediction models over 15-minute and 5-minute intraday intervals. In the 15-minute interval, the FinPool-Multitier model demonstrates a clear lead in performance. The FinPool-Stacked model also shows a positive but more varied trend. This is reflective of its performance in 2021 during the half-hourly interval, where it experiences a downturn in the first half of the year before rebounding, indicating lesser robustness compared to the FinPool-Multitier model. Other models, including LSTM, MLP, PCA, RF, SVM, Transformer and the pooling baselines exhibit mixed results with some instability.

When examining the 5-minute interval, the upward trend of the FinPool-Multitier model is particularly pronounced, signaling strong profitability at this granular level of frequency. The FinPool-Stacked model displays a similar pattern of growth, ensuring it remains a strong contender amidst the high-frequency trading conditions.

5.4.3 Discussions and implications

Overall, the standout performance of our proposed FinPool models, which rely exclusively on the stock features of constituent companies without incorporating the index feature information, underscore the efficacy of hierarchical graph pooling in predicting index movements. This approach effectively aggregates the nuances of market dynamics at different levels of the industry hierarchy.

The comparison with the PCA baseline model, which shares the same MLP classification head architecture with our FinPool models, further illuminates our findings. The PCA model's relatively inferior performance highlights its limitations in integrating hierarchical features for index predictions. This disparity suggests that the PCA model may not adequately integrate the complex, multi-tiered information inherent in the S&P 500 constituents.

The comparisons with graph pooling baselines, including DiffPool, MinCut, and SAGPool, further emphasize the superiority of the FinPool models. SAGPool, which exhibits the weakest performance among the pooling baselines, highlights the limitations of node-dropping clustering in company graphs. The rigid node ranking and elimination mechanism inherent to SAGPool does not align well with the hierarchical structure of financial networks, leading to significant underperformance in this context. In contrast, the better performance of DiffPool and MinCut underscores the importance of soft assignments in scenarios like ours. By allowing for flexible cluster assignments and more nuanced feature aggregation, these methods outperform SAGPool, suggesting that soft assignment mechanisms are critical for effectively capturing the interconnected relationships within the S&P 500 constituents.

Additionally, the robustness of FinPool-Multitier compared to FinPool-Stacked, particularly evident in Figure 5.4c and Figure 5.5a, validates the advantages of Multi-tier Attention discussed in Section 5.2.3. This Multi-tier Attention mechanism allows the model to concurrently process information from different representational subspaces corresponding to various GICS tiers. This attribute significantly mitigates the potential for information loss that can occur in a stacked, layer-by-layer information propagation approach, as is the case with FinPool-Stacked. This comparative analysis highlights the nuanced and sophisticated capability of FinPool-Multitier in capturing a more comprehensive and detailed picture of the market index dynamics.

Another key observation concerns the influence of trading frequencies. When aligning the 15-minute trading simulation's data period with that of the 5-minute in-

terval, the FinPool models demonstrate reduced profitability. Specifically, FinPool-Multitier achieves a Sharpe Ratio (SR) of 3.71 and a PnL of 0.14, while FinPool-Stacked achieves an SR of 3.23 and a PnL of 0.11. These results are notably lower than their respective performances at the 5-minute frequency (SR of 4.90 and 4.17), though both still outperform the market baseline. We attribute the performance disparity to variations in market efficiency regarding information diffusion. At higher frequencies, market information is less fully absorbed, leaving greater opportunities for profitable trading decisions.

More importantly, the superior performance of our models over the two state-of-the-art deep learning baselines in time-series predictions, LSTM and Transformer, both renowned for their temporal position-aware capabilities, raises critical questions about the utility of domestic index price features for capturing trading signals. Since our models rely exclusively on the constituent feature information and do not incorporate any direct features from the domestic index sequences, this data difference reveals a key insight: the S&P 500 index price does not fully reflect all available public information. Specifically, our empirical results indicate that half-hourly historical price information from component stocks is not timely assimilated into the current index quotes in the US stock market.

This observation presents a significant challenge to the Efficient Market Hypothesis (EMH) as proposed by Fama (1970), from the perspective of information diffusion (Grossman and Stiglitz, 1980; Merton et al., 1987; Babus and Kondor, 2018; Di Maggio et al., 2019). EMH asserts that market index prices should encapsulate all publicly available information, suggesting that no excess profits can be derived from historical public information of component stocks. However, our findings indicate a 'lagged constituent information diffusion' effect, where markets do not immediately integrate information from individual components into the overall index's value. This lag in information diffusion highlights potential inefficiencies in market dynamics, offering opportunities for trading strategies that exploit this phenomenon. By capitalizing on the lagged diffusion of constituent information, investors could potentially gain alphas, achieving excess returns beyond standard market predictions.

5.5 Conclusion

This chapter marks a significant stride in the field of intraday financial market index prediction. By reorienting the focus from classical machine learning methods to a

graph-based classification framework, this research underscores the untapped potential of hierarchical graph pooling methods, specifically the FinPool with the Semantic Company Relationship Graph, in forecasting market indices.

The introduction of Stacked FinPool and Multi-tier Attention FinPool as novel market index prediction frameworks has proven effectiveness in exploiting the multi-tiered structure of the Global Industry Classification Standard. The empirical results of FinPool-Multitier showcase a notable enhancement in trading profits and risk-adjusted returns, thereby setting a new benchmark in this domain. The robustness of FinPool-Multitier over FinPool-Stacked also proves the effectiveness of attention-based joint hierarchical feature aggregation, which can avoid information loss during the layer-by-layer stacked information propagation.

One notable limitation of this study is the use of a static graph pooling framework that incorporates DiffPool and SCRG. As the components of the S&P 500 index undergo periodic changes each year, relying on a static framework means our analysis is confined to a fixed subset of intersecting index components. Consequently, this approach may overlook insights from the remaining components, which in this study account for 12% of the market capitalization. To address this limitation, future research could explore the development of a new graph pooling framework capable of processing the dynamically changing index components to enable a more comprehensive analysis that incorporates all constituents over time.

It is important to note that the primary aim of our study is to investigate the existence of the effect of lagged intraday constituent information diffusion. Extending our methodology to efficiently utilize such effect by fully capturing the dynamically-changing components of S&P 500 for better trading performance, while valuable, falls outside the scope of this research. However, this direction represents a promising area for further exploration, potentially offering deeper insights into the dynamics of index movements and the efficiency of markets.

Chapter 6

Discussions and Implications

This chapter provides a critical evaluation of the thesis’s contributions, contextualizing them within the broader academic and practical landscape of financial modeling and prediction. By reflecting on the innovations and implications of each contribution, this chapter highlights their significance in the current research literature. The key contributions evaluated include the Semantic Company Relationship Graph (SCRG), the Non-IID Spatial-Temporal Graph Neural Network (NIST-GNN) for portfolio selection, the Non-IID Spatial-Temporal Chebyshev Network (NIST-Cheb) for sentiment inference, and the FinPool framework for intraday index prediction. Together, these contributions not only address fundamental challenges related to information diffusion in financial markets but also offer significant implications for advancing research in this area.

6.1 Discussions of Contributions

Throughout the thesis, we have presented four major contributions to the research topic of Company Relationship Modeling and Graph Neural Networks for Financial Market Forecasting.

The first leading contribution is the Semantic Company Relationship Graph (SCRG), an innovative framework for establishing company relationship connections by mining semantic business co-occurrence patterns from financial news articles. The collective patterns are aggregated into a co-occurrence matrix, which is then transformed into company embeddings using the GloVe algorithm (Pennington et al., 2014). Recognizing that company relationships are often multi-dimensional and implicit (Xiang et al., 2022), such as the stronger correlations typically observed among companies within

the same sector (Chen et al., 2018), the learning of the company embeddings incorporates context words, ensuring that hidden relationships not directly evident from company name co-occurrence can be effectively captured. The learned embeddings reside within a unified semantic relational space, so the weighted relational strengths between companies can be retrieved through cosine similarity (Lund and Burgess, 1996).

Compared to widely adopted methods such as the correlation graph (Li et al., 2020a; Keskin et al., 2021; Yin et al., 2021; Xiang et al., 2022) and the VAE graph (Hou et al., 2021), SCRG offers a clear advantage by capturing direct relational connections through real-world business events. In contrast, correlation graphs rely on indirect patterns derived from stock price co-movements, while VAE graphs infer relationships from similarities in company fundamentals. Furthermore, unlike the WikiData graph, SCRG is not limited to a predefined set of relationship types (Feng et al., 2019; Kim et al., 2019; Sawhney et al., 2020). It also provides continuous weights to represent connection strengths, offering a more nuanced and flexible representation compared to the binary edge weights typically found in WikiData graphs.

The second contribution, Non-IID Spatial-Temporal Graph Neural Network (NIST-GNN), extends the traditional Spatial-Temporal Graph Neural Networks for the financial prediction applications. Conventional temporal machine learning methods typically treat financial time-series data as independent and identically distributed (IID), an assumption that fails to reflect the interconnected realities of the financial market (Xiang et al., 2022). To address this limitation, NIST-GNN innovatively assigns each company node in the SCRG its own set of LSTM weights, in contrast to models like GCRN (Seo et al., 2018), which share temporal weights across all stocks. This design allows NIST-GNN to account for the non-IID nature of financial data. Other innovations of NIST-GNN include the use of GCN spatial filter and node multiplication for speeding up the training process.

When compared to state-of-the-art GNN methods in Section 3.5.6 (Feng et al., 2019; Sawhney et al., 2020; Kim et al., 2019; Xu et al., 2021; Cheng and Li, 2021; Xiang et al., 2022), NIST-GNN demonstrates superior performance, achieving the highest risk-adjusted returns. This underscores its ability to balance returns and risks effectively, delivering consistent and stable performance over different market conditions over time. In comparison to advanced temporal deep learning models (Section 3.5.2), NIST-GNN's leading results highlight the value of incorporating neighboring information in stock movement predictions. Furthermore, ablation studies (Section 3.5.5) emphasize the significance of SCRG over the correlation graph, as evidenced by NIST-

GNN’s superiority over NGNN-Corr. The comparison against GCN-LSTM verifies the effectiveness of our approach for tackling temporal non-IIDness. All these empirical findings have solidified NIST-GNN’s position as a robust solution for financial prediction tasks.

The third contribution, Non-IID Spatial-Temporal Chebyshev Network (NIST-Cheb) builds on the NIST-GNN for absent news sentiment inference. It is known that investor sentiment has predictive power towards asset pricing, but the direct measurement of investor sentiment is not applicable (Huang et al., 2015). News sentiment is a widely adopted proxy for investor sentiment (Tetlock, 2007; García-Laencina et al., 2010). However, our statistics of news absent rates finds out that even the S&P 500 companies have an average news absent rate of 50.67% across the trading days. We consider that absent news sentiment does not necessarily equate to absent investor sentiment (Erdogan and Yezegel, 2009), as investors often rely on information from related companies to make decisions. Based on this consideration, we propose NIST-Cheb to infer absent investor sentiment using the news sentiment of related companies as a proxy. NIST-Cheb employs a Chebyshev spatial filter to aggregate sentiment information from companies up to three hops away. The built-in non-IID mechanism ensures that each company’s unique temporal sentiment process is individually modeled.

Both error-based and trading simulation evaluations confirm the effectiveness of NIST-Cheb compared to conventional temporal MVI methods. Specifically, as shown in Section 4.5.2, five stock movement classifiers equipped with our inferred sentiment scores consistently outperform those using benchmark sentiment inference methods and demonstrate robustness across different years. The ablation study in Section 4.5.3 indicates that the major performance improvement arises from leveraging neighboring sentiment, aligning with our consideration regarding the importance of interconnected company information.

The final contribution, FinPool for intraday index movement predictions, redefines index prediction by framing it as a graph classification task. Traditional index prediction research treats index quotes as standalone time-series, neglecting the wealth of insights embedded in the constituent industrial clusters (Hong et al., 2007; Bhandari et al., 2022; Hoseinzade and Haratizadeh, 2019; Wang et al., 2022; Tao et al., 2024). However, financial markets exhibit inherent hierarchical structures (Mantegna, 1999), such as sectors and industries. These hierarchical layers offer unique perspectives on market dynamics, unveiling layer-specific market microstructures that are essential for understanding price discovery and liquidity (Bhojraj et al., 2003; Schumaker and Chen,

2009).

In order to effectively capture these layer-specific market microstructures, FinPool builds upon DiffPool (Ying et al., 2018b) by eliminating redundant node sampling and incorporating skip-connections across layers. These designs ensure more efficient and comprehensive graph representation. Additionally, we introduce two distinct deployment frameworks: FinPool-Stacked and FinPool-Multitier, to explore optimal practices for utilizing market hierarchies. Among these, FinPool-Multitier stands out by incorporating an attention mechanism that adaptively assigns weights to different market layers during graph feature aggregation. This innovation enhances the model's ability to extract meaningful features, further improving the robustness of index movement predictions.

The effectiveness of our FinPool designs is validated through empirical trading experiments, where they outperform state-of-the-art graph pooling methods (Ying et al., 2018b; Lee et al., 2019; Bianchi et al., 2020) and advanced temporal models such as Transformer (Wang et al., 2022). The superior performance compared to graph pooling methods, as demonstrated in Section 5.4, highlights the value of both our architectural innovations, specifically tailored for financial prediction tasks, and the two FinPool deployment frameworks. Furthermore, the consistent outperformance of FinPool over temporal models underscores the importance of incorporating hierarchical market microstructural information in index predictions. The robustness of our FinPool methods is further affirmed through evaluations at varying intraday frequencies, as detailed in Section 5.4.2.

6.2 Summary of Implications

Together, these contributions have made a significant impact on the understanding of market inefficiency from the angle of information diffusion in the financial markets.

Lo and MacKinlay (1990) was one of the first to explore the phenomenon of slow market diffusion, demonstrating that the lagged weekly returns of large companies could help explain the current weekly returns of smaller companies, indicating delayed market responses. Subsequent studies suggested that economic connections between companies, such as supplier-customer relationships or competitive ties, might indicate delays in information diffusion, often due to the limited attention of investors and the complexity of analyzing interrelated companies (Cohen and Frazzini, 2008; Sharifkhani and Simutin, 2021; Yu et al., 2024).

In addition to the company level, the delays in information diffusion are also prevalent within market hierarchies. Hong et al. (2007) demonstrates that stock markets react with a delay to information contained in industry returns about their fundamentals and that information diffuses only gradually across markets. Specifically, Hong et al. (2007) finds that industry returns can significantly predict broader market movements for up to two months, with similar patterns observed in the eight largest non-US stock markets. Extending this understanding, Menzly and Ozbas (2010) provides a theoretical explanation for this phenomenon, attributing it to investor specialization and market segmentation. According to their findings, these factors contribute to the gradual diffusion of value-relevant information in financial markets, as specialized investors focus on specific industries or sectors, delaying the incorporation of cross-industry information into market prices.

Building on these foundational insights, the Semantic Company Relationship Graph (SCRG) provides explicit prior knowledge of the paths through which information diffuses between companies. This is achieved by directly modeling the numerical interconnections that emerge from their frequent co-mentioning in business events described in financial news articles.

Drawing on the SCRG, the empirical findings of NIST-GNN for portfolio selection illustrate that public information from cross-correlated stocks in the US market typically experiences a minimum one-day lag in diffusion, in line with the finds of Cohen and Frazzini (2008). This insight has significant implications for investment strategies and market analysis, suggesting that these lags can be exploited for profitable trading opportunities, thus challenging traditional notions of market efficiency.

Moreover, the empirical outcomes of NIST-Cheb for high-quality news sentiment inference underscore the information spillover effect of news sentiment in the financial market, representing a sophisticated instance of information diffusion (Manela, 2014; Li et al., 2020c). Prior research has explored various channels of information spillover, including public announcements (Kim, 2003; Lang and Stulz, 1992), perceptions of fear and risk (Tsai, 2014; Long et al., 2021; Wang et al., 2021a), as well as price dynamics and volatility (Singh et al., 2010; Engelberg et al., 2018; Wang et al., 2021a). Building on this foundation, our study reveals that news sentiment from directly and indirectly interconnected companies (up to three hops) provides critical insights for inferring absent sentiment towards less-covered firms. The success of NIST-Cheb in deriving accurate inferred news sentiment scores that effectively guide stock price predictions positions news sentiment as a novel and influential form of spillover informa-

tion. These findings not only expand the understanding of information spillover mechanisms but also highlight their practical significance in enhancing predictive models and decision-making in financial markets.

In Chapter 5, our analysis shifts to examining the delayed information diffusion within market hierarchies. Utilizing our proposed FinPool models, which rely solely on the hierarchical feature representations of the markets and exclude direct inputs from domestic index sequences, we uncover a critical insight: the market index quote does not fully integrate all publicly available information from the industrial sectors, aligning with the observations of Hong et al. (2007). Specifically, our empirical findings reveal that half-hourly historical price data from component stocks and market sub-sectors are not promptly incorporated into current index quotes in the US stock market. This highlights a notable lag in the assimilation of information, with significant implications for understanding the real-time dynamics of index pricing and its role in developing effective trading strategies.

Overall, this thesis challenges the Efficient Market Hypothesis (EMH) by demonstrating persistent inefficiencies in information diffusion across related companies and market hierarchies. Our empirical evidence shows that public information, including price-related dynamics and news sentiment, from interconnected companies and hierarchical market structures is not fully absorbed into market prices immediately, contradicting the EMH's assertion of instant price adjustments (Fama, 1970). These delays, observed at both daily and intraday frequencies, suggest that financial markets are slower and less efficient than traditionally assumed. Our findings advocate for a more nuanced understanding of market efficiency, where delayed assimilation of information across networks and hierarchies enables opportunities for more effective trading strategies and improved financial modeling.

Chapter 7

Conclusions and Future Work

7.1 Conclusions

In this thesis, we explored the research topic of Company Relationship Modeling and Graph Neural Networks for Financial Market Forecasting. We addressed the initial research problem of Company Relationship Modeling by developing the Semantic Company Relationship Graph (SCRG). Building on the foundational knowledge of the SCRG, we then investigated the application of Graph Neural Networks to three areas of market forecasting: 1) stock movement prediction, 2) absent news sentiment inference, and 3) intraday market index movement prediction.

Specifically, in Chapter 2, the SCRG introduced leverages a GloVe-based novel company vector embedding scheme, which effectively captured investors' perceptions of company relationships based on real-world financial events. This approach addressed the limitations associated with indirect relationship modeling inherent in traditional correlation graphs and the discrete edge weights in WikiData graphs. Experimental results from various GNNs for financial market forecasting demonstrated the effectiveness of the SCRG in company relationship modeling.

In Chapter 3, we introduced the Non-IID Spatial-Temporal Graph Neural Network (NIST-GNN) as a non-IID extension of conventional ST-GNNs for financial predictions. The NIST-GNN was designed to efficiently handle the non-IID nature among different stock sequences, offering an innovative solution to a complex problem in financial analytics. Our comprehensive evaluation demonstrated that the NIST-GNN, in conjunction with the SCRG (NGNN-SCRG framework), not only outperformed leading benchmark models but also significantly improved the ability to identify profitable stocks while offering superior risk-adjusted returns.

In Chapter 4, we introduced a new research question regarding the assessment of investor sentiment on media silent days. To address this question, we investigated the use of the Non-IID Spatial-Temporal Chebyshev Network (NIST-Cheb) for inferring absent news sentiment, given that news sentiment is a crucial proxy for investor sentiment. Our statistics on news incompleteness showed that even highly visible companies lack daily news coverage on nearly half of the trading days. Traditional missing value imputation techniques fell short in the financial sector due to the complex spatial and temporal dependencies among companies. The NIST-Cheb we proposed leverages these dependencies to infer absent news sentiment. Its effectiveness has been verified through both error based metrics and real-world trading evaluations, focusing on risk-adjusted profits.

In Chapter 5, we introduced a new perspective on the task of predicting market index movements by relying on market microstructural information from the hierarchical representations of industrial clusters. Specifically, we first proposed a FinPool graph pooling operator for hierarchical feature integration of industrial clusters. To better deploy FinPool for market index movement predictions, we further developed the Stacked FinPool and Multi-tier FinPool frameworks. Empirical analysis has revealed significant enhancements in trading profits and risk-adjusted returns, signifying a substantial advancement over existing benchmark models.

7.2 Future work

Although our company relationship embedding scheme and the graph neural network innovations have demonstrated significant success in various financial market forecasting applications, several limitations remain.

First, our Semantic Company Relationship Graph (SCRG) is static, whereas company relationships are inherently dynamic. Future work could include modeling the SCRG dynamically to better capture evolving company interactions.

Second, the SCRG and graph neural network combinations such as NGNN-SCRG, NIST-Cheb, Stacked FinPool, and Multi-tier Attention FinPool are trained separately, with the SCRG used as a preprocessing step. An integrated, end-to-end training approach that allows for gradient propagation through both the SCRG and the various GNNs could enable more precise tuning of the SCRG with additional market information.

Another limitation is the binary nature of our stock and index movement predic-

tions. While these predictions offer valuable insights, they simplify the complex dynamics of price movements. Future research could extend to continuous predictions, which would provide a more detailed understanding of stock market dynamics by capturing not only the direction but also the magnitude of price changes.

Addressing these limitations could greatly enhance the adaptability of the SCRG framework to continuously evolving market conditions and various economic regimes. This would also allow GNNs to gain a deeper understanding of the information diffusion process through more nuanced company connections.

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